

Regulations and Curriculum for Master of Computer Applications (MCA)

Version 2025.01



(Established under Section 3 of UGC Act, 1956)
Placed under Category 'A' by MHRD, GoI | Accredited with 'A+' Grade by NAAC

Regulations and Curriculum for

Master of Computer Applications (MCA)

Choice Based Credit System (CBCS)

Applicable to the batch admitted from AY 2025-26 and onwards

Effective from AY 2025-26



Deemed to be University under Section 3 of UGC Act, 1956)
(Placed under Category 'A' by MHRD, Govt. of India, Accredited with 'A+' Grade by NAAC)
University Enclave, Medical Sciences Complex, Deralakatte,
Mangaluru – 575 018, Karnataka INDIA
Tel: +91-824-2204300/01/02/03, Fax: 91-824-2204305
Website: www.nitte.edu.in E-mail: info@nitte.edu.in

VISION

To build a humane society through excellence in education and healthcare

MISSION

To develop

Nitte (Deemed to be University)

*As a centre of excellence imparting quality education,
Generating competent, skilled manpower to face the scientific and social
challenges with a high degree of credibility, integrity,
ethical standards and social concern*

MCA SYLLABUS

Effective from
Academic Year
2025-26

Applicable to the batch admitted from AY 2025-26 and onwards

Version 2025.01

With Scheme of Teaching & Examination

REGULATIONS: 2025.01
for
Master of Computer Applications Program
UNDER OUTCOME BASED EDUCATION (OBE)
AND
CHOICE BASED CREDIT SYSTEM (CBCS)

Applicable to the batch admitted from AY 2025-26 and onwards.

Key Information

Program Title	Master of Computer Applications
Short Description	Two-year, Four Semester Choice Based Credit System (CBCS) type of postgraduate program as per NEP 2020, with English as medium of Instruction
Program Code	NMAMIT: 22CMPA20D2
	NMIT: 22CMPA14D3
Revision version	2025.01 These regulations may be modified from time to time as mandated by the policies of the University. Revisions are to be recommended by the Board of Studies for Computer Applications and approved by the Academic Council.
Effective from	01-08-2025
Approvals	Approved in the 61st Academic Council meeting of NITTE (Deemed to be University), held on 13-06-2025 and vide Notification Ref: N(DU)/REG/AC/2024-25/1395 dated 02.07.2025.
Program offered at	1. NMAM Institute of Technology, N(DU) Off -Campus Centre, Nitte, 574110, Karkala Taluk 2. Nitte Meenakshi Institute of Technology N(DU) Off -Campus Centre, Bangalore P.B.No.6429, Yelahanka, Bangalore - 560064
Grievance and dispute resolution	All disputes arising from this set of regulations shall be addressed to the Executive Council. The decision of the Executive Council is final and binding on all parties concerned. Further, any legal disputes arising out of this set of regulations shall be limited to jurisdiction of Courts of Mangalore only.

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**Regulations for Master of Computer Applications (MCA)
Nitte (Deemed to be University)**

PREAMBLE

Nitte (Deemed to be University) currently offers two-year MCA. The program is recognized by the AICTE and offered at NMAM Institute of Technology, Nitte & Nitte Meenakshi Institute of Technology, Off Campus Centers of Nitte (Deemed to be University) at Nitte and Bangalore respectively.

MCA is an educational program in Computer Applications leading to the award of Degree. It involves events/ activities, comprising of lecturers/ tutorials/ laboratory work/ field work, outreach activities/ project work/ vocational training/ viva/ seminars /internship/ assignments/ presentation/ quiz/ self-study, etc., or a combination of some of these. The core subjects of this program are designed to introduce students to the various areas of Computer Science and Engineering. The departmental laboratories provide good work ambiance with state-of-the-art computing facilities to enable the students to carry out projects and acquire expertise in emerging technologies. The students are offered special inputs in aptitude, soft skills, and overall personality development regularly to enhance their employability.

1. INTRODUCTION

Regulations for the MCA program under Nitte (Deemed to be University) govern the policies and procedures including selection, admission, imparting of instructions, conduct of examinations, evaluation and certification of candidate's performance and all amendments thereto, leading to the award of MCA degree. The regulations shall come into effect from the academic year 2025-26 and are applicable to the batch admitted from 2025-26 and onwards. This set of regulations shall be binding on all the candidates undergoing the said program.

2. ELIGIBILITY FOR ADMISSION

2.1 Entry requirements

Admission options	Entry requirements
Admission into two-year MCA Program	Passed BCA/ Bachelor's degree in Computer Science Engineering or equivalent Degree. OR Passed B.Sc./ B.Com./ B.A. with Mathematics at 10+2 level or at Graduation Level (with additional bridge courses as per the norms of the concerned University). Obtained at least 50% marks in the qualifying Examination.

- * Eligibility will be determined by the University based on evaluation of equivalency of qualification.
- * The candidates should have appeared for NUCAT Entrance Examination conducted by Nitte (Deemed to be University).

2.2 Qualifications from foreign countries

Candidates with qualifications from educational institutions outside of India, may be admitted to the program(s) subject to establishment of equivalence by the university. The Program Committee will evaluate and establish the eligibility of such candidates.

3. COURSE

A “Course” is defined as a unit of learning that typically lasts one semester, led by one or more teachers, for a fixed roster of students. Often referred to as a “subject” or “paper”, a course has identified course outcomes, modules / units of study, specified teaching-learning methods and assessment schemes. A course may be designed to include lectures, tutorials, practical, laboratory work, field work, project work, internship experiences, seminars, self-study components, online learning modules etc. in any combination of some of these.

3.1 Types of Courses

The following types of courses are included in the MCA Program:

3.1.1 Professional Core Courses (PCC): These are the professional Core Courses, relevant to the chosen specialization/ branch. The core courses shall be compulsorily studied by students, and it is mandatory to complete them to fulfill the requirements of a Program.

3.1.2 Professional Elective Courses (PEC): These are Professional Electives, relevant to the chosen specialization/branch. It shall be supportive to the discipline providing extended scope/enabling exposure to some to other discipline /domain and nurturing student proficiency skills.

3.1.3 Seminar (SEM): Each student has to present the seminar on a specific topic chosen from the relevant field /list provided by the department under the supervision of a faculty coordinator.

3.1.4 Internship (INT): The internship (a form of experimental learning) program is a workplace based professional learning experience that offers supervised exposure to real life work experience in an area related to field of study or career interest. An internship may be undertaken at a workplace such as industry/ R&D organization / Government organization, or any other reputed organization / institution recognized for the purpose by the University. The internship program not only helps fresh pass-outs in gaining professional know-how but also benefits corporate sectors. The internship also enhances the employability skills of the student passing out from Technical Institutions.

3.1.5 Project Work (PROJ): Provide experiential learning opportunities for students. Students are required individually, or in a small group, select and complete a project that may include review, design, development, curation, analysis, etc., with application of skills and knowledge relevant to area of study. Mini-project and Project work carried out at the parent institution, or any university / Government recognized organization/ Industry without affecting the regular class work are such courses.

3.1.6 Research Methodology (RM): Helps to understand the meaning, objectives and characteristics of research. Undergoing the course enables a student to select and define research problems. It helps students understand research designs, methods and also ethics in research and publications. The course also provides information on Intellectual property rights.

3.1.7 Mandatory Non-Credit Course (MNC): This course is mandatory for the students who have completed their bachelor's degree program in non-computer disciplines.

Typically, MCA program has the following component of courses.

SN	Category	Minimum Credits to be earned for the MCA degree
1	Core Courses (PCC)	57 (43+14 Lab)
2	Elective Courses (PEC) (Specializations)	15
3	Research Methodology (RM)	03
4	Seminar (SEM)	02
5	Mini Project (PROJ)	04
6	Internship (INT)	04
7	Major Project (PROJ)	15
	TOTAL	100

4. ACADEMIC YEAR

Refers to the sessions of two consecutive semesters (odd followed by an even) including periods of vacation.

5. COURSE REGISTRATION

The faculty advisors will guide the students in registering for courses. Course Registration refers to formal registration for the Courses of a semester by every student under the supervision of a faculty advisor (also called Mentor, Counselor etc.,) in each semester for the Institution to maintain proper record.

6. CREDITS

Refers to a unit by which the Course work is measured. It determines the number of hours of instructions required per week. One credit is equivalent to one hour of lecture or two hours of laboratory/ practical courses/ tutorials/ field work per week etc.

Credit Limits	Regular Semester
Minimum credits per Semester	16
Maximum credits per Semester	28

7. REGISTRATION FOR ELECTIVES

- 7.1 Elective options will be offered based on the specialization a student would like to pursue.
- 7.2 A student must take every course available in the specialization they have chosen.
- 7.3 Registration to elective options(specializations) will be on first-come first-serve basis.
- 7.4 Registration will be subject to minimum and maximum enrolment requirements specified by the department. Typically, the minimum enrolment requirement is 15.
- 7.5 If a course option is not offered due to lack of minimum enrolments, students who pre-register for it will be transferred to other course options, based on their preference, subject to availability.

8. REQUIREMENTS FOR PROGRESSION

- 8.1 A student may progress from Semester 1 to 2, from Semester 3 to 4 irrespective of the grades obtained in the courses of these odd Semesters.
- 8.2 A student may carry over not more than 4 courses from 1st & 2nd Semester (put together) to progress to 3rd Semester.

9. PROGRAM STRUCTURE

I SEMESTER														
Sl. No	Course Type and Course Code		Course Title	Teaching Department	Contact Hours/Week				SL#	Examination				Credits
					L	T	P	J		Duration in Hours	CIE Marks	SEE Marks	Total Marks	
1.	PCC	25MCA101	Data Structures with Algorithms	MCA	3	0	0		2	3	50	50	100	3
2.	PCC	25MCA102	Advanced Database Systems	MCA	3	0	0		2	3	50	50	100	3
3.	PCC	25MCA103	Data Communication and Networks	MCA	3	0	0		2	3	50	50	100	3
4.	PCC	25MCA104	Mathematical Foundation for Computer Applications	MCA	3	1	0		2	3	50	50	100	4
5.	PCC	25MCA105	Research Methodology and Publication Ethics	MCA	3	0	0		1	3	50	50	100	3
6.	PCC	25MCA106	Advanced Operating Systems with UNIX	MCA	3	1	0		2	3	50	50	100	4
7.	PCC	25MCA107	Data Structures with Algorithms Lab	MCA	0	0	4		2	3	50	50	100	2
8.	PCC	25MCA108	Advanced Database Systems Lab	MCA	0	0	4		2	3	50	50	100	2
9.	PCC	25MCA109	Data Communication and Networks Lab	MCA	0	0	4		2	3	50	50	100	2
Total					18	2	12	-	-	27	450	450	900	26
# Supervised Learning (SL): Additional Academic activities beyond class hours					-	-	-	-	17					
Learning hours per week					18	2	12	-	17					
Total notional learning hours (L+T+P+J+SL)			Per week		49									
			Per semester		49x16 = 784									

Bridge Course*

Sl. No	Course Type and Course Code		Course Title	Teaching Department	Contact Hours/Week				SL#	Examination				Credits
					L	T	P	J		Duration in Hours	CIE Marks	SEE Marks	Total Marks	
10.	MNC	25MCA110	Fundamentals of Programming – Bridge Course*	MCA	0	0	4			--	50	--	50	0

*The bridge course 25MCA110, Fundamentals of Programming is a mandatory noncredit Bridge Course only for the students who have completed their Bachelor's degree program in non-computer disciplines.

II SEMESTER														
Sl. No	Course Type and Course Code		Course Title	Teaching Department	Contact Hours/Week				S L #	Examination				Credits
					L	T	P	J		Duration in Hours	CIE Marks	SEE Marks	Total Marks	
1.	PCC	25MCA201	Enterprise Java	MCA	3	0	0		2	3	50	50	100	3
2.	PCC	25MCA202	Advanced Web Technology	MCA	3	0	0		2	3	50	50	100	3
3.	PCC	25MCA203	Software Engineering and Testing	MCA	3	1	0		2	3	50	50	100	4
4.	PCC	25MCA204	Data Warehousing and Data Mining	MCA	3	1	0		2	3	50	50	100	4
5.	PCC	25MCA205	Professional Communication Skills	HUM	1	1	0		2	3	50	50	100	2
6	PE C	FSD*	25MCA211	MCA	3	0	0		2	3	50	50	100	3
		DS**	25MCA221											
		IT***	25MCA231											
7	PE C	FSD*	25MCA212	MCA	3	0	0		2	3	50	50	100	3
		DS**	25MCA222											
		IT***	25MCA232											
8.	PCC	25MCA206	Enterprise Java Lab	MCA	0	0	4		2	3	50	50	100	2
9.	PCC	25MCA207	Advanced Web Technology Lab	MCA	0	0	4		2	3	50	50	100	2
10.	SEM	25MCA208	Technical Seminar and Report Writing	MCA	0	0	4		1	--	50	--	50	2
Total					19	3	12	-	-	27	500	450	950	28
# Supervised Learning (SL): Additional Academic activities beyond class hours					-	-	-	-	19					

Learning hours per week			19	3	12	-	19
Total notional learning hours (L+T+P+J+SL)	Per week		53				
	Per semester		53x16 = 848				

Audit Course

Sl. No	Course Type and Course Code		Course Title	Teaching Department	Contact Hours/Week				SL#	Examination				Credits
					L	T	P	J		Duration in Hours	CIE Marks	SEE Marks	Total Marks	
10.	AU	25MCAAU21/ 25MCAAU22	Online Course / Introduction Japanese Language	Any	2	0	2			--	50	--	50	0

* Specialization with Full Stack Development

** Specialization with Data Science

*** Specialization with Information Technology

Elective/Specialization Tracks:

Full Stack Development	
Code	Subject
25MCA211	Full Stack Development using Django
25MCA212	Web Development Using Ruby on Rails & PostgreSQL

Data Science	
Code	Subject
25MCA221	Applied Data Science and Analytics
25MCA222	Digital Image Processing and Pattern Recognition

Information Technology	
Code	Subject
25MCA231	Mobile Application Development using Android and Flutter
25MCA232	Block Chain Technology

III SEMESTER														
Sl. No	Course Type and Course Code		Course Title	Teaching Department	Contact Hours/Week				SL#	Examination				Credits
					L	T	P	J		Duration in Hours	CIE Marks	SEE Marks	Total Marks	
1.	PCC	25MCA301	Full Stack Web Development using MERN	MCA	3	0	0		2	3	50	50	100	3
2.	PCC	25MCA302	Artificial Intelligence and Machine Learning	MCA	3	0	0		1	3	50	50	100	3
3.	PCC	25MCA303	Cloud Computing and Big Data Analytics	MCA	3	1	0		1	3	50	50	100	4
4	PEC	FSD* 25MCA311	React Native and Kotlin for Mobile Application Development	MCA	3	0	0		1	3	50	50	100	3
		DS** 25MCA321	Health Care Analytics											
		IT*** 25MCA331	Industrial and Medical IoT											
5	PEC	FSD* 25MCA312	Angular and Spring Boot	MCA	3	0	0		1	3	50	50	100	3
		DS** 25MCA322	Generative AI and Prompt Engineering											
		IT*** 25MCA332	.NET Framework and C#											
6	PEC	FSD* 25MCA313	UI/UX Design & Product Management	MCA	3	0	0		2	3	50	50	100	3
		DS** 25MCA323	Natural Language Processing											
		IT*** 25MCA333	Digital and Social Media Marketing											
7.	PCC	25MCA305	Full Stack Web Development using MERN Lab	MCA	0	0	4		2	3	50	50	100	2
8.	PCC	25MCA306	Artificial Intelligence and Machine Learning Lab	MCA	0	0	4		2	3	50	50	100	2
9.	PRJ	25MCA307	Mini Project	MCA	0	0	0	12		3	50	50	100	4
Total					18	1	8	12	-	27	450	450	900	27
# Supervised Learning (SL): Additional Academic activities beyond class hours					-	-	-	-	12					
Learning hours per week					18	1	08	12	12					
Per week					51									

Total notional learning hours (L+T+P+J+SL)	Per semester		51x16 = 816
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- * Specialization with Full Stack Development
- ** Specialization with Data Science
- *** Specialization with Information Technology

Elective/Specialization Tracks:

Full Stack Development	
Code	Subject
25MCA311	React Native and Kotlin for Mobile Application Development
25MCA312	Angular and Spring Boot
25MCA313	UI/UX Design & Product Management

Data Science	
Code	Subject
25MCA321	Health Care Analytics
25MCA322	Generative AI and Prompt Engineering
25MCA323	Natural Language Processing

Information Technology	
Code	Subject
25MCA331	Industrial and Medical IoT
25MCA332	.NET Framework and C#
25MCA333	Digital and Social Media Marketing

IV SEMESTER												
Sl. No	Course Type and Course Code		Course Title	Teaching Department	Contact Hours	SL#	Examination					Credits
							Duration in Hours	CIE Marks	SEE Marks		Total Marks	
									Evaluation	Viva Voce		
1.	INT	25MCA401	Industry Internship ^	MCA	Full Time (4 weeks) 160 hours	-	3	50	--	50	100	4
2.	PRJ	25MCA402	Major Project ^^	MCA	Full Time (15 weeks) 600 Hours	-	3	100	100	100	300	15
Total						-	06	150	100	150	400	19
# Supervised Learning (SL): Additional Academic activities beyond class hours						-						
Total notional learning hours (L+T+P+J+SL)			Per Semester		160+600 = 760							

^ The internship should be completed before the commencement of the major project in the fourth semester. Total of 160 hours. The internship can also be done during the vacation period of the first or second semester.

^^The Major Project should be carried out preferably in the industry for a minimum duration of 15 weeks with total 600 Learning hours.

Note: PCC: Professional Core Course, PEC: Professional Elective Course
 L – Lecture, T – Tutorial, P- Practical, SL# – Supervised Learning
 CIE: Continuous Internal Evaluation
 SEE: Semester End Examination.

10. ATTENDANCE

- 10.1 MCA is a full-time program.
- 10.2 Students are not permitted to enroll in any other program offered by this or other University without prior permission.
- 10.3 All students must attend every lecture, tutorial and practical classes. To account for approved leave of absence (e.g., representing the University in sports, games or athletics, placement activities, NCC/NSS activities etc.) and/ or any other such contingencies like medical emergencies etc., the attendance requirement shall be a minimum of 85% of the classes conducted.
- 10.4 A candidate having shortage of attendance in one or more subjects shall have to repeat those courses. Such students shall re-register for the same subjects in the subsequent semester/ academic year. Semesters are subjected to maximum permissible credits.
- 10.5 A candidate, who does not satisfy the attendance requirement mentioned as above, in minimum number of credit requirement shall not be eligible to appear for the examination of that semester and not promoted to higher semester.

The candidate shall be required to repeat that semester along with regular students later.

- 10.6 If a candidate, for any reason, discontinues the course in the middle, he/she may be permitted to register to continue the course along with subsequent batch, subject to the condition that he/she shall complete the class work, lab work and seminar including the submission of dissertation within maximum stipulated period. Such candidate is not eligible to be considered for the award of rank.
- 10.7 The Head of the Department shall notify regularly, the list of such candidates who fall short of attendance. The list of the candidates falling short of attendance shall be sent to the principal with a copy to Controller of Examinations.

11. ABSENCE DURING THE SEMESTER

- 11.1 Leave of Absence: If the period of leave is more than two days and less than three weeks, prior application for leave shall have to be submitted to the Head of the Department concerned, with the recommendation of the faculty-Advisor stating fully the reasons for the leave request along with supporting documents. It will be the responsibility of the student to intimate the course instructors, Head of the Department and also Chief Warden of the hostel, regarding his/her absence before availing leave.
- 11.2 Absence during Mid-Semester Examinations: A student who has been absent from a Mid-Semester Examination (MSE) due to illness and other contingencies may give a request for additional MSE within two working days of such absence to the office of the respective Head of the Department (HOD) with necessary supporting documents and certification from authorized personnel. The HOD may consider such requests depending on the merits of the case and may permit an additional Mid-Semester Examination for the concerned student.
- 11.3 Absence during Semester End Examination: In case of absence for a Semester End Examination, on medical grounds or other special circumstances the student can apply for 'I' grade in that course with necessary supporting documents and certifications by authorized personnel to the Controller of Examination through Chairman of The Department. The Controller of Examination may consider the request depending on the merits of the case and permit the make-up Semester End Examination for the concerned student. The student may subsequently complete all course requirements within the date stipulated by DPGC (which may be extended till first week of next semester under special circumstances) and 'I' grade will then be converted to an appropriate letter grade. If such an application for the 'I' grade is not made by the student, then a letter grade will be awarded based on his in-semester **performance.**

12. WITHDRAWAL FROM THE PROGRAM

12.1 Temporary Withdrawal

A student who has been admitted to a Postgraduate Degree program of the College may be permitted to withdraw temporarily, for a period of one semester or more on the grounds of prolonged illness or grave calamity in the family etc. The student should abide by the applicable rules and regulations of the college/University at the time of Temporary Withdrawal.

12.2 Permanent Withdrawal

Any student who withdraws admission before the closing date of admission for the Academic Session is eligible for the refund of the deposits only. Fees once paid will not be refunded on any account. Once the admission for the year is closed, the following conditions govern withdrawal of admissions: a) A student who wants to leave the College for good, will be permitted to do so (and can take Transfer Certificate from the College, if needed), only after remitting the Tuition fees as applicable for all the remaining semesters and clearing all other dues, if any. b) Those students who have received any scholarship, stipend or other forms of assistance from the College shall repay all such amounts in addition to those mentioned in (a) above. The decision of the Principal of the Institute regarding withdrawal of a student is final and binding.

13. EVALUATION

13.1 Course Evaluation

For all courses evaluation will be based on both formative assessment (Continuous Internal Evaluation, CIE) and summative assessment (Semester End Evaluation, SEE). CIE and SEE will carry 50 % and 50% respectively, to enable each course to be evaluated for 100 marks, irrespective of its credits. Weightage for CIE and SEE will be 50% each.

13.2 Continuous Internal Evaluation (CIE)

CIE refers to the evaluation of students' achievement in the learning process. The course instructor will perform the Continuous Internal Evaluation (CIE) for 50 marks which includes tests, homework, problem solving, group discussion, quiz, mini-project and seminar throughout the semester, with weightage for the different components being fixed at the University level as follows.

- | | | |
|---|---|----------|
| 1. Quizzes, tutorials, assignments, etc., | - | 20 marks |
| 2. Mid Semester Examination | - | 30 marks |

An additional MSE may be conducted for those students absent for valid reasons/ with prior permission.

For those students who could not score minimum required CIE marks (25 marks), an additional MSE may be conducted, however the maximum CIE marks shall be restricted to 25 out of 50.

13.3 Semester End Examinations (SEE)

Refers to examination conducted at the University level covering the entire Course Syllabus. The Syllabi is modularized and SEE questions are selected from each module, with a choice confined to the concerned module only. SEE is also termed as university examination.

13.4 For Technical Seminar and Fundamentals of Programming, the evaluation will be based on formative assessment (Continuous Internal Evaluation) only.**13.5 For the major project, the dissertation will be evaluated by two examiners, one of the examiners shall be the guide of the candidate and the other examiner shall be preferably an external expert in the area of the dissertation being evaluated. The evaluation of the dissertation shall be made independently by each examiner.**

- a) Examiners shall evaluate the dissertation normally within a period of not more than two weeks from the date of receipt of dissertation through email.
- b) The examiners shall independently submit the marks for the dissertation during the viva-voce examination date.
- c) Sum of marks awarded by the two examiners shall be the final evaluation marks for the dissertation.
- d) Viva-Voce examination of the candidate shall be conducted, if the dissertation work and the reports are accepted by the external examiner.
- e) If the external examiner finds that the dissertation work is not up to the expected standard and the minimum passing marks cannot be awarded, the dissertation shall not be accepted for SEE.
- f) If the dissertation is rejected during the Project work, then the second examiner (external) will be appointed by the COE against whom the candidate has to re-present the same dissertation. The decision of the Second Examiner (external) will be final.
- g) If the second examiner (external) accepts the dissertation, then the viva-voce examination of the candidate shall be conducted as per the norms. If the second examiner (external) rejects the dissertation, then the student has to take an extension for a minimum period of 3 months and re-work on the project. After the completion of the extension period, viva-voce examination of the candidate shall be conducted as per the norms, if the dissertation work is accepted by the external examiner.
- h) Viva-voce examination of the candidate shall be conducted jointly by the external examiner and internal examiner/guide at a mutually convenient date.
- i) The relative weightages for the evaluation of dissertation and the performance at the viva-voce shall be as per the scheme of teaching and examination.
- j) The marks awarded by both the examiners at the viva-voce examination shall be sent jointly to the office of the Controller of Examination immediately after the examination.

- k) The candidates who fail to submit the dissertation work within the stipulated time have to apply for the extension of the Project duration through the Guide and the head of the department to the office of the Controller of Examination. Such candidate is not eligible to be considered for the award of rank.

14. THE TRANSITIONAL GRADES

'I', 'W' and 'X' would be awarded in the following cases. These would be converted into one or the other of the letter grades (O to F) after the student completes the course requirements.

- 14.1 **Grade “I”:** To a student having attendance $\geq 85\%$ and CIE $\geq 70\%$, in a course, but remained absent from SEE for valid & convincing reasons acceptable to the College, like:
- i. Illness or accident, which disabled him/her from attending SEE.
 - ii. A calamity in the family at the time of SEE, which required the student to be away from the College.
 - iii. However, the committee chaired by the Principal is authorized to relax the requirement of CIE $\geq 70\%$ if the student is hospitalized or advised to have long term rest after discharge from the hospital by the Doctor.
 - iv. Students who remain absent for Semester End Examinations due to valid reasons and those who are absent due to health reasons are required to submit the necessary documents along with their request to the Controller of Examinations to write Make up Examinations within 2 working days of that examination for which he or she is absent, failing which they will not be given permission.
- 14.2 **Grade “W”:** To a student having satisfactory attendance at classes but withdrawing from that course before the prescribed date in a semester as per faculty advice.
- 14.3 **Grade “X”:** To a student having attendance $\geq 85\%$ and CIE $\geq 70\%$, in a course but SEE performance could result in a 'F' grade in the course. (No “F” grade awarded in this case, but student's performance record will be maintained separately).
- 14.4 The Make Up Examination facility would be available to students who may have missed attending the SEE of one or more courses in a semester for valid reasons and given the 'I' grade. Also, students having the 'X' grade shall also be eligible to take advantage of this facility. The makeup examination would be held as per dates notified in the Academic Calendar. However, it should be made possible to hold a make-up examination at any other time in the semester with the permission of the Academic Council of the College. In all these cases, the standard of SEE would be the same as the normal SEE.
- Make Up examination will be conducted for the candidates who has a CIE ≥ 35 marks and may have missed to attend the SEE covering the entire course syllabus. The standard of the Make Up Examination is same as that of the SEE.

- All the 'W' grades awarded to the students would be eligible for conversion to the appropriate letter grades only after the concerned students re-register for these courses in a main/summer semester and fulfill the passing standards for their CIE and (CIE+SEE).
- The suggested passing standards are CIE to have $\geq 50\%$ and CIE+SEE to have a grade better or at least equal to C. For maintaining high standards, the students scoring less than 50% in CIE are advised to withdraw and to re-register for the course when offered next. The letter grade 'W' to be entered in the grade card against the subject and not to be taken into account while calculating SGPA & CGPA

14.5 Summer / Fast Track Semester

- i) The students who have satisfied CIE and attendance requirements for the course/s and obtained an F grade in SEE are permitted to appear directly in ensuing examination/s as backlog paper/s. The students need not re-register for such course/s in the summer / fast track semester. In case the student wishes to improve CIE he/she has to re-register for the summer / regular semester as and when offered next.
- ii) The student who obtains required attendance and CIE in the summer semester, but obtains an 'F' grade in SEE; is permitted to appear for SEE subsequently as backlog course/s. The student need not repeat the course for Attendance and CIE.
- iii) The course/s for which the student does not possess satisfactory attendance and CIE score shall be marked as 'N' on the grade sheet. Such students are not permitted to SEE for the courses marked as 'N' on the grade sheet. The students have to re-register only for course/s marked as 'N' in the summer /subsequent semester whenever that course is offered and obtain the required CIE and attendance. Subsequently, they are eligible to appear for SEE in such course/s.
- iv) Courses with Transitional Grades viz "W", "I" and "X" are also eligible to register in the summer semester in case they wish to improve their score in CIE.
- v) All courses may not be offered in the summer semester. It is the discretion of the University to offer the courses based on the availability of resources. The institutes shall notify timetable for the summer semester well in advance.
- vi) Summer semester is optional; it is for the student to make the best use of the opportunity.
- vii) A student is permitted to register for maximum of 16 credits in the Summer / fast track semester.
- viii) A student has to choose those courses which are offered by the Institution in a given summer semester.
- ix) In the summer semester, each course needs to be offered for the required number of lectures/ tutorial/ laboratory hours as prescribed in the syllabus.

14.6 SUPPLEMENTARY EXAMINATION: Refers to the examination conducted to assist slow learners and/or failed students through make up courses for duration of 8 weeks. This comprises of both the CIE & SEE and will be conducted after the completion of First year MCA even semester.

15. QUALIFYING STANDARD

Sessional (CIE): Score: $\geq 50\%$ (≥ 25 marks)

Terminal (SEE): Score: $\geq 40\%$ (≥ 20 marks)

For securing a final Pass:

Total 50 % of the Course maximum marks (100) i.e., sum of the CIE and SEE marks prescribed for the Course is desired.

16. GRADING SYSTEM: ABSOLUTE GRADING

The performance of a candidate in a course shall be evaluated according to a Letter Grading System, based on both CIE and SEE. The letter grades (O, A+, A, B+, B, C and F) indicate the level of academic achievement assessed on a 10-point scale (0 to 10) (See Table below).

Letter grade system and corresponding marks range

Marks Range (%)	Grade Point	Letter Grade	Descriptor
90 & above	10	O	Outstanding
80-89	9	A+	Excellent
70-79	8	A	Very Good
60-69	7	B+	Good
55-59	6	B	Above Average
50-54	5	C	Average
Below 50	0	F	Fails
Absent	0	F	Ab

CGPA	Classification
7.00-& above	First Class with Distinction
6.00-6.99	First Class
5.00-5.99	Second Class
CGPA < 5.00	Fails

- 16.1 A candidate shall be considered to have completed a course successfully and earned the credits assigned, if he/she secures an acceptable letter grade in the range O to C.
- 16.2 The letter grade 'F' in any course implies failure in that course and no credit is earned.
- 16.3 A candidate having satisfactory attendance at classes and meeting the passing standard at CIE in a course but remained absent from SEE shall be awarded 'I' grade in that course. No credit is earned in such a case.
- 16.4 Grade point averages: The overall performance of a candidate will be indicated by Grade Point Average (GPA).
- 16.5 Semester Grade Point Average (SGPA) is computed as follows.

$$SGPA = \frac{[\sum(CourseCredit) \times (GradePoint)]}{[\sum(CourseCredit)]}$$

All courses of the semester for which a letter grade has been awarded, including F, will be included in this calculation.

- 16.6 Cumulative Grade Point Average (CGPA) is computed as follows.

$$CGPA = \frac{[\sum(CourseCredit) \times (GradePoint)]}{[\sum(CourseCredit)]}$$

All courses of the all the semesters for which a letter grade has been awarded, excluding F, will be included in this calculation.

- 16.7 Grade Card: Based on the secured letter grades, grade points, SGPA and CGPA, a grade card for each semester shall be issued. On specific request on paying prescribed fee, a transcript indicating the performance in all semesters may be issued.
- 16.8 Conversions of Grades into Percentage and Class Equivalence
 Conversion formula for the conversion of CGPA into percentage is given below:
 Percentage of marks secured, 'P' = CGPA Earned $\times$ 10
 Illustration: for A CGPA of 8.18:
'P' = CGPA Earned 8.18 $\times$ 10 = 81.8 %

17. AWARD OF CLASS

- 17.1 The candidate, who has passed all the courses prescribed, shall be declared to have passed the program.
- 17.2 A candidate who secures $CGPA \geq 7.00$ and above shall be declared to have passed in 'First Class with Distinction'.
- 17.3 A candidate who secures $CGPA \geq 6.00$ or more but less than 7.00 shall be declared to have passed in 'First Class'.
- 17.4 A candidate who secures $CGPA \geq 5.00$ or more but less than 6.00 shall be declared to have passed in 'Second Class'.
- 17.5 An attempt means the appearance of a candidate in one or more courses either in part or full in a particular re-examination including supplementary semester's examinations.
- 17.6 A candidate who fails in the main examination and passes one or more subjects/courses or all subjects/courses in the supplementary examination, such candidate's attempts shall be considered as multiple attempts.
- 17.7 If a candidate submits application for appearing for the regular examination but does not appear for any of the courses/subjects in the regular University examination, he can appear for supplementary examination provided other conditions such as attendance requirement, internal assessment marks, etc. are fulfilled and his appearing in the supplementary examination shall be considered as the first attempt.

18. MERIT CERTIFICATES AND UNIVERSITY GOLD MEDALS

Merit Certificates and University Gold Medals will be awarded on the basis of overall CGPA.

- 18.1 Only those candidates who have completed the MCA Program and fulfilled all the requirements in the minimum number of years prescribed (i.e., 2 years for master's degree) and who have passed each semester in the first attempt is eligible for the award of Merit Certificates and /or University Gold Medals.
- 18.2 Award of University Gold Medals, if any, are governed by the specific selection criteria that may be formulated by the University for such Medals / Awards
- 18.3 Candidates with W, N, I, X and F grades and passes the courses in the supplementary examinations are not eligible for the award of Gold Medal or Merit Certificate.

19. RULES FOR GRACE MARKS

Grace marks up to 1% of the maximum total marks of the courses for which he/she is eligible and have registered (non-credit courses excluded) in the examination or 10 marks whichever is less shall be awarded to the failed course(s), (with a restriction of a maximum of 5 marks per course) provided on the award of such grace marks the candidate passes in that course(s).

20. CHALLENGE EVALUATION

If a student is not satisfied with the marks allotted to him/her in the semester end examinations, he/she could apply for challenge evaluation within the prescribed time specified. In such cases the answer papers will be valued by the DPGC committee and marks secured by the students in the challenge evaluation will be final.

21. AWARD OF DEGREE REQUIREMENTS

The Degree requirements of a student for the MCA Degree program are as follows:

- 21.1 The maximum duration for a student to comply with the Degree requirements is 8 semesters from the date of first registration for his first semester.
- 21.2 A student shall be declared to have completed the Degree of Master of Computer Applications, provided the student has undergone the stipulated course work as per the regulations and has earned the prescribed credits, as per the scheme of teaching and examination of the program.
- 21.3 A student shall be declared successful at the end of the program for the award of Degree only on obtaining $CGPA \geq 5.00$, with none of the courses remaining with F grade.
- 21.4 In case the CGPA falls below 5.00, the student shall be permitted to appear again for SEE for required number of courses (other than seminar, practical, internship and project) subject to the provision of university, to make up $CGPA \geq 5.0$. The student should reject the SEE results of previous attempt and obtain written permission from the Controller of Examinations to reappear to the subsequent SEE.

22. TERMINATION FROM THE PROGRAM/RE-ADMISSION

A student shall be required to leave the college without the award of the degree, under the following circumstances:

- i. Failing to complete the degree requirements in double the duration of the program.
- ii. Based on the disciplinary actions suggested by the Academic Council/Governing Council

23. GRADUATION REQUIREMENTS AND CONVOCATION

- 23.1 A student shall be declared to be eligible for the award of the Degree if he has Fulfilled Degree Requirements
- 23.2 No Dues to the College, Departments, Hostels, Library Central Computer Centre and any other center
- 23.3 No disciplinary action pending against him.
- 23.4 The award of the Degree must be recommended by the Academic council and approved by Governing Council of Nitte (DU)

- 23.5 **Convocation:** Degree will be awarded in person to the students who have graduated during the preceding academic year. Degrees will be awarded in absentia to such students who are unable to attend the convocation. Students are required to apply for the convocation along with the prescribed fees, after having satisfactorily completed all the degree requirements within the specified date in order to arrange for the award of the degree during convocation.

24. CONDUCT AND DISCIPLINE

- 24.1 Students shall conduct themselves within and outside the premises of the Institute, in a manner befitting the students of an Institution of National Importance.
- 24.2 As per the order of Honorable Supreme Court of India, ragging in any form is considered as a criminal offence and is banned, any form of ragging will be severely dealt with.
- 24.3 The following acts of omission/ or commission shall constitute gross Violation of the code of conduct and are liable to invoke disciplinary measures:
- Ragging
 - Lack of courtesy and decorum; indecent behavior anywhere within or outside the campus.
 - Willful damage or stealthy removal of any property /belongings of the Institute /Hostel or of fellow students/ citizens
 - Possession, consumption or distribution of alcoholic drinks or any kind of hallucinogenic drugs.
 - Mutilation or unauthorized possession of Library books.
 - Noisy and unseemly behavior, disturbing studies of fellow Students.
 - Hacking in computer systems (such as entering into other Person's area without prior permission, manipulation and/or Damage of computer hardware and software or any other Cybercrime etc.,).
 - Plagiarism of any nature.
 - Any other act of gross indiscipline as decided by the University from time to time.
 - Smoking in College Campus and use of any tobacco products.
 - Unauthorized fund raising and promoting sales
- 24.4 Commensurate with the gravity of offense, the punishment may be: reprimand, expulsion from the hostel, debarment from an examination, disallowing the use of certain facilities of the College, rustication for a specified period or even outright expulsion from the College, or even handing over the case to appropriate law enforcement authorities or the judiciary, as required by the circumstances.
- For an offence committed in
 - A hostel
 - A department or in a classroom
 - Elsewhere,the Chief Warden, the Head of the Department and the Dean (Students Welfare), respectively, shall have the authority to reprimand or impose fine.
 - All cases involving punishment shall be reported to the Head of the institution.
- 24.5 Cases of adoption of unfair means and/or any malpractice in an examination shall be reported to the Controller of Examination.

- **Note:** Students are required to be inside the examination hall 20 minutes before the commencement of examination. This is applicable for all examinations (Semester end/Supplementary/makeup) henceforth. Students will not be allowed inside the examination hall after the commencement, under any circumstances.

25. PROGRAM OUTCOMES

By the end of the program the student will be able to acquire:

- 25.1 **PO1 (Foundation Knowledge):** Apply knowledge of mathematics, programming logic and coding fundamentals for solution architecture and problem solving.
- 25.2 **PO2 (Problem Analysis):** Identify, review, formulate and analyze problems for primarily focusing on customer requirements using critical thinking frameworks.
- 25.3 **PO3 (Development of Solutions):** Design, develop and investigate problems with as an innovative approach for solutions incorporating ESG/SDG goals.
- 25.4 **PO4 (Modern Tool Usage):** Select, adapt and apply modern computational tools such as development of algorithms with an understanding of the limitations including human biases.
- 25.5 **PO5 (Individual and Teamwork):** Function and communicate effectively as an individual or a team leader in diverse and multidisciplinary groups. Use methodologies such as agile.
- 25.6 **PO6 (Project Management and Finance):** Use the principles of project management such as scheduling, work breakdown structure and be conversant with the principles of Finance for profitable project management.
- 25.7 **PO7 (Ethics):** Commit to professional ethics in managing software projects with financial aspects. Learn to use new technologies for cyber security and insulate customers from malware
- 25.8 **PO8 (Life-long learning):** Change management skills and the ability to learn, keep up with contemporary technologies and ways of working.

26. PROGRAM SPECIFIC OUTCOMES

- 26.1 To empower the students to exemplify themselves in the field of Information Technology ensuring credibility, integrity and ethical standards.
- 26.2 Adopt research practices and trending technologies such as Artificial Intelligence and Machine Learning, Internet of Things, Cloud Computing and Data Analytics to solve real world problems.

DATA STRUCTURES WITH ALGORITHMS			
Course Code	25MCA101	Course Type	PCC
Teaching Hours/Week (L: T: P: S)	3:0:0:0	Credits	03
Total Teaching Hours	40	CIE + SEE Marks	50+50
Teaching Department: Master of Computer Applications			
Course Objectives:			
1.	Explain the fundamental concepts of data structures, including primitive and non-primitive data structures, abstract data types, and the basics of algorithm analysis with asymptotic notations		
2.	Implement and manipulate arrays, linked lists, stacks, and queues using C programming to perform operations such as insertion, deletion, traversal, and searching efficiently.		
3.	Apply recursion and stack-based techniques for problem-solving, including evaluating postfix expressions, infix-to-postfix conversion, and solving problems like factorial, Fibonacci, and Tower of Hanoi.		
4.	Analyze and implement binary trees and graphs, perform various tree traversal techniques, and apply graph algorithms such as BFS, DFS, Prim's, Kruskal's, and Dijkstra's algorithms for real-world applications.		
5.	Develop and compare sorting algorithms like Bubble Sort, Quick Sort, Merge Sort, and Shell Sort, and implement searching algorithms including Sequential Search, Binary Search, and Interpolation Search to optimize data retrieval.		
UNIT-I			
Introduction			04 Hours
Introduction to data structure, Primitive and Non-Primitive Data Structures, Abstract data type, Arrays, Pointers			
Algorithm Analysis			04 Hours
Fundamentals of Algorithm Design and Analysis, Analysis Framework, Time Complexity and Space Complexity, Asymptotic Notations: Big O, Omega (Ω), Theta (Θ). Basic efficiency classes,			
UNIT-II			
Lists			08 Hours
Introduction to Linked lists, Types of linked lists, C implementation of Singly Linked Lists, Doubly linked lists, Circular lists.			
UNIT-III			
Stack			06 Hours
Definition and Primitive operations, Representing and implementing stack using Arrays and Linked Lists, Applications of stacks: Evaluating a postfix expression, converting an expression from infix to postfix, Recursion: definition and processes, Implementation of recursion program in C, Examples (Factorial function, Multiplication of natural numbers, Fibonacci sequence, Binary search, Towers of Hanoi problem).			
Queues			02 Hours
Definition and Primitive operations, Implementation of ordinary queues and circular queues using Arrays and Linked lists			
UNIT-IV			
Binary trees			05 Hours

Basic Terminology, Operations on binary trees, Applications of binary trees. Binary tree representation: Node representation of binary tree, Array representation of binary trees, Binary tree traversals and related properties, Threaded binary trees.											
Graphs										03 Hours	
Definitions, Representation: Adjacency Matrix, Adjacency List, Graph Traversal Techniques: BFS, DFS. Application of graphs - Greedy Technique, Prim's Algorithm, Kruskal's Algorithm, Dijkstra's Algorithm.											
UNIT-V											
Sorting										05 Hours	
Exchange Sort: Bubble Sort, Selection Sort, Quick Sort. Divide-and-Conquer: Merge sort. Decrease-and-Conquer - Insertion Sort, Shell sort. Time Complexity Comparison.											
Searching										03 Hours	
Sequential search, Indexed sequential search, Binary search, Interpolation Search.											
Course Outcomes: At the end of the course student will be able to											
1.	Explain the fundamental concepts of Data Structures and the basics of Algorithm Analysis.										
2.	Implement Linear Data Structures such as Arrays, Linked lists.										
3.	Apply and Implement Recursive and Non-Recursive Techniques.										
4.	Analyze and Implement Binary Trees and Graphs										
5.	Develop, Implement and Compare different Searching and Sorting Techniques.										
Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes											
Program Outcomes→		1	2	3	4	5	6	7	8	PSO	PSO
↓ Course Outcomes										1	2
25MCA101.1		3	3	-	-	-	-	-	-	3	2
25MCA101.2		3	3	3	2	3	-	-	-	3	2
25MCA101.3		3	3	3	3	3	-	-	-	3	2
25MCA101.4		3	3	3	3	3	-	-	-	3	2
25MCA101.5		3	3	3	3	3	-	-	-	3	2
1: Low 2: Medium 3: High											
TEXTBOOKS:											
1.	Yedidyah Langsam and Moshe J Augenstein and Aaron M Tenenbaum: Data Structures using C and C++, 2 nd Edition, Pearson Education Asia										
2.	Reema Thareja – "Data Structures Using C", 2 nd Edition, Oxford University press										
3.	Jean-Paul Tremblay: An Introduction to Data Structures with applications, Tata McGraw Hill										
4.	Ellis Horowitz and Sartaj Sahni, Fundamentals of Data Structures in C, 2 nd Edition, Universities Press.										
5.	Introduction to the Design and Analysis of Algorithms. Anany Levitin, Pearson Education, 2 nd Edition.										
6.	Horowitz E., Sahani S., Rajashekharan S: Computer Algorithm, Galgotia Publications.										
REFERENCE BOOKS:											
1.	Robert L Kruse : Data Structures and Program Design , 3 rd Edition, Prentice – Hall of India.										
2.	Mark Allen Weiss : Data Structures and Algorithm Analysis in C, 2 nd Edition, Pearson Education Asia.										
3.	Yashavant Kanetkar – "Data Structures Through C", BPB Publications.										

ADVANCED DATABASE SYSTEMS				
Course Code		25MCA102	Course Type	PCC
Teaching Hours/Week (L: T: P:S)		3:0:0:0	Credits	03
Total Teaching Hours		40	CIE + SEE Marks	50+50
Teaching Department: Master of Computer Applications				
Course Objectives:				
1.	Understand the importance of database management systems and have thorough understanding of terminologies used.			
2.	Implement concepts of relational model using SQL.			
3.	Use the features of PL/SQL to write procedural programs.			
4.	Design the databases and to use different levels of normalization.			
5.	Understand the working of NoSQL, and implement features of MongoDB.			
UNIT-I				
Essentials of Databases and ER Modeling				08 Hours
Introduction, Characteristics of Database approach, Advantage of using DBMS approach, Disadvantages of DBMS. Data Models, Three Schema Architecture, Entity Types, Entity Sets, Attributes and Keys, Relationship types, Relationship Sets, Roles and Structural Constraints, Weak Entity Types, Refining the ER Design.				
UNIT-II				
SQL - The Relational Database Standard :				08 Hours
SQL Data Definition and Data Types, Specifying basic constraints in SQL, DDL, DML, DQL statements in SQL, Aggregate functions, Sub-queries and Co-related sub-queries, LIKE, Inner and Outer Joins, Views in SQL.				
UNIT-III				
Procedural Programming with MySQL:				08 Hours
Introduction stored programs, Writing procedural code, COMMIT, ROLLBACK and save points, Implicit and explicit locking, Stored Procedures, Stored functions, Triggers, Cursors.				
UNIT-IV				
Structured Database Design and Transaction Processing				08 Hours
Informal Design Guidelines for Relation Schema's, Functional Dependencies, Normal Forms (1NF, 2NF and 3NF), Closure sets, Decomposition. Introduction to Transaction Processing, Transaction States, Desirable Properties of Transactions.				
UNIT-V				
Foundations of NoSQL and MongoDB				08 Hours
Introduction to NoSQL, Types of NoSQL databases, Advantages of NoSQL, Comparison of SQL and NoSQL. What is MongoDB? Why MongoDB? Using JSON, Creating or generating unique key, Support for dynamic queries, Terms used in RDMS and MongoDB, Data types in MongoDB, MongoDB Query Language: Inserting, updating and deleting documents, Retrieving documents, Dealing with Null values, Count, Limit, Sort and Skip methods.				
Course Outcomes: At the end of the course student will be able to				
1.	Analyze the characteristics of database systems and apply Entity-Relationship (ER) modeling to design structured databases.			

2.	Construct and manipulate relational databases using SQL statements, including constraints, joins, views, and aggregate functions.
3.	Develop stored procedures, functions, triggers, and cursors in MySQL to implement procedural database programming.
4.	Evaluate functional dependencies and apply normalization techniques to design efficient and consistent relational schemas.
5.	Differentiate between SQL and NoSQL databases, and demonstrate the ability to perform CRUD operations in MongoDB using JSON-based queries.

Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes

Program Outcomes→	1	2	3	4	5	6	7	8	PSO	PSO
↓ Course Outcomes									1	2
25MCA102.1	3	-	3	2	3	2	-	3	3	-
25MCA102.2	3	-	3	3	2	3	-	3	3	-
25MCA102.3	3	-	2	3	2	3	-	3	3	-
25MCA102.4	3	-	2	3	2	3	-	3	3	-
25MCA102.5	3	-	3	3	3	3	-	3	3	-

1: Low 2: Medium 3: High

TEXTBOOKS:

1. Elmasri and Navathe: Fundamentals of Database Systems, 7th Edition 2016
2. Joel Murach: Murach's MySQL, 4th Edition 2023
3. Seema Acharya, Subhashini Chellappan, "Big Data Analytics", 2nd Edition, Wiley, 2016

REFERENCE BOOKS

1. Silberschatz, Korth and Sudarshan: Database Systems Concepts, 6th Edition, McGraw-Hill
2. Raghu Ramakrishnan and Johannes Gehrke : Database Management Systems, 6th Edition, McGraw-Hill
3. Ivan Bayross : Commercial Application Development using Oracle Developer 2000

E Books / MOOCs/ NPTEL

1. NOC : Fundamentals of Database Systems(Course sponsored by Aricent), IIT Kanpur, Dr. Arnab Bhattacharya : <https://nptel.ac.in/courses/106104135>
2. MySQL® Notes for Professionals book: <https://www.goalkicker.com/MySQLBook/>
3. Learning MongoDB: https://www.dbooks.org/learning-mongodb-5605980453/#google_vignette
4. Oracle PL/SQL Language Pocket Reference, 5th Edition: A Guide to Oracle's PL/SQL Language Fundamentals.
<https://www.pdfdrive.com/oracle-plsql-language-pocket-reference-5th-edition-a-guide-to-oracles-plsql-language-fundamentals-d167357800.html>

DATA COMMUNICATION AND COMPUTER NETWORKS			
Course Code:	25MCA103	Course Type:	PCC
Teaching Hours/Week (L: T: P: S)	3:0:0:0	Credits:	03
Total Teaching Hours:	40	CIE + SEE Marks:	50+50
Teaching Department: Master of Computer Applications			
Course Objectives: This Course will enable students to			
1.	To introduce fundamental concepts of data communication, networking architectures, protocols, encoding techniques, and error detection mechanisms for reliable enterprise communication.		
2.	To provide an understanding of data link control mechanisms, flow control, error handling techniques, and key network services like email, DNS, and HTTP for efficient and reliable communication.		
3.	To explore transport layer services and protocols, focusing on process-to-process communication, flow and error control, congestion control, and key transport protocols such as UDP, TCP, and SCTP.		
4.	To understand congestion control techniques, traffic management strategies, and congestion handling mechanisms in packet-switched networks, including TCP and Datagram Congestion Control Protocol (DCCP).		
5.	To introduce security protocols, cryptographic algorithms, and multimedia information processing techniques, including data compression and image/video coding for secure and efficient communication.		
UNIT-I			
Introduction:			02 Hours
Data Communication and Networking for today's Enterprise, A communication Model, Data Communication, Network Types: LAN, WAN, Internet, Internet Architecture.			
Networks Protocols and Architecture:			02 Hours
The Need for a protocol Architecture, A simple protocol Architecture, TCP/IP Protocol Architecture : The TCP/IP layers.			
Data Encoding and Communication Interfaces:			04 Hours
Digital Data & Signals: NRZ, Manchester, Multilevel binary, Biphasic, Modulation rate, Differential Encoding, Scrambling Techniques, Analog & Digital Signal Encoding: Modulation Techniques, Pulse Code Modulation, Delta Modulation, Types of errors, Error Detection & Correction: Parity Check, CRC, Forward error correction.			
UNIT-II			
Data Link Control:			04 Hours
Flow Control: Stop and Wait Sliding Window Flow Control; Error Control: Stop and Wait ARQ, GO-Back-N ARQ, Selective-Reject ARQ; High-level Data Link Control: Basic Characteristics, Frame Structure.			

Application Layer:										04 Hours																																																								
SMTP, MIME- Internet Mail Architecture, POP, IMAP, Internet Directory Services- DNS, The DNS Database, DNS operations, Web Access and HTTP: HTTP Overview, Messages, Request Message, Response Messages, Entities.																																																																		
UNIT-III																																																																		
The Transport Layer:										08 Hours																																																								
Transport Layer Services: Process-to-Process Communication, Addressing: Port Numbers, Encapsulation and Decapsulation, Multiplexing and Demultiplexing, Flow control, Error control, Combination of flow control and error control, Congestion control, Connectionless and connection-Oriented protocol.																																																																		
Transport Layer Protocols: Services, Port Numbers, UDP services, TCP Services, TCP Features, Segment, A TCP Connection, State Transition Diagram, SCTP, SCTP services, SCTP Features, Packet format, An SCTP Association, Flow Control and error control.																																																																		
UNIT-IV																																																																		
Network layer:										08 Hours																																																								
Network layer design issues: Store and Forward packet Switching, Services Provided to the Transport Layer, Implementation of Connection less Service, Implementation of Connection-Oriented Service, Comparison of Virtual Circuit and Datagram Subnets; Routing algorithms: The Optimality Principle, Shortest Path Routing, Flooding, Distance Vector Routing, Link state Routing, Hierarchical Routing, Broadcast Routing, Multicast Routing.																																																																		
UNIT-V																																																																		
Congestion Control:										08 Hours																																																								
Effects of congestion, Congestion Control, Traffic Management, Congestion Control in Packet Switching Networks, TCP Congestion control, Datagram Congestion Control Protocol																																																																		
Course Outcomes: At the end of the course student will be able to																																																																		
1.	Demonstrate an understanding of network models, protocols, data encoding, and error correction techniques to analyze and optimize communication systems.																																																																	
2.	Analyze and apply data link control protocols, error control mechanisms, and network services like DNS, email, and HTTP to enhance data communication efficiency..																																																																	
3.	Analyze and apply transport layer mechanisms, including flow control, error handling, congestion management, and transport protocols (UDP, TCP, SCTP) for efficient data communication.																																																																	
4.	Analyze network layer design issues, including packet switching and service implementations, and apply various routing algorithms to evaluate their efficiency and scalability in network communication.																																																																	
5.	Analyze and apply congestion control strategies in network communication, including traffic management, TCP congestion control, and DCCP, to optimize network performance.																																																																	
Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes																																																																		
<table><tr><td>Program Outcomes→</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>PSO</td><td>PSO</td></tr><tr><td>↓ Course Outcomes</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>1</td><td>2</td></tr><tr><td>25MCA103.1</td><td>2</td><td>3</td><td>2</td><td>-</td><td>-</td><td>-</td><td>-</td><td>3</td><td>1</td><td>-</td></tr><tr><td>25MCA103.2</td><td>2</td><td>3</td><td>2</td><td>-</td><td>-</td><td>-</td><td>-</td><td>3</td><td>1</td><td>-</td></tr><tr><td>25MCA103.3</td><td>2</td><td>2</td><td>2</td><td>-</td><td>-</td><td>-</td><td>-</td><td>3</td><td>1</td><td>-</td></tr></table>												Program Outcomes→	1	2	3	4	5	6	7	8	PSO	PSO	↓ Course Outcomes									1	2	25MCA103.1	2	3	2	-	-	-	-	3	1	-	25MCA103.2	2	3	2	-	-	-	-	3	1	-	25MCA103.3	2	2	2	-	-	-	-	3	1	-
Program Outcomes→	1	2	3	4	5	6	7	8	PSO	PSO																																																								
↓ Course Outcomes									1	2																																																								
25MCA103.1	2	3	2	-	-	-	-	3	1	-																																																								
25MCA103.2	2	3	2	-	-	-	-	3	1	-																																																								
25MCA103.3	2	2	2	-	-	-	-	3	1	-																																																								

	25MCA103.4	2	3	2	-	-	-	-	3	1	-
	25MCA103.5	2	3	2	-	-	-	-	3	1	-
1: Low 2: Medium 3: High											
TEXTBOOKS:											
1.	William Stallings: Data and Computer Communication, 10 th Edition, Pearson Education, 2022										
2.	Forouzan B. A., Data Communications and Networking (6e), Tata McGraw Hill 2022.										
3.	Computer Networks, Andrew S. Tanenbaum & David J. Wetherall.										
REFERENCE BOOKS:											
1.	James F. Kurose & Keith W. Ross – Computer Networking: A Top-Down Approach (8 th Edition, Pearson)										
2.	Douglas E. Comer – Internetworking with TCP/IP: Principles, Protocols, and Architecture										
3.	Larry Peterson & Bruce Davie – Computer Networks: A Systems Approach (6 th Edition, Morgan Kaufmann)										
E Books / MOOCs/ NPTEL:											
1.	NPTEL- https://onlinecourses.nptel.ac.in/noc25_cs15/preview										
2.	https://freecomputerbooks.com/An-Introduction-to-Computer-Networks-by-Peter-Dordal.html										
3.	NPTEL - https://onlinecourses.nptel.ac.in/noc25_cs15/preview										

MATHEMATICAL FOUNDATION FOR COMPUTER APPLICATIONS			
Course Code	25MCA104	Course Type	PCC
Teaching Hours/Week (L: T: P: S)	3:1:0:0	Credits	04
Total Teaching Hours	50	CIE + SEE Marks	50+50
Teaching Department: Master of Computer Applications			
Course Objectives:			
1.	Understand operations on set theory, Solve example problems on set operations		
2.	Understand and solve problems on addition subtraction and multiplication of matrices.		
3.	Solve problems on eigen values and eigen vectors		
4.	Find mean, variance and covariance of two discrete and continuous random variable		
5.	Solve problems related to Graphs		
UNIT-I			
Set theory and operations on sets			05 Hours
Introduction, Venn Diagrams, subsets, The size of a set, power sets, cartesian products, Using set notations with quantifiers, Truth sets and quantifies			
Operations on sets : Introduction, set identities, generalizes unions and intersections, computer representations of sets			
Cardinality, Matrices, Eigen values and eigen vectors			05 Hours
Cardinality : Introduction, countable sets, an uncountable set			
Inclusion - exclusion principle : Introduction, the principle of inclusion exclusion			
Matrices : Introduction, matrix arithmetic, transposes and powers of matrices, Zero-one matrices			
Eigen values and eigen vectors :			
Finding Eigen values and Eigen vectors : Introduction, Procedure to find Eigen values, Procedure to find Eigen Vectors			
UNIT-II			
Propositional Logic, Applications of Propositional Logic			05 Hours
Propositional Logic : Introduction, Propositions, Conditional statements, Converse, Contrapositive and Inverse statements, Biconditional statements, Truth tables of Compound propositions, Precedence of logical operators, logic and bit operations			
Applications of Propositional Logic :			
Introduction, Translating English sentences, System Specifications, Logic Circuits, Logic Puzzles			
Propositional Equivalences, Predicates and Quantifiers			05 Hours
Propositional Equivalences :			
Introduction, Logical Equivalences, Using De Morgan's Laws, Constructing new logical equivalences, Propositional satisfiability, solving satisfiability problems			
Predicates and Quantifiers :			
Introduction, Predicates, Preconditions and post conditions, Quantifiers : Universal quantifier and existential quantifiers and uniqueness quantifiers			
UNIT-III			
Relations and their Properties, n-ary Relations and their Applications			05 Hours
Relations and their Properties :			
Functions as relations, relations on a set, properties of relations, combining relations			
n-ary Relations and their Applications :			
Introduction, n-ary relations, databases and relations, operations on n-ary relations, SQL			

Representing Relations, Closures of Relations, Equivalence relations										05 Hours	
Representing Relations : Introduction, representing relations using Matrices, representing using diagraphs											
Closures of Relations : Introduction, closures, paths in directed graphs, transitive closures, Warshall's algorithm.											
Equivalence relations : Introduction, Equivalence relations, equivalence classes, equivalence classes and partitions											
UNIT-IV											
The concept of probability, The axioms of probability										05 Hours	
The concept of probability : Introduction, Sample space and events											
The axioms of probability : Introduction, Properties of Probability, Bayes' Theorem, Basics of Probability, Concept of random variable											
Probability Distributions										05 Hours	
Discrete probability distributions : Introduction, Probability distribution for discrete random variable, expected values, Mean, variance and co-variance											
Continuous probability distributions : Introduction, Probability density function, expected values, Mean, variance and co-variance Binomial and normal distribution											
UNIT-V											
Graph Models Graph Terminology and Special Types of Graphs										05 Hours	
Introduction, Basic Terminology, some special simple graphs, Bipartite Graphs, Bipartite Graphs and matchings, some applications of special types of graphs, new graphs from old											
Representing Graphs and Graph Isomorphism, Euler and Hamilton Paths										05 Hours	
Representing Graphs and Graph Isomorphism : Introduction, Representing Graphs, Adjacency Matrices, Incidence Matrices, Isomorphism of graphs, Determining whether two simple graphs are isomorphic											
Euler and Hamilton Paths : Introduction, Euler paths and circuits, Hamilton paths and circuits, Applications of Hamilton Circuits											
Course Outcomes: At the end of the course student will be able to											
1.	Solve problems effectively on set operations.										
2.	Illustrate and solve problems on addition subtraction and multiplication of matrices.										
3.	Outline translation of sentences based on mathematical logic and solve problems related to applications of mathematical logic.										
4.	Find mean, variance and covariance of discrete and continuous random variable.										
5.	Demonstrate and solve problems related to Graphs.										
Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes											
Program Outcomes→		1	2	3	4	5	6	7	8	PSO	PSO
↓ Course Outcomes										1	2

	25MCA104.1	3	3	3	3	1	1	3	3	3	2
	25MCA104.2	3	3	3	3	1	1	3	3	3	3
	25MCA104.3	3	3	3	3	1	1	3	3	3	3
	25MCA104.4	3	3	3	3	1	1	3	3	3	3
	25MCA104.5	3	3	3	3	1	1	3	3	3	2
1: Low 2: Medium 3: High											
TEXTBOOKS:											
1.	Kenneth H Rosen, “Discrete Mathematics and its Applications”, McGraw Hill Publications, 8 th Edition, 2021.										
2.	Wolpole Myers Ye “Probability and Statistics for engineers and Scientist” Pearson Education, 9 th Edition, 2024										
REFERENCE BOOKS:											
1.	Richard A Johnson and C.B Gupta “Probability and statistics for engineers” Pearson Education, 9 th Edition.										
2.	J.K Sharma “Discrete Mathematics”, Mac Millian Publishers India, 4 th Edition, 2022.										
E Books / MOOCs/ NPTEL											
1.	H. Pishro-Nik, "Introduction to probability, statistics, and random processes", available at https://www.probabilitycourse.com , Kappa Research LLC, 2014.										

RESEARCH METHODOLOGY AND PUBLICATION ETHICS				
Course Code		25MCA105	Course Type	PCC
Teaching Hours/Week (L:T:P:S)		3:0:0:0	Credits	03
Total Teaching Hours		40	CIE + SEE Marks	50+50
Teaching Department: Master of Computer Applications				
Course Objectives:				
1.	To understand the Meaning, Objectives and Characteristics of research			
2.	To understand how to select and define the research problem effectively.			
3.	To explore Research design and methodologies and data collection techniques			
4.	To analyze ethical considerations, research integrity, and plagiarism detection.			
5.	To understand Intellectual Property Rights and scientific publication practices.			
UNIT - I				
Introduction to Research:				08 Hours
Meaning, Objectives and Characteristics of research, Research methods Vs Methodology, Types of research, Descriptive Vs. Analytical, Applied Vs. Fundamental, Quantitative Vs. Qualitative, Conceptual Vs. Empirical, Research process, Criteria of good research, Importance of AI and ML in Research Analysis.				
UNIT – II				
Research Problem Identification & Literature Review:				08 Hours
Developing a research plan. Defining and Selecting a Research Problem - Necessity of defining the problem -Techniques involved in defining the problem - Importance of literature review in defining a problem - Survey of literature - Primary and secondary sources ,Use of Digital Libraries, Google Scholar and AI-based Research Assistance (ChatGPT, Zotero,Mendeley)				
UNIT – III				
Research design, Data Collection & Analysis:				08 Hours
Research design, Basic Principles, Need of research design, Features of good design, Important concepts relating to research design.				
Sampling design, Steps in sampling design, Sampling Techniques, Characteristics of a good sample design, Types of sample designs, Measurement and scaling techniques, Methods of data collection, Collection of primary data, Data collection instruments, Statistical Tools for Data Analysis.				
Unit – IV				
Ethics in Research & Scientific Writing:				08 Hours
Philosophy and Ethics, Scientific Conduct: Ethics: definition, moral philosophy, nature of moral judgments and reactions. Scientific Conduct Ethics with respect to science and research, Plagiarism and Detection Tools (Turnitin ,Grammarly), Intellectual Honesty and Research Integrity, Scientific Paper Writing, Journal Selection, and Peer Review Process.				
Unit - V				
Intellectual Property Rights & Research Publications:				08 Hours
Intellectual Property Rights(IPR) : Patent ,Copyrights, Trademarks ,and Trade Secrets. IPR Laws in India and International Context (TRIPS Agreement), Open Access Vs Paid Journals, Predatory Journals Awareness. Research Funding, Grant Proposal Writing and Patent Filing Process.				
Course Outcomes: At the end of the course student will be able to				

1.	Outline the Meaning, Objectives and Characteristics of research.
2.	Demonstrate how to select and define a research problem with a literature review.
3.	Design appropriate research methodologies and analyze data effectively.
4.	Assess ethical considerations, plagiarism detection and research integrity.
5.	Demonstrate knowledge of intellectual property rights and scientific publishing

Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes

Program Outcomes→	1	2	3	4	5	6	7	8	PSO	PSO
↓ Course Outcomes									1	2
25MCA105.1	3	2	1	1	-	-	-	2	2	1
25MCA105.2	2	3	2	3	-	-	-	2	3	2
25MCA105.3	2	2	3	3	-	-	-	2	3	3
25MCA105.4	-	-	-	-	-	-	3	2	2	3
25MCA105.5	-	-	-	-	-	3	2	3	3	2

1: Low 2: Medium 3: High
TEXT BOOKS:

1.	Kothari, C. R., 1990. Research Methodology : Methods and Techniques. New Age International. 418 p.
2.	Dipankar Deb • Rajeeb Dey, Valentina E. Balas “Engineering Research Methodology”, ISSN 1868- 4394 ISSN 1868-4408 (electronic), Intelligent Systems Reference Library, ISBN 978-981-13- 2946-3 ISBN 978-981-13-2947-0 (eBook), https://doi.org/10.1007/978-981-13-2947-0

REFERENCE BOOKS:

1.	Sinha, S. C. and Dhiman, A. K., 2002. Research Methodology, EssEss Publications. 2 volumes.
2.	Garg, B.L., Karadia, R., Agarwal, F. and Agarwal, U.K., 2002. An Introduction to Research Methodology, RBSA Publishers.
3.	Anderson, T. W., An Introduction to Multivariate Statistical Analysis, Wiley Eastern Pvt., Ltd., New Delhi
4.	Day, R.A., 1992. How to Write and Publish a Scientific Paper, Cambridge University Press.
5.	Fink, A., 2009. Conducting Research Literature Reviews: From the Internet to Paper. Sage Publications
6.	Coley, S. M. and Scheinberg, C. A., 1990, "Proposal Writing", Sage Publications.
7.	Intellectual Property Rights in the Global Economy : Keith Eugene Maskus, Institute for International Economics, Washington, DC, 2000
8.	Subbarau N R – Handbook on Intellectual Property Law and Practice-S Viswanathan Printers and Publishing Private Limited. 1998

ADVANCED OPERATING SYSTEMS WITH UNIX			
Course Code	25MCA106	Course Type	PCC
Teaching Hours/Week (L: T: P:S)	3:1:0:0	Credits	04
Total Teaching Hours	50	CIE + SEE Marks	50+50
Teaching Department: MCA			
Course Objectives:			
1.	Understand Advanced Operating System Concepts		
2.	Gain In-Depth Knowledge of UNIX Internals		
3.	Analyze Distributed and Multiprocessor OS Mechanisms		
4.	Learn Security and Performance Optimization Techniques		
5.	Explore Emerging Trends in Operating Systems		
UNIT-I			
Fundamentals of Advanced Operating Systems:			05 Hours
Overview of Advanced Operating Systems. Types of Advanced OS: Distributed, Multiprocessor, Real-Time, Cloud-Based.			
Operating System Structure:			06 Hours
Process Synchronization & Communication (IPC, Semaphores, Monitors, Message Passing), Deadlock Detection, Avoidance, and Recovery, Case Study: Real-World Advanced Operating Systems (Unix, Linux,Android).			
UNIT-II			
UNIX Operating System Internals:			10 Hours
Evolution & Architecture of UNIX, Process Management in UNIX (Fork, Exec, Signals), File System in UNIX (Structure, Inodes, System Calls) .Memory Management & Virtual Memory in UNIX, UNIX System Performance & Optimization			
UNIT-III			
Distributed Operating Systems:			11 Hours
Introduction to Distributed Systems and Their Architecture, Distributed Process Scheduling & Load Balancing, Distributed File Systems (NFS, HDFS) & Remote Procedure Calls (RPC), Synchronization in Distributed Systems			
UNIT-IV			
Security and Performance in UNIX & Advanced OS			10 Hours
Security Mechanisms in UNIX (Access Control, Firewalls, Authentication), Intrusion Detection & Prevention, Malware Protection, Performance Optimization Techniques in UNIX, Resource Management & Scheduling in Multi-core Systems			
UNIT-V			
Emerging Trends and Case Studies in Operating Systems			08 Hours
Cloud-based & Virtualized Operating Systems (Docker, Kubernetes), Real-Time Operating Systems (RTOS) & Embedded OS, AI and Machine Learning Integration in OS.			
Course Outcomes: At the end of the course student will be able to			

1.	Demonstrate a Strong Understanding of Advanced OS Architectures.
2.	Analyze and Work with UNIX Operating System Internals.
3.	Develop an Understanding of Distributed and Multiprocessor Systems.
4.	Apply Security and Performance Optimization Techniques in UNIX.
5.	Evaluate Emerging Trends and Future Directions in Operating Systems.

Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes

Program Outcomes→	1	2	3	4	5	6	7	8	PSO	PSO
↓ Course Outcomes									1	2
25MCA106.1	3	3	2	2	-	-	-	2	3	3
25MCA106.2	3	3	3	3	-	-	-	2	3	2
25MCA106.3	3	3	2	3	-	-	-	3	3	2
25MCA106.4	3	3	3	2	2	-	-	2	3	2
25MCA106.5	3	3	3	3	2	2	2	3	3	2

1: Low 2: Medium 3: High

TEXTBOOKS:

1. Mukesh Singhal, Niranjana G. Shivaratri, Advanced Concepts in Operating Systems, McGraw-Hill -2017
2. Robert Love, Linux System Programming, 2nd edition, 2013
3. Andrew S. Tanenbaum, Modern Operating Systems, 4th Edition, Pearson, 2014
4. William Stallings, Operating Systems: Internals and Design Principles, 9th Edition, Pearson, 2019

REFERENCE BOOKS

1. Abraham Silberschatz, Operating System Concepts, 10th Edition, Wiley
1. UNIX: The Complete Reference: Kenneth Rosen et al, Osborne/McGraw Hill, 2010
2. Maurice J. Bach, *The Design of the UNIX Operating System*, Pearson
3. Simson Garfinkel, Gene Spafford, and Alan Schwartz, Practical UNIX and Internet Security, 3rd edition

E Books / MOOCs/ NPTEL

1. <https://www.oreilly.com/library/view/learning-the-unix/1565923901/>

DATA STRUCTURES WITH ALGORITHMS LAB			
Course Code	25MCA107	Course Type	PCC
Teaching Hours/Week (L: T: P: S)	0:0:4:0	Credits	02
Total Teaching Hours	26	CIE + SEE Marks	50+50
Teaching Department: Master of Computer Applications			
Course Objectives:			
1.	Implement the primitive data structures such as arrays and structures.		
2.	Implement Stack, Queue data structures statically using arrays and dynamically using linked lists.		
3.	Implement recursive and non recursive programs.		
4.	Implement the non linear data structures such as Linked list, Binary Trees, Graphs.		
5.	Implement different sorting methods and searching methods		
List of Experiments			
1.	C programs using pointers.		
2.	C programs using Arrays and Structures.		
3.	C programs to demonstrate the operations of stack. (using arrays and structures).		
4.	C programs to convert the valid infix arithmetic expression to postfix and prefix form.		
5.	Evaluate postfix expression.		
6.	Recursive C programs to i) Find the Factorial of a number. ii) Find Fibonacci of a number. iii) Solve Towers of Hanoi Problem. iv) Find the GCD of two numbers. v) Find the multiplication of two numbers.		
7.	C programs to simulate the working of simple queue, circular queue and priority queue using arrays and structures.		
8.	C programs to demonstrate the operations of singly & doubly linked list. (operations like insert a node at the front, at the back, at the specified position; delete a node from the front end, from the back end, from the specified position; search a node, if the info field is specified; display all the nodes in the list).		
9.	Implement stack and queues dynamically using Linked lists.		
10.	C programs on binary trees. (Construct a binary search tree and Traverse the tree using all the methods i.e., inorder, preorder, postorder).		
11.	C programs to demonstrate the different searching techniques over a list of integers. (Linear search, Binary search).		
12.	C programs to sort a list of elements using different sorting techniques. (Bubble sort selection sort, quick sort, simple insertion sort, shell sort, merge sort, Heap sort, Binary tree sort).		
Course Outcomes: At the end of the course student will be able to			
1.	Implement static and dynamic Stack and Queue data structures. Demonstrate the concept of Recursive programs and the implementation details of linked lists.		
2.	Explain the implementation details of Binary tree and Graph data structures. Implement and Illustrate various Sorting and Searching Techniques		

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Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes											
Program Outcomes→	1	2	3	4	5	6	7	8	PSO	PSO	
↓Course Outcomes									1	2	
25MCA107.1	3	3	3	3	3	-	-	-	3	2	
25MCA107.2	3	3	3	3	3	-	-	-	3	2	
1: Low 2: Medium 3: High											
REFERENCE BOOKS:											
1.	Yedidyah Langsam and Moshe J Augenstein and Aaron M Tenenbaum: Data Structures using C and C++, 2 nd Edition, Pearson Education Asia, 2002.										
2.	Reema Thareja – "Data Structures Using C", 2 nd Edition, Oxford University press										
3.	Jean-Paul Tremblay: An Introduction to Data Structures with Applications, Tata McGraw Hill										
4.	Ellis Horowitz and Sartaj Sahni, Fundamentals of Data Structures in C, 2 nd Edition, Universities Press.										
5.	Introduction to the Design and Analysis of Algorithms. Anany Levitin, Pearson Education, 2 nd Edition.										
6.	Horowitz E., Sahani S., Rajashekharan S: Computer Algorithm, Galgotia Publications.										

ADVANCED DATABASE SYSTEMS LAB													
Course Code				25MCA108		Course Type			PCC				
Teaching Hours/Week (L: T: P:S)				0:0:4:0		Credits			02				
Total Teaching Hours				26		CIE + SEE Marks			50+50				
Teaching Department: Master of Computer Applications													
Course Objectives:													
1.	Create and query the databases using SQL DML, DDL and DQL commands.												
2.	Use the features of MySQL to write procedural programs including stored procedures, cursors, and triggers.												
3.	Create and query NoSQL database using MongoDB.												
Course Outcomes: At the end of the course students will be able to													
1.	Apply SQL and procedural programming concepts in SQL and MySQL to query, update, and manage relational databases effectively.												
2.	Develop CRUD operations in a NoSQL database like MongoDB to manage unstructured and semi-structured data.												
List of Experiments													
1.	Exercise on creating tables.												
2.	Exercise on altering tables, dropping tables.												
3.	Exercise on giving table level constraints, field level constraints.												
4.	Exercise on insertion, retrieval, deletion and modification of data values.												
5.	Exercise on ORDER BY, GROUP BY, HAVING clauses.												
6.	Exercise on Aggregate functions in SQL.												
7.	Exercise on Joins, Unions, Sub queries, Nested Sub queries, Co-related sub-queries.												
8.	Exercise on creating views, dropping views.												
9.	Exercise on MySQL : control structures, connecting tables.												
10.	Exercise on MySQL : Procedures, Cursors, Triggers												
11.	Exercise on MongoDB.												
Course Outcomes: At the end of the course students will be able to													
1.	Use database language statements to query, update, and manage a database using SQL and PL/SQL.												
2.	Apply the concepts used in performing CRUD operations using NoSQL database like MongoDB.												
Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes													
Program Outcomes→				1	2	3	4	5	6	7	8	PSO	PSO
↓ Course Outcomes												1	2
25MCA108.1				3	-	3	2	3	2	-	3	3	-
25MCA108.2				3	-	3	3	2	3	-	3	3	-
1: Low 2: Medium 3: High													
TEXTBOOKS:													
1.	Elmasri and Navathe: Fundamentals of Database Systems. 7 th Edition 2016												

2.	Ivan Bayross : Commercial Application Development using Oracle Developer 2000
3.	Seema Acharya, Subhashini Chellappan, “Big Data Analytics”, 1 st Edition, Wiley, 2015
4.	Raghu Ramakrishnan and Johannes Gehrke : Database Management Systems, 6 th Edition, McGraw-Hill

REFERENCE BOOKS

1.	Silberschatz, Korth and Sudarshan: Database Systems Concepts, 6 th Edition, McGraw-Hill
2.	Alexis Leon, Mathews Leon: Database Management Systems, Vikas Publishing House
3.	Connolly: Database Systems: A Practical Approach To Design Implementation And Management, 3 rd Edition, Person Education

E Books / MOOCs/ NPTEL

1.	NOC : Fundamentals of Database Systems(Course sponsored by Aricent), IIT Kanpur, Dr. Arnab Bhattacharya https://nptel.ac.in/courses/106104135
2.	Oracle PL/SQL Language Pocket Reference, 5th Edition: A Guide to Oracle's PL/SQL Language Fundamentals. https://www.pdfdrive.com/oracle-plsql-language-pocket-reference-5th-edition-a-guide-to-oracles-plsql-language-fundamentals-d167357800.html
3.	MySQL® Notes for Professionals book: https://www.goalkicker.com/MySQLBook/
4.	Learning MongoDB: https://www.dbooks.org/learning-mongodb-5605980453/#google_vignette

DATA COMMUNICATION AND NETWORKS LAB													
Course Code				25MCA109				Course Type				PCC	
Teaching Hours/Week (L: T: P: S)				0:0:4:0				Credits				02	
Total Teaching Hours				26				CIE + SEE Marks				50+50	
Teaching Department: Master of Computer Applications													
Course Objectives:													
1.	To get practical knowledge of working principles of various communication protocols												
2.	To analyze the various routing algorithms and To know the concept of data/packet transfer between nodes.												
3.	Analyse structure and formats of TCP/IP layer protocols using network tools like network simulators.												
4.	Implementing various network algorithms such as error control, error detection, routing and security related algorithms.												
List of Experiments													
1.	Implementation of error detection mechanisms												
2.	Implementation of the congestion control algorithms.												
3.	Implementation of routing algorithms.												
4.	Implementation of client / server programs using TCP and UDP.												
5.	Implementation of RSA algorithm for encryption/Decryption of message between client & server.												
6.	Implementation of frame sorting logic, to order frames received from multiple senders.												
7.	Implementation of TELNET.												
8.	Simulation of Networking components using any Network Simulation software												
9.	Implementation of wireless technology												
10.	Use of Open NMS and TCL tool.												
Course Outcomes: At the end of the course student will be able to													
1.	Interpret the practical approach to network communication protocols.												
2.	Classify the applications of various Routing Protocols/Algorithms.												
Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes													
Program Outcomes→				1	2	3	4	5	6	7	8	PSO	PSO
↓Course Outcomes												1	2
25MCA109.1				3	3	3	3	2	-	3	3	3	1
25MCA109.2				3	3	2	2	3	-	3	3	3	1
1: Low 2: Medium 3: High													
REFERENCE BOOKS:													
1.	Cisco CCNA Study Guide v2.71, Aaron Balchunas												
2.	CCNA, Study Guide, 6 th Edition, TodLammle												
E Resources													
1.	https://www.isi.edu/nsnam/ns/												
2.	https://www.geeksforgeeks.org/network-simulator-3/												

FUNDAMENTALS OF PROGRAMMING – BRIDGE COURSE											
Course Code				25MCA110				Course Type		MNC	
Teaching Hours/Week (L: T: P: S)				0:0:4:0				Credits		00	
Total Teaching Hours				26				CIE		50	
Teaching Department: Master of Computer Applications											
Course Objectives:											
1.	Learn the basics of programming structure and module.										
2.	Study the concept of decision-making statements, loop controlling structures.										
3.	Learn and execute programs on arrays and structures.										
4.	Gain knowledge about pointers and execute the programs using pointers.										
List of Experiments											
	Simple C programs with input – output statements.										
	C programs with decision making statements.										
	C programs using loop control structures.										
	C programs using arrays.										
	C programs using structures.										
	C programs using pointers.										
Course Outcomes: At the end of the course student will be able to											
1.	Explain the basics of programming structure and module. Demonstrate the concept of decision-making statements, loop controlling structures. Execute simple programs, programs using arrays and structures.										
2.	Interpret the concepts of functions and subroutine, execute the programs. Summarize the pointer concepts and execute the programs using pointers.										
Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes											
Program Outcomes→		1	2	3	4	5	6	7	8	PSO	PSO
↓ Course Outcomes										1	2
25MCA110.1		3	3	3	3	2	-	3	3	3	2
25MCA110.2		3	3	2	2	3	-	2	3	3	2
1: Low 2: Medium 3: High											

ENTERPRISE JAVA			
Course Code	25MCA201	Course Type	PCC
Teaching Hours/Week (L:T:P:S)	3:0:0:0	Credits	03
Total Teaching Hours	40	CIE + SEE Marks	50+50
Prerequisites	Basic Programming Knowledge		
Teaching Department: Master of Computer Applications			
Course Objectives:			
1.	To understand Input/Output using Java and use Class fundamentals, Objects, Methods, Polymorphism,		
2.	To understand Inheritance, Abstraction, Package creation and Exceptions handling in Java		
3.	To understand Multithreaded Programming concepts using Java, develop GUI Applications in Java using Swing packages and Event handling.		
4.	To understand and use Networking concepts, Distributed computing using JDBC for Database connectivity.		
5.	To understand Web application development, Using Servlets, Spring Boot.		
UNIT - I			
			10 Hours
Introduction to JAVA :			
Introducing classes: Class fundamentals; Declaring objects; Assigning object reference variables; Introducing methods; Constructors; The this keyword; Overloading methods; Introducing access control; Understanding static: Introducing final; introducing nested and inner classes.			
Stream Classes:			
Reading and writing files; Stream classes; Byte Streams (InputStream; OutputStream; FileInputStream; FileOutputStream; DataInputStream, DataOutputStream); Character streams(Reader; Writer, FileReader, FileWriter, BufferedReader, BufferedWriter); Serialization: Serializable, ObjectOutputStream, ObjectInputStream.			
UNIT -II			
			06 Hours
Object Oriented Fundamentals and Exception Handling			
Inheritance :			
Inheritance basics; Using super; Creating a multilevel hierarchy; Method overriding; Dynamic method dispatch; Using Abstract class; Using final with inheritance, Aggregation, Packages; Access protection; Importing packages; Interfaces.			
Exception handling :			
Exception- handling fundamentals; Exception types; Uncaught exceptions; Using try and catch; Multiple catch clauses; Nested try statements; throw; throws; finally; Java’s built-in exceptions; Creating your own exception sub-classes;			
UNIT - III			
			10 Hours
The Collection Framework, Multithreaded Programming and Event Handling, AWT and Swings:			
The Collection Framework:			
The Collection interfaces: The Collection interface, The List interface, The Set interface; The collection classes: The ArrayList class, The LinkedList class, The Hashset class, The Linked HashSet class.			
Multithreaded programming:			

Java Thread model; Main thread; Creating a thread; Creating multiple threads; Using `isAlive()` and `join()`; Synchronization; Interthread communication

Event Handling, AWT and Swings :

Event Handling:

Two event handling mechanisms; Delegation event model; Event classes; Sources of events; `EventListener` interfaces; Using the Delegation Event Model

Swings :

`JApplet`; `JFrame`; Labels; Text fields; Buttons; Check boxes; `RadioButtons`; Adapter classes

Unit - IV

06 Hours

Networking , Database Connectivity, Java Beans:

RMI :

Remote Method Invocation (RMI): Remote Method Invocation concept; Server side; Client side.

Networking basics: Java and the net; `InetAddress`; TCP/IP client sockets; URL: `URLConnection`; TCP/IP server sockets; `Datagrams`.

DataBase Connectivity:

JDBC objects: Concept of JDBC; JDBC packages; Brief overview of the JDBC process; Database connection; Associating the JDBC/ODBC bridge with the database; Statement objects; `ResultSet`.

Java Beans :

Introduction to Java Beans; Advantages of Java Beans; JAR files; Developing a simple Bean; Using bound properties; Using `BeanInfo` Interface; Introduction to Constrained properties.

Unit - V

08 Hours

Servlets and Spring Boot :

Servlets :

Background; Life cycle of a Servlet; Simple Servlet; Servlet API; `javax.servlet` package; Reading Servlet parameter; `javax.servlet.http` package; Handling HTTP requests and responses; Using Cookies and sessions.

JSP:

JSP elements, Request and Response, Introduction to JSP implicit objects

Spring Boot :

Introduction to Spring Boot, Essential Key Components of spring Boot, Setting up a Spring Boot workspace, Developing first spring boot application.

Course Outcomes: At the end of the course student will be able to

1.	Analyze and Apply Input/Output operations in Java, leveraging Class fundamentals, Objects, Methods, and Polymorphism to design efficient and reusable software solutions.
2.	Evaluate and Implement advanced Object-Oriented Programming principles, including Inheritance, Abstraction, Package creation, and Exception handling, to enhance code modularity and robustness.
3.	Design and Develop multithreaded applications in Java, integrating GUI components using Swing and applying event-driven programming for responsive user interfaces.
4.	Apply and Implement networking concepts and database connectivity using JDBC to develop secure and efficient data-driven applications.
5.	Develop and Enhance web applications using Servlets and Spring Boot, applying best practices in web development and software architecture.

Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes

	Program Outcomes→	1	2	3	4	5	6	7	8	PSO	PSO
	↓ Course Outcomes									1	2
	25MCA201.1	3	3	3	3	3	3	-	3	3	-
	25MCA201.2	3	3	3	3	2	2	-	3	3	-
	25MCA201.3	3	3	3	3	2	2	-	3	3	-
	25MCA201.4	3	2	2	2	2	3	-	3	3	-
	25MCA201.5	3	3	3	2	3	3	-	3	3	-
1: Low 2: Medium 3: High											
TEXT BOOKS:											
1.	Herbert Schildt : Java: The Complete Reference, 12 th Edition, Tata McGraw Hill., 2019										
2.	Mastering Spring Boot 2.0: Build modern, cloud-native, and distributed systems using Spring Boot, Dinesh Rajput, Pack Publishing Ltd., 2018.										
REFERENCE BOOKS:											
1.	Dr. Donald Doherty and Rick Leinecker : JavaBeans Unleashed										
2.	James Goodwill : Developing Java Servlets										
3.	Steven Holzner : Java 2 Black Book										
4.	Ed Roman : Mastering Enterprise JavaBeans										
5.	Jim Keogh : The Complete Reference J2EE, Tata McGraw Hill, 2008.										

ADVANCED WEB TECHNOLOGIES			
Course Code	25MCA202	Course Type	PCC
Teaching Hours/Week (L:T:P:S)	3-0-0	Credits	03
Total Teaching Hours	40	CIE + SEE Marks	50+50
Prerequisite	Fundamentals of Programming		
Teaching Department: Master of Computer Applications			
Course Objectives:			
1.	To understand and apply the principles of XHTML, CSS, HTML5, and Bootstrap and develop the skills to create structured web pages, implement effective styling, leverage the features of HTML5 for modern web development, and utilize Bootstrap for responsive and visually appealing design, and overview of XML key components,		
2.	To understand PHP scripting language and its applications in web development.		
3.	To understand JavaScript and its programming concepts, and. JavaScript execution environment and event handler techniques		
4	To understand the JQuery concept and its effects and provide an introduction to AngularJS.		
5.	To develop an ability to design and implement static and dynamic website and client-server architecture to develop a web application		
UNIT – I			
XHTML:			02 Hours
Standard Structure, Basic Text Markup, Lists, Tables, Forms, Frames.			
CSS:			02 Hours
Introduction, Levels of Style Sheets, Selector Forms, Property Value Forms, Font Properties, List Properties, Color, Alignment of Text, The Box Model, The and <div> tags.			
HTML5:			02 Hours
Introduction of html5, iframes, layout, responsive web design(view port),computer code, new semantics elements in HTML5, Migration, canvas, svg, input types, new form elements and attribute, google map, media(audio, video)			
BOOTSTRAP:			02 Hours
Introduction of Bootstrap, First Web Page with Bootstrap, Bootstrap Grid system, CDN, Tables, Images, Jumbotron and Page Header, Wells, Alerts, Buttons, Badges and Labels, Progress Bars, List Groups, Navigation Bar, Forms, Form Inputs, Media Objects, Carousel Plug-in			
UNIT –II			
PHP:			04 Hours
Overview of PHP and uses of PHP, General Syntactic Characteristics, Primitives, Operations and expressions, Control Statements, Arrays, Functions, Pattern Matching, Form Handling, Files.			
Advanced PHP:			04 Hours
Cookies, Session tracking, Database Access with PHP and MySQL.			

UNIT – III	
JavaScript	04 Hours
Overview of JavaScript, Object Orientation and JavaScript, Syntactic Characteristics, Screen Output and Keyboard Input, Control Statements, Object Creation and Modification, Arrays, Functions, Constructors, Pattern Matching.	
Dynamic Object Model (DOM)	04 Hours
JavaScript Execution Environment, The Document Object Model, Element Access in JavaScript, Events and Event Handling, Handling Events from the Body Elements, Button Elements, Text Box and Password Elements, The Navigator Object.	
UNIT – IV	
Dynamic Documents with Java Script	04 Hours
Introduction to dynamic documents, Positioning Element, Moving Elements, Element Visibility, Changing Colors and Fonts, Dynamic Content, Stacking Elements, Locating the Mouse Cursor, Reacting to a Mouse Click, Slow Movement of Elements, Dragging and Dropping Elements.	
JQUERY:	02 Hours
Introduction, Syntax, selectors, events, JQuery HTML, Effects	
Introduction to Angular JS:	02 Hours
What is AngularJS? Angular Directives, AngularJS Expression, Angular Expression, AngularJS Module, AngularJS Controller, AngularJS Filter.	
UNIT – V	
Web 2.0 and Building Rich Internet Applications with AJAX	02 hours
Web 2.0 What is Web 2.0?, Folksonomies and Web 2.0, Software As a Service(SaaS), Data and Web 2.0, Convergence, Iterative development, Rich User experience, Multiple Delivery Channels, Social Networking. Web Services: SOAP, RPC Style SOAP, Document style SOAP, WSDL, REST services, JSON format, What is JSON?, Array literals, Object literals, Mixing literals, JSON Syntax, JSON Encoding and Decoding, JSON versus XML.	
Building Rich Internet Applications with AJAX	04 Hours
Building Rich Internet Applications with AJAX: Limitations of Classic Web application model, AJAX principles, Technologies behind AJAX, Examples of usage of AJAX, Dynamic web applications through Hidden frames for both GET and POST methods. IFrames, Asynchronous communication and AJAX application model, XMLHttpRequest Object – properties and methods, handling different browser implementations of XMLHttpRequest, The same origin policy, Cache control, AJAX Patterns (Only algorithms – examples not required): Predictive fetch pattern, Submission throttling pattern, Periodic refresh, Multi stage download.	
Building Advanced Web 2.0 applications	02 Hours
Definition of mash up applications, Mash up Techniques, Building a simple mash up application with AJAX,	
Course Outcomes: At the end of the course student will be able to	
1.	Create well-structured web pages using XHTML, apply CSS styling techniques, utilize HTML5 features for modern web development, and design responsive websites using Bootstrap, demonstrating proficiency in web development with these technologies as well as gain a solid understating of XML

2.	Develop proficiency in PHP programming by demonstrating knowledge of its manipulate arrays, create functions, handle form data, manage files, utilize cookies and session tracking, and interact with MySQL databases.
3.	Develop proficiency in JavaScript programming and creating dynamic web documents.
4.	Effectively use jQuery to manipulate HTML elements and enabling to apply jQuery for dynamic web development and explore further possibilities with AngularJS.
5.	Learn advanced Web 2.0 applications and overall concepts of Rich Internet Applications with AJAX

Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes

Program Outcomes→	1	2	3	4	5	6	7	8	PSO	PSO
↓ Course Outcomes									1	2
25MCA202.1	3	3	3	2	3	-	3	3	3	-
25MCA202.2	3	3	2	2	3	-	3	3	3	-
25MCA202.3	3	3	3	3	2	-	3	3	3	-
25MCA202.4	3	3	2	3	3	-	3	3	3	-
25MCA202.5	3	3	3	3	2	-	3	3	3	-

1: Low 2: Medium 3: High
TEXT BOOKS:

1.	Robert W. Sebesta: Programming with World Wide Web, 8 th Edition, Pearson Education, 2014.
2.	Eric Van derVlist et al: Professional Web 2.0 Programming, Wiley India, 2007.

REFERENCE BOOKS:

1.	Internet & World Wide Web: How to Program, 5 th Edition by M. Deitel et al.
2.	Chris Bates: Web Programming Building Internet Applications, 3 rd Edition, Wiley India, 2007.
3.	Xue Bai et al: The Web Warrior Guide to Web Programming, Cengage Learning, 2001
4.	Thomas A. Powel: Ajax The Complete Reference, McGraw Hill, 2008.
5.	Gottfried Vossen, Stephan Hagemann: Unleashing Web 2.0 From Concepts to Creativity, Elsevier, 2007.
6.	Steven Holzner : Ajax Bible Wiley India , 2007.
7.	Justin Gethland et al: A Web 2.0 primer Pragmatic Ajax, SPD Publications, 2006.
8.	Jake Spurlock: Bootstrap- Responsive Web Development, O'Reilly Media, 2014.

SOFTWARE ENGINEERING AND TESTING			
Course Code	25MCA203	Course Type	PCC
Teaching Hours/Week (L:T:P:S)	3:1:0:0	Credits	04
Total Teaching Hours	50	CIE + SEE Marks	50+50
Teaching Department: Master of Computer Applications			
Course Objectives:			
1.	Provide an overview of Agile development techniques		
2.	Know about system models and software designs.		
3.	Introduce Advanced Software Engineering Concepts.		
4.	Develop Proficiency in Advanced Software Testing Techniques.		
5.	Provide Experience in Performance, Security, and Scalability Testing.		
UNIT - I			
Agile Software development:			04 Hours
Agile methods, Agile development techniques.			
Software requirements:			06 Hours
Functional and non-functional requirements; Requirements engineering Process, Requirement elicitation, Requirement specification, Requirement Validation.			
UNIT-II			
System models:			05 Hours
Context models; Interaction models; sequence diagram Structured methods; class diagram, generalization, Aggregation. Behavioural Models-Data driven modelling; Event driven modelling, Model driven engineering.			
Software Design:			05 Hours
Object-oriented design using the UML-System context and interactions, Architecture design, Object class identification, Design models.			
UNIT-III			
Advanced Software Engineering Concepts			10 Hours
Software Engineering Paradigms & Models, Introduction to DevOps, and Continuous Integration/Deployment (CI/CD), Software Architecture & Design Patterns, Microservices and Serverless Architectures			
UNIT - IV			
Advanced Software Testing Techniques			10 Hours
Software Testing Fundamentals & Testing Levels (Unit, Integration, System), Automated Testing Tools (Selenium, JUnit, TestNG, Cypress), Static & Dynamic Testing (Code Analysis, Mutation Testing), Model-Based Testing (MBT) & AI-Driven Testing.			
UNIT - V			
Performance, Security & Scalability Testing			10 Hours
Performance Testing (Load, Stress, Endurance, Volume), Security Testing (Penetration Testing, OWASP Top 10, Threat Modelling), Scalability Testing in Cloud Environments, Testing in DevOps			

(Shift-Left & Shift-Right Testing).

Course Outcomes: At the end of the course students will be able to

1.	Understand Agile development and Requirement Engineering Process.
2.	Apply the different System Models and Software Designs on software engineering.
3.	Understand and Apply Advanced Software Engineering Principles.
4.	Summarize the relevance of Software Testing Techniques.
5.	Perform Performance, Security, and Scalability Testing.

Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes

Program Outcomes→	1	2	3	4	5	6	7	8	PSO	PSO
↓ Course Outcomes									1	2
25MCA203.1	3	2	2	2	-	-	-	3	3	2
25MCA203.2	3	3	3	3	-	-	-	3	3	2
25MCA203.3	3	2	3	3	-	-	-	3	3	2
25MCA203.4	3	3	3	3	2	-	-	2	3	2
25MCA203.5	3	3	3	3	3	2	2	3	3	2

1: Low 2: Medium 3: High
TEXT BOOKS:

1.	Ian Somerville: Software Engineering, 10 th Edition, Pearson Education Ltd
2.	Paul C. Jorgensen, Software Testing: A Craftsman's Approach, 4 th Edition, CRC Press
3.	Elfriede Dustin, <i>Effective Software Testing: 50 Specific Ways to Improve Your Testing</i> , Addison-Wesley

REFERENCE BOOKS:

1.	William Perry: Effective Methods for Software Testing, 2 nd Edition, John Wiley.
2.	Bezier B: Software Testing Techniques, 2 nd Edition, Wan Nostrand Reinluold.
3.	Myers G J: The Art of Software Testing, John Wiley.
4.	Roger S. Pressman: Software Engineering – A Practitioners Approach, 7 th Edition, McGraw-Hill

DATA WAREHOUSING AND DATA MINING				
Course Code		25MCA204	Course Type	PCC
Teaching Hours/Week (L:T:P:S)		3:1:0:0	Credits	04
Total Teaching Hours		50	CIE + SEE Marks	50+50
Prerequisites		Advanced Database Systems Mathematical Foundation for Computer Applications		
Teaching Department: Master of Computer Applications				
Course Objectives:				
1.	Provide the student with an understanding of the concepts of data warehousing, its architecture and modelling			
2.	Understand the Knowledge Discovery Process and get the notion of Data Mining			
3.	Learn the Preprocessing techniques required for Data Mining			
4.	Learn the data mining tasks of Association Analysis, Classification and Clustering and also study their well-known techniques			
5.	Test real data sets using popular data mining tools			
UNIT – I				
Data Warehousing & modelling:				10 Hours
Basic Concepts: Data Warehousing: A multitier Architecture, Data warehouse models: Enterprise warehouse, Data mart and virtual warehouse, Extraction, Transformation and loading, Data Cube: A multidimensional data model, Stars, Snowflakes and Fact constellations: Schemas for multidimensional Data models, Dimensions: The role of concept Hierarchies, Measures: Their Categorization and computation, Typical OLAP Operations				
UNIT – II				
Introduction to Data mining:				10 Hours
Introduction, What is data mining, Motivating Challenges, The Origins of Data Mining, Data Mining Tasks Data: Types of Data - Attributes and Measurements, types of Data Sets – Record data, graph based data, ordered data, Data Quality, Data Preprocessing – aggregation, sampling, dimensionality reduction, feature subset selection, feature creation, discretization and binarization, variable transformation, Measures of Similarity and Dissimilarity – Euclidean distance measures, similarity measures for binary data, simple matching coefficient, Jaccard coefficient and cosine similarity				
UNIT – III				
Association Analysis:				10 Hours
Problem Definition, Frequent Item set Generation, Rule generation. Alternative Methods for Generating Frequent Item sets, FP Growth Algorithm, Subjective and Objective measures for Evaluating Association Patterns				
UNIT – IV				
Classification:				10 Hours
Preliminaries, General Approach to solving classification Problem, Decision Tree Induction, Rule Based Classifiers, Nearest Neighbor Classifiers, Bayesian Classifiers, Evaluating the performance of classifier.				

UNIT – V											
Clustering Analysis:										10 Hours	
Overview, K-Means, Agglomerative Hierarchical Clustering, DBSCAN, Density-based Clustering, Graph-based clustering, Cluster Evaluation- Classification oriented measures, Similarity oriented measures											
Course Outcomes: At the end of the course student will be able to											
1.	Articulate the architecture of Data Warehouses and compare design schemas										
2.	Apply various preprocessing tasks and define Data Mining										
3.	Understand Association rule mining and apply association algorithms										
4.	Learn to apply various classification techniques and algorithms and evaluate										
5.	Interpret and Understand various clustering techniques and algorithms										
Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes											
Program Outcomes→		1	2	3	4	5	6	7	8	PSO	PSO
↓ Course Outcomes											
25MCA204.1		3	3	3	3	3	3	1	1	2	3
25MCA204.2		3	3	3	3	1	1	1	3	2	3
25MCA204.3		3	3	3	3	3	1	1	3	2	3
25MCA204.4		3	3	3	3	3	1	1	3	2	3
25MCA204.5		3	3	3	3	3	1	1	3	2	3
1: Low 2: Medium 3: High											
TEXT BOOKS:											
1.	Pang-Ning Tan, Michael Steinbach, Vipin Kumar: Introduction to Data Mining, Pearson, Fourth impression, 2018.										
2.	Jiawei Han, Micheline Kamber, Jian Pei: Data Mining -Concepts and Techniques, 3 rd Edition, Morgan Kaufmann Publisher										
REFERENCE BOOKS:											
1.	Arun K. Pujari : Data Mining Techniques, 3 rd Edition, Universities Press, 2013.										
2.	Sam Anahory, Dennis Murray: Data Warehousing in the Real World, Pearson, 10 th Impression, 2012.										
3.	Michael J. Berry, Gordon S. Linoff: Mastering Data Mining, Wiley Edition, 3 rd Edition, 2012										
E Books / MOOCs/ NPTEL											
1.	NPTEL Course on Data Mining, IIT Kharagpur, 2024										

PROFESSIONAL COMMUNICATION SKILLS			
Course Code	25MCA205	Course Type	PCC/Practical
Teaching Hours/Week (L:T:P: S)	1:1:0:0	Credits	02
Total Teaching Hours	25	CIE + SEE Marks	50+50
Teaching Department: Master of Computer Applications			
Course Objectives:			
1.	Comprehend the process of communication and enhance presentation abilities for professional and social contexts.		
2.	Cultivate interpersonal skills, time management, and stress management for personal and professional growth.		
3.	Gain proficiency in effective writing by understanding its dynamics and structuring content clearly.		
4.	Recognize ethical considerations and responsibilities in professional practice.		
UNIT - I			
			11 Hours
Communication Skills			
Communication, Methods of Communication Effective Communication Skills			
One-way and Two-way Communication, Interpersonal & Social Skills			
Effective Presentation Strategies			
Defining the purpose of presentation, preparation and understanding nuances of delivery Formal Presentation, Sensitivity towards multi-cultural workspaces			
UNIT - II			
			10 Hours
Writing Skills			
Application letters and Resume writing, Formal E-mails, Email etiquette, Job Cover Letter			
Self-Enhancement			
Types of Personality, Johari window and SWOT Analysis and Interpersonal Skills, General Etiquettes			
UNIT - III			
			04 Hours
Ethics and Professional Responsibilities			
Rights and responsibilities regarding intellectual property, Workplace rights and responsibilities, Responsibilities for environment and society			
Time Management and Stress			
Time Management, Identifying the Stress, Signs of Stress, Healthier ways to deal with Stress.			

Course Outcomes: At the end of the course student will be able to

1	An ability to communicate effectively.
2	A recognition of the need for and an ability to engage in lifelong learning.
3	An ability to function effectively as an individual/member/leader on multidisciplinary teams and understanding of professional and ethical responsibility.

Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes

Program Outcomes→	1	2	3	4	5	6	7	8	PSO	PSO
↓ Course Outcomes									1	2
25MCA205.1	-	-	-	3	2	-	2	2	2	2
25MCA205.2	-	-	-	-	-	-	2	3	2	2
25MCA205.3	2	2	2	-	3	-	3	3	2	2

1: Low 2: Medium 3: High

TEXT BOOKS:

1.	Meenakshi Raman and Sangeeta Sharma: Technical Communication - Principles and Practices, Oxford University Press 2004.
2.	George Reynolds: Ethics in Information Technology, Thomson Course Technology, 2003.
3.	Ray French: Cross Culture Management, Universities Press
4.	Urmila Rai: Business Communication, Himalaya Publishing House
5.	Gerard Assey: Professional Business Etiquette & Grooming, https://www.barnesandnoble.com/w/professional-business-etiquette-grooming-gerard-assey eBook

REFERENCE BOOKS:

1.	M. Ashraf Rizvi: Effective Technical Communication by, 1 st Edition, Tata McGraw Hill.
2.	Mike W Martin and Ronald Scherzinger: Ethics in Engineering, 3 rd Edition Tata McGraw Hill.
3.	C.S Rayudu: Communication, Himalaya Publishing House
4.	Ronald B Adler & Jeanne Marquardt Elmhurst; Communicating at work – Principles and Practices for Business and the Professions McGraw-Hill College; Sixth Edition.
5.	Stephen P Robbins, Organizational Behaviour, Prentice Hall, India.
6.	Fred Luthans; Organizational Behaviour McGraw Hill International Edition. Carnegie, D. How to Win Friends and Influence People, New York: Gallery Books, 2019.
7.	Gaur, R. R., Sangal, R., & Bagaria, G. P. “Human Values and Professional Ethics”, New Delhi, India: Excel Books, 2010.
8.	Sarvesh Gulati: Corporate Grooming and Etiquette, Rupa Publisher 2016
9.	Dale Carnegie: Quick & Easy Way To Effective Speaking, Simon & Schuster Publication, 1990.

FULL STACK DEVELOPMENT USING DJANGO			
Course Code	25MCA211	Course Type	PEC
Teaching Hours/Week (L:T:P: S)	3:0:0:0	Credits	03
Total Teaching Hours	40	CIE + SEE Marks	50+50
Teaching Department: Master of Computer Applications			
Course Objectives:			
1.	To introduce students to the Django web framework and its Model-View-Template (MVT) architecture.		
2.	To enable students to design and manage databases using Django’s Object-Relational Mapping (ORM).		
3.	To provide hands-on experience in creating dynamic web pages using Django templates and forms.		
4.	To develop skills in implementing user authentication, session management, and security best practices.		
5.	To familiarize students with REST APIs, AJAX-based interactivity, and deployment of Django applications on cloud platforms.		
UNIT - I			
Django Basics & MVT Architecture:			08 Hours
What is a Web Framework? Introduction to Django, MVC vs MVT Architecture, Creating and Importance of virtual environments (In VS Code), Installing and setting up Django projects and apps, Project structure explained (settings.py, urls.py, views.py), Creating views and mapping URLs, HTTP request-response cycle in Django, Debugging with error pages, Concept of loose coupling in Django, Using virtual environments and pip, Working with development vs production settings			
UNIT - II			
Models, ORM & Database Integration :			08 Hours
Introduction to Django ORM and relational databases, Creating models and fields, Model relationships (OneToOne, OneToMany, ManyToMany), Making migrations and applying them, The QuerySet API: creating, retrieving, filtering, updating, deleting records, Adding __str__, Meta, ordering, and verbose_name, Understanding model managers, Schema evolution and database migrations, Using PostgreSQL, MySQL, and SQLite with Django, Custom model methods			
UNIT - III			
Django Templates, Static & Forms :			08 Hours
Template system basics and directory structure, Template inheritance and blocks, Template tags, filters, and context processors, Working with static files (CSS, JS) and media files (uploads), Creating Django forms (Form, ModelForm), Custom validation (clean methods), Rendering forms manually vs {{form.as_p }}, Manual vs auto form rendering, CSRF protection in Django, Built-in form field types and widgets, Integrating Bootstrap with templates			
Admin Interface, CBVs, State Management :			08 Hours
Activating and using the Django admin site, Customizing model admin (list_display, search_fields, readonly_fields), Introduction to Function-Based Views (FBV) and Class-Based Views (CBV), Using ListView, DetailView, CreateView, UpdateView, DeleteView, Form handling using CBVs, Sessions, cookies, and state management. Authentication system: login, logout, signup, password			

reset, Permissions, groups, and decorators (login_required), Handling file uploads (images, PDFs), Overriding default user model (basic introduction)

APIs, AJAX, Security & Deployment :
08 Hours

Introduction to REST APIs and serializers (conceptual), Using fetch and axios for frontend integration, AJAX basics and jQuery in Django, Sending/receiving JSON data in views, CSRF and CORS in AJAX, Generating and downloading CSV and PDF files, Introduction to deployment on PythonAnywhere/Heroku, Django project structure for production, Django settings for production (debug, allowed hosts, static files), Security best practices: XSS, SQL Injection, clickjacking prevention

Course Outcomes: At the end of the course student will be able to

1	Understand Django's Architecture and Core Concepts
2	Develop Models and Integrate Databases using Django ORM
3	Design Dynamic Web Pages Using Templates and Forms
4	Implement Secure User Authentication and State Management
5	Integrate Frontend Logic, Generate Non-HTML Output and Deploy Projects

Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes

Program Outcomes→	1	2	3	4	5	6	7	8	PSO	PSO
↓ Course Outcomes									1	2
25MCA211.1	3	2	--	--	--	--	--	--	3	--
25MCA211.2	3	3	2	2	--	--	--	--	3	2
25MCA211.3	2	3	3	2	2	--	--	--	3	3
25MCA211.4	2	2	3	3	2	2	--	--	3	2
25MCA211.5	3	2	3	3	3	3	2	2	3	3

1: Low 2: Medium 3: High

TEXT BOOKS:

1. Adrian Holovaty, Jacob Kaplan Moss, The Definitive Guide to Django

REFERENCE BOOKS:

1. William Vincent – Django for Beginners
2. Antonio Mele – Django 4 by Example
3. Arun Ravindran – Django Design Patterns and Best Practices

WEB DEVELOPMENT USING RUBY ON RAILS & POSTGRESQL			
Course Code:	25MCA212	Course Type	PEC
Teaching Hours/Week (L: T: P: S)	3:0:0:0	Credits	03
Total Teaching Hours	40	CIE + SEE Marks	50+50
Prerequisite			
Teaching Department: Master of Computer Applications			
Course Objectives:			
1.	Learn how websites work and how Ruby on Rails helps build them.		
2.	Build web pages and connect them using routes, controllers, and views		
3.	Understand how data is stored and managed using PostgreSQL		
4.	Connect Rails apps with PostgreSQL to save and use data		
5.	Learn how to deploy web apps online and keep them running		
UNIT-I			
Web Fundamentals & Getting Started with Rails			08 Hours
Basics of web architecture (HTTP, request/response cycle), MVC architecture in Rails, Introduction to Ruby, Rails project structure			
Installing Rails and PostgreSQL, Creating first Rails project, Scaffold generation, Running server			
UNIT-II			
Routing, Controllers, and Views			08 Hours
Routing in Rails, Controllers and actions, View rendering with ERB, Flash messages, Layout templates			
Dynamic pages, HTML forms in Rails, Navigation, Handling user input			
UNIT-III			
Models, Associations & User Management			08 Hours
ActiveRecord ORM, Migrations, Validations and Callbacks, Model Relationships, Devise Authentication.			
Creating models, Form validations, Sign-up/Login, Simple blog with users and posts			
UNIT-IV			
PostgreSQL Basics and SQL Queries			08 Hours
PostgreSQL architecture, Data types, SQL basics (SELECT, INSERT, etc.), Joins, Indexing.			
Creating/querying databases, SQL commands, Using pgAdmin/CLI, Fetching related data			
UNIT-V			
PostgreSQL in Rails & Deployment Concepts			08 Hours
Connecting Rails to PostgreSQL, Schema management, Indexes, Transactions, Deployment (Heroku)			
Rails-PostgreSQL integration, Migrations, Seed data, Deploy to Heroku			
Course Outcomes: At the end of the course student will be able to			
1.	Describe the structure of a web application and explain the role of Ruby on Rails in web development.		
2.	Develop basic web applications using controllers, views, models, and routes in Rails.		
3.	Write SQL queries to create, retrieve, update, and delete data using PostgreSQL.		
4.	Integrate a PostgreSQL database with a Rails application for data-driven features		
5.	Deploy a full-stack web application to a live server (like Heroku) with database support.		

Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes											
	Program Outcomes→	1	2	3	4	5	6	7	8	PSO	PSO
	↓ Course Outcomes									1	2
	25MCA212.1	3	--	--	2	--	--	--	2	3	2
	25MCA212.2	3	2	3	3	2	--	--	2	2	2
	25MCA212.3	3	2	--	2	--	--	--	2	2	2
	25MCA212.4	3	3	2	3	2	--	--	2	2	3
	25MCA212.5	--	2	3	3	3	2	2	2	2	2
1: Low 2: Medium 3: High											
TEXTBOOKS:											
1.	Agile Web Development with Rails 7 by Sam Ruby, Pragmatic Bookshelf (2023), ISBN: 9781680509293										
2.	The Odin Project – Full Stack Ruby on Rails Path (Free, Online) URL: https://www.theodinproject.com/paths/full-stack-ruby-on-rails										
3.	PostgreSQL: Up and Running by Regina O. Obe & Leo S. Hsu, O'Reilly Media, 3rd Ed (2021), ISBN: 9781491963418										
4.	Learning PostgreSQL 11 by Salahaldin Juba and Andrey Volkov, Packt Publishing (2019), ISBN: 9781789535464										
REFERENCE BOOKS:											
1.	The Rails 5 Way, Obie Fernandez, Addison-Wesley Professional (2017), 9780134657674, Deep dive into professional Rails programming practices										
2.	Ruby on Rails Tutorial: Learn Web Development with Rails (6th Edition), Michael Hartl, Addison-Wesley Professional (2022), ISBN: 9780138051836 Step-by-step tutorial ideal for beginners, includes testing, authentication, and deployment										
3.	Learning PostgreSQL 11, Salahaldin Juba, Andrey Volkov. Packt Publishing (2019) ISBN: 9781789535464										
4.	Mastering PostgreSQL in Application Development, Dimitri Fontaine, The SQL Company (2019), ISBN:9782956601402										
E Books / MOOCs/ NPTEL											
1.	NPTEL Course: Programming, Data Structures and Algorithms using Python, By Prof. Madhavan Mukund IIT Madras.										
2.	MOOC: Ruby on Rails Web Development Specialization, Johns Hopkins University, Coursera, 2023.										

APPLIED DATA SCIENCE AND ANALYTICS			
Course Code:	25MCA221	Course Type	PEC
Teaching Hours/Week (L:T:P:S)	3:0:0:0	Credits	03
Total Teaching Hours	40	CIE + SEE Marks	50+50
Teaching Department: Master of Computer Applications			
Course Objectives:			
1.	Understand the overview of the Data Science Process.		
2.	Apply the knowledge of Data Collection and Preprocessing to acquire the data and clean the data for further training.		
3.	Understand how to perform Exploratory Data Analysis using matplotlib, pandas and Seaborn.		
4.	Use Statistical Inference to gain key information from the dataset.		
5.	Use machine learning to create a data model and evaluate the model w.r.t its accuracy, precision and recall.		
UNIT - I			
Introduction to Data Science			04 Hours
What is Data Science? Data Science vs. Data Analytics vs. Business Intelligence, Data Science in the Big Data World, Types of Data- Structured, Unstructured, Natural Language, Machine Generated, Graph Based, Audio, Videos, Images & Streaming Data, Quantitative Vs Categorical Data.			
The Data Science Process- Defining Research Goals, Retrieving Data, Cleansing, integrating and transforming data, Exploratory Analysis, Build the models, presenting findings and building applications.			
Python for Data Science:			04 Hours
Data Types-int, float, bool, str, list, tuple, dictionary, set, Functions. Exploring NumPy Library- Indexing, Slicing, One dimensional Array, Multi-Dimensional Array, Concatenation and Splitting Arrays, Arithmetic operations, Aggregation, Sorting, Partitioning.			
UNIT - II			
Data Collection and Preprocessing			04 Hours
Getting Data: Reading Files, Scraping from Web, Using API's. Data cleaning: missing values, duplicates, encoding, scaling.			
Data Manipulation using Pandas			04 Hours
Loading Data- read_csv, read_excel, read_html. Data Cleaning and Preparation- Filtering out missing data, Filling in Missing Data, Removing Duplicates, Replacing Values, Detecting and filtering Outliers, String Manipulation.			
UNIT - III			
Exploratory Data Analysis (EDA)			04 Hours
Elements of Structured data, Estimates of Location-Mean, Median, Mode, Outliers, Estimates of Variability- Standard Deviation, Percentiles, Boxplots, Frequency Tables and Histograms, Density Plots. Exploring Binary and Categorical data- Bar Charts, Pie Charts. Correlation, Scatterplots			
Plotting using Matplotlib, Pandas & Seaborn			04 Hours
Line plots, Scatter Plots, Density and Contour plots, Histogram, Bar plots, Corelation Matrix, Heat Maps.			

UNIT - IV											
Statistical inference for Data Science										04 Hours	
Data and Sampling Distributions - Random Sampling and Sample Bias, Sample Mean Vs Population Mean, Selection Bias, Bootstrapping, Resampling Vs Bootstrapping. A/B Testing, Hypothesis Tests.											
Data Analysis using Python										04 Hours	
Perform Data analysis on Iris and Titanic Dataset.											
UNIT - V											
Data Modelling & Data Ethics										04 Hours	
What is a model? What is Machine Learning? Supervised, Unsupervised and Reinforcement Learning, Overfitting and Underfitting, Model Evaluation -Accuracy, Precision & Recall, The Bias-Variance Trade-off.											
What is Data Ethics, Building bad data products, Trading off accuracy and fairness, Recommendations, Biased Data, Data Protection											
Data Modelling using Python										04 Hours	
Perform Data Modelling on Titanic Dataset using Logistic Regression and evaluate the model.											
Course Outcomes: At the end of the course student will be able to											
1.	Describe the concept of the Data Science Process.										
2.	Apply the knowledge of Data Collection and Preprocessing to prepare the dataset for further training.										
3.	Perform Exploratory Data Analysis to identify critical information from data for further analysis.										
4.	Apply Statistical inference to back the data science model.										
5.	Create data models and evaluate them for their accuracy, precision and recall.										
Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes											
	Program Outcomes→	1	2	3	4	5	6	7	8	PSO	PSO
	↓ Course Outcomes									1	2
	25MCA221.1	3	3	3	3	-	-	-	3	3	3
	25MCA221.2	3	3	3	3	-	-	-	3	3	3
	25MCA221.3	3	3	3	3	-	-	-	3	3	3
	25MCA221.4	3	3	3	3	-	-	-	3	3	3
	25MCA221.5	3	3	3	3	-	-	3	3	3	3
1: Low 2: Medium 3: High											
TEXT BOOKS:											
1.	Introducing Data Science- Big data, machine learning, and more, using Python tools, Davy Cielen, Arno D. B. Meysman, and Mohamed Ali, Manning Publications, 2016.										
2.	Practical Statistics for Data Scientists, by Peter Bruce, Andrew Bruce, O'Reilly Media, Inc., May 2017										
3.	Python Data Science Handbook, by Jake VanderPlas, O'Reilly Media, Inc., 2nd Edition										
REFERENCE BOOKS:											
1.	Python for Data Analysis, by Wes McKinney, O'Reilly Media, Inc., 3rd Edition										
2.	Data Science from Scratch, by Joel Grus, O'Reilly Media, Inc., 2nd Edition.										
3.	Doing Data Science, by Cathy O'Neil, Rachel Schutt, O'Reilly Media, Inc.										
E Books / MOOCs/ NPTEL											
1.	Python for Data Science - By Prof. Ragunathan Rengasamy IIT Madras NPTEL, 2025										

DIGITAL IMAGE PROCESSING & PATTERN RECOGNITION			
Course Code	25MCA222	Course Type	PEC
Teaching Hours/Week (L: T: P: S)	3:0:0:0	Credits	03
Total Teaching Hours	40	CIE + SEE Marks	50+50
Prerequisite	Data Structures with Algorithms Mathematical Foundation for Computer Applications		
Teaching Department: Master of Computer Applications			
Course Objectives:			
1.	To understand the fundamentals of digital image processing.		
2.	To implement spatial and frequency domain techniques for image enhancement.		
3.	To apply image restoration and segmentation techniques.		
4.	To understand various pattern recognition methods.		
5.	To implement different image classification Clustering techniques and to develop real-world applications using image processing and pattern recognition		
UNIT-I			
Fundamentals of Digital Image Processing			03 Hours
Introduction to Digital Image processing, Elements of visual perception, Image formation & human vision system; Applications in various domains (medical, agriculture, security, etc.)			
Image Representation & Basic Operations			04 Hours
Digital image representation; Image acquisition, sampling & quantization; Image transformations.			
UNIT-II			
Image Enhancement in Spatial Domain			04 Hours
Histogram processing; Point processing; Spatial filtering: smoothing (mean, median, Gaussian); Sharpening (Laplacian, high-boost)			
Image Enhancement in Frequency Domain			04 Hours
Fourier Transform (DFT, FFT); filtering in frequency domain; Low-pass & high-pass filtering (Gaussian, Butterworth filters)			
UNIT-III			
Image restoration and Reconstruction			04 Hours
Noise models, restoration in the presence of noise, inverse filtering.			
Morphological Image Processing			03 Hours
Preliminaries, erosion and dilation, opening and closing, basic morphological algorithms.			
Image Segmentation			03 Hours
Fundamentals, point, line and edge detection, thresholding, region-based segmentation.			
UNIT-IV			
Pattern Recognition Techniques			04 Hours
Introduction to Pattern Recognition; Supervised vs. unsupervised learning; Feature space & decision boundaries			
Statistical & Probabilistic Models			03 Hours
Bayes decision theory; Naïve Bayes classifier; Linear Discriminant Analysis (LDA)			

UNIT-V											
Image Classification										04 Hours	
Principal Component Analysis (PCA); Support Vector Machines (SVM); K-Nearest Neighbors (KNN).											
Clustering										04 Hours	
Overview of clustering; Different distance functions and similarity measures, K-means clustering; Hierarchical clustering; DBSCAN.											
Course Outcomes: At the end of the course student will be able to											
1.	Apply fundamental concepts of digital image processing to explain the image representation.										
2.	Implement spatial and frequency domain techniques for image enhancement.										
3.	Perform image restoration and segmentation for feature extraction.										
4.	Analyze various pattern recognition techniques and their applications.										
5.	Implement different image classification Clustering techniques. Develop real-world applications of digital image processing and pattern recognition.										
Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes											
	Program Outcomes→	1	2	3	4	5	6	7	8	PSO	PSO
	↓ Course Outcomes									1	2
	25MCA222.1	3	3	3	3	3	2	2	2	3	3
	25MCA222.2	3	3	3	3	3	2	2	2	3	3
	25MCA222.3	3	3	3	3	3	2	2	2	3	3
	25MCA222.4	3	3	3	3	3	2	2	2	3	3
	25MCA222.5	3	3	3	3	3	2	2	2	3	3
1: Low 2: Medium 3: High											
TEXT BOOKS:											
1.	Rafael C. Gonzalez & Richard E. Woods : Digital Image Processing, 4th Edition, PHI										
2.	R.O. Duda, P.E. Hart and D.G. Stork, Pattern Classification, John Wiley.										
REFERENCE BOOKS:											
1.	Anil K. Jain : Fundamentals of Digital Image Processing, Prentice Hall.										
2.	William K. Pratt, John Wiley & Sons Inc : Digital Image Processing, 3 rd Edition.										
3.	Statistical Pattern Recognition; K. Fukunaga; Academic Press.										
4.	S. Theodoridis and K. Koutroumbas, Pattern Recognition, 4 th Edition, Academic Press.										
5.	Robert Schalkoff, Pattern Recognition: Statistical, Structural and Neural Approaches, John Wiley & Sons.										

MOBILE APPLICATION DEVELOPMENT USING ANDROID AND FLUTTER				
Course Code		25MCA231	Course Type	PEC
Teaching Hours/Week (L: T: P: S)		3-0-0	Credits	03
Total Teaching Hours		40	CIE + SEE Marks	50 + 50
Teaching Department: Master of Computer Applications				
Course Objectives: Students can learn mobile application development by understanding the following tools and components.				
1.	Understand the Fundamentals of Mobile Application Development.			
2.	Develop Android Applications Using Core Components.			
3.	Design Interactive User Interfaces and Implement Location-Based Features.			
4.	Implement Messaging, Networking, and Services in Android.			
5.	Build Cross-Platform Applications Using Flutter and iOS Tools.			
UNIT-I				
Introduction				8 Hours
Preliminary Considerations – Cost of Development, Importance of Mobile Strategies in the Business World, Effective use of Screen Real Estate, Understanding Mobile Applications Users, Understanding Mobile Information Design, Understanding Mobile Platforms, Using the Tools of Mobile Interface Design.				
UNIT-II				
Getting Started with Android Programming				8 Hours
What is Android? Obtaining the required tools, Anatomy of an Android Application, Components of Android Applications, Activities, Fragments, Utilizing the Action Bar.				
UNIT-III				
Android UI Design and Location Based Services				8 Hours
Views and View Groups, Basic Views, Displaying Maps, Getting Location Data, Preparing for Publishing, Deploying APK Files.				
UNIT-IV				
Android Messaging and Networking				8 Hours
SMS Messaging, Sending Email, Networking, Downloading Binary Data, Text files, Accessing Web Services, Performing Asynchronous Call, Creating your own services.				
UNIT-V				
Flutter and IOS				8 Hours
Flutter: Introduction, Installation, Creating Simple Application in Android Studio, Architecture Application, Introduction to Dart Programming, Widgets, Layouts, Gestures. IOS: Obtaining the tools and SDK, Components of XCODE, Architecture of iOS, Objective C and Swift Programming Languages				

Case Studies:

Design and develop a mobile application to display user profiles with 3 UI activities using intents, design an application that contains Phone Contacts in a vertical linear manner. Selected contact appears at the top of the list with a large, italicized font and a blue background, create an application that uses Layout Managers and Event Listeners, develop a mobile application to list the tourist places of Karnataka using List View, develop a mobile application to register a form in first activity and display the registered information in second activity using intents, develop a standard calculator application to perform basic calculations like addition, subtraction, multiplication and division, devise an application that draws basic graphical primitives (rectangle, circle) on the screen, build a mobile application that creates, saves, updates and deletes data in a database.

Course Outcomes: Upon completion of the course, the students will be able to:

1.	Illustrate effective user interfaces that leverage evolving mobile device capabilities.
2.	Develop applications using different SDK frameworks and toolkits.
3.	Establish various methods to integrate database and server-side technologies.
4.	Design and develop open-source software based mobile applications.
5.	Build and deploy competent mobile development solutions.

Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes

Program Outcomes→	1	2	3	4	5	6	7	8	PSO	PSO
↓ Course Outcomes									1	2
25MCA231.1	3	2	-	3	3	-	3	3	3	3
25MCA231.2	3	2	-	3	3	-	3	3	3	3
25MCA231.3	3	2	-	3	3	-	3	3	3	3
25MCA231.4	3	3	-	3	-	-	3	3	3	3
25MCA231.5	3	3	-	3	-	-	3	3	3	3

1: Low 2: Medium 3: High

TEXTBOOKS:

1.	Jeff McWherter and Scott Gowell, "Professional Mobile Application Development", 1 st Edition, 2012.
2.	Wei-Meng Lee, "Beginning Android Application Development", Wiley 2011.

REFERENCE BOOKS:

1.	Reto Meier, "Professional Android 4 Application Development", Wrox Publications 2012.
2.	Rap Payne, "Beginning App Development with Flutter: Create Cross-Platform Mobile Apps", 1 st Edition, Publications 2019.

BLOCK CHAIN TECHNOLOGY			
Course Code	25MCA232	Course Type	PEC
Teaching Hours/Week (L: T: P:S)	3:0:0:0	Credits	03
Total Teaching Hours	40	CIE + SEE Marks	50+50
Teaching Department: Master of Computer Applications			
Course Objectives:			
1.	To understand Block Chain- Blockchain Architecture and its Applications.		
2.	To understand crypto currencies- Types and Applications		
3.	To understand Concept of Double Spending- Hashing- Mining- payment verification- Resolving Conflicts and Creation of Blocks		
4.	To understand Crypto currency wallets and conversion to Fiat Currency.		
5.	To understand Smart contracts- usage- application- working principle- law and regulations.		
UNIT-I			
Introduction			08 Hours
Introduction to Blockchain, How Blockchain works, Blockchain vs Bitcoin, Practical applications, public and private key basics, pros and cons of Blockchain, Myths about Bitcoin.			
UNIT-II			
Blockchain Architecture			08 Hours
Blockchain Architecture, versions, variants, use cases, Life use cases of blockchain, Blockchain vs shared Database, Introduction to crypto currencies, Types, Applications.			
UNIT-III			
Double Spending			08 Hours
Concept of Double Spending, Hashing, Mining, Proof of work. Introduction to Merkel tree, Privacy, payment verification, Resolving Conflicts, Creation of Blocks.			
UNIT - IV			
Introduction to Bitcoin			08 Hours
Introduction to Bitcoin, key concepts of Bitcoin, Merits and De Merits Fork and Segwits, Sending and Receiving bitcoins, choosing bitcoin wallet, Converting Bitcoins to Fiat Currency.			
UNIT-V			
Introduction to Ethereum			08 Hours
Introduction to Ethereum, Advantages and Disadvantages, Ethereum vs Bitcoin, Introduction to Smart contracts, usage, application, working principle, Law and Regulations. Case Study			
Course Outcomes: At the end of the course student will be able to			
1.	Explain the core concepts of blockchain technology and differentiate it from traditional databases and systems.		
2.	Analyze blockchain architecture and assess its real-world applications, including the role of cryptocurrencies.		
3.	Demonstrate an understanding of security mechanisms like hashing, proof of work, and Merkle trees in blockchain.		
4.	Describe Bitcoin’s working principles, including transactions, wallets, and conversion to fiat currency.		
5.	Evaluate Ethereum and smart contracts by examining their applications, advantages, and legal considerations.		

Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes											
	Program Outcomes→	1	2	3	4	5	6	7	8	PSO	PSO
	↓ Course Outcomes									1	2
	25MCA232.1	-	3	2	-	-	-	2	-	3	1
	25MCA232.2	-	3	2	-	-	-	2	-	3	1
	25MCA232.3	-	3	2	-	-	-	2	-	3	1
	25MCA232.4	-	3	2	-	-	-	2	-	3	1
	25MCA232.5	-	3	2	-	-	-	2	-	3	1
1: Low 2: Medium 3: High											
TEXTBOOKS:											
1.	Bikramaditya Singhal, Gautam Dhameja, Priyanshu Sekhar Panda, , “Beginning Blockchain: A Beginner's Guide to Building Blockchain Solutions”, APress, 2018 (Reprint year 2024)										
2.	Arshdeep Bahga, Vijay Madiseti, “Blockchain Applications: A Hands-On Approach”. 2018										
3.	Melanie Swan, “Blockchain- Blueprint for a new economy”, 2015, Oreilly										
4.	Daniel Drescher , “Blockchain Basics-A Non-Technical Introduction in 25 steps”, APress, Second Edition 2020.										
REFERENCE BOOKS:											
1.	Aravind Narayan. Joseph Bonneau, “Bitcoin and Crypto currency Technologies”, Princeton										

ENTERPRISE JAVA LAB												
Course Code				25MCA206			Course Type			PCC		
Teaching Hours/Week (L: T: P: S)				0:0:4:0			Credits			02		
Total Teaching Hours				26			CIE + SEE Marks			50+50		
Teaching Department: Master of Computer Applications												
Course Objectives:												
1.	To understand and implement Object Oriented Programming in Java, handling of exceptions and multithreaded programs.											
2.	To understand and implement network and distributed programming in Java											
3.	To understand Java APIs to connect to a backend database and development of Graphic User Interface for Java Applications.											
4.	To understand and implement Java components as well as develop web applications in Java.											
List of Experiments												
1.	Programs to implement encapsulation, abstraction, inheritance and polymorphism											
2.	Programs to handle Exceptions											
3.	Implementation of Multithreading Programs.											
4.	Implementation of Stream classes.											
5.	Network programming using Sockets and Datagrams											
6.	Implementation of Swing components											
7.	Programs to connect to backend databases using JDBC											
8.	Programs to Implement Java Beans.											
9.	Development of Server sided programs using servlets and JSP.											
10.	Implementation of Spring Boot 2.0											
Course Outcomes: At the end of the course student will be able to												
1.	Develop Java applications using Object Oriented Programming features, exception handling, multithreading and networking features.											
2.	Develop Java applications that connect to a backend database, provide Graphic User Interface, develop Custom components as well as web applications.											
Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes												
	Program Outcomes→		1	2	3	4	5	6	7	8	PSO	PSO
	↓ Course Outcomes										1	2
	25MCA206.1		3	3	3	3	-	3	-	3	3	-
	25MCA206.2		3	3	3	3	-	3	-	3	3	-
1: Low 2: Medium 3: High												
REFERENCE BOOKS:												
1.	Herbert Schildt : Java: The Complete Reference, 11 th Edition, Tata McGraw Hill., 2019											
2.	Mastering Spring Boot 2.0: Build modern, cloud-native, and distributed systems using Spring Boot, Dinesh Rajput, Pack Publishing Ltd., 2018.											
3.	Dr. Donald Doherty and Rick Leinecker : JavaBeans Unleashed											
4.	James Goodwill : Developing Java Servlets											
5.	Steven Holzner : Java 2 Black Book											
6.	Ed Roman : Mastering Enterprise JavaBeans											
7.	Jim Keogh : The Complete Reference J2EE. Tata McGraw Hill, 2008.											

ADVANCED WEB TECHNOLOGIES LAB

Course Code	25MCA207	Course Type	PCC
Teaching Hours/Week (L: T: P: S)	0:0:4:0	Credits	02
Total Teaching Hours	26	CIE + SEE Marks	50+50
Prerequisite	Fundamentals of Programming		

Teaching Department: Master of Computer Applications

Course Objectives:

1.	To develop an ability to design and implement static and dynamic website
2.	Use appropriate client-side or Server-side applications
3.	Develop basic programming skills using Javascript and JQuery
4.	Learn the language of web : HTML,CSS, Bootstrap, PHP, JSON

List of Experiments

1.	Develop and demonstrate XHTML document that illustrate the use of images, tables, links, formatting tools, lists, forms, frames and style sheets
2.	Develop and demonstrate HTML5 document that illustrate the use of new semantics elements, Migration, canvas, svg, input types, new form elements and attribute, google map, media(audio, video).
3.	Develop and demonstrate XHTML document that includes javascript for the following a. I/O statements b. Control statements c. Arrays d. Objects e. Functions f. Constructors g. Event Handling through... - Body elements - Button elements - Text box and password elements h. Element accessing, positioning and moving h. Stacking and slow movements to elements h. Dragging and dropping the elements h. Element visibility
4.	Develop and Demonstrate XHTML document that includes javascript for Pattern Matching Using Regular Expressions
5.	Design a XML document to store information and display the XML document.
6.	Execute simple programs using PHP, Create a Web Application using PHP
7.	Database programs using PHP
8.	Simple programs using JQuery, JSON and AJAX, Angular
9.	AJAX applications using various GUI components, Implementation of “Shopping cart” application using AJAX
10.	AJAX application to keep track of user data and retrieve the session data
11.	AJAX application to verify client side and server side data
12.	Simple Flex applications, Flex applications using actionsript.
13.	Web applications that use RPC
14.	Building RESTful Web Services with JAX-RS
15.	Simple SOAP application.

Course Outcomes: At the end of the course student will be able to

- | | |
|----|---|
| 1. | Design and implement static and dynamic websites with good aesthetic sense of designing and latest technical know-how's |
| 2. | Analyze and apply the role of languages like HTML, Bootstrap, CSS, XML, JavaScript, PHP and protocols in the workings of the web applications and web Services. |

Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes

Program Outcomes→	1	2	3	4	5	6	7	8	PSO	PSO
↓Course Outcomes									1	2
25MCA207.1	3	3	3	3	2	-	3	3	3	-
25MCA207.2	3	3	2	2	3	-	3	3	3	-

1: Low 2: Medium 3: High

REFERENCE BOOKS:

- | | |
|----|---|
| 1. | Programming the World Wide Web" is the 8th edition, published by Pearson. |
| 2. | Internet & World Wide Web: How to Program" by Paul Deitel, Harvey Deitel, and Abbey Deitel. |

E Resources

- | | |
|----|---|
| 1. | https://onlinecourses.swayam2.ac.in/nou22_cs03/preview |
| 2. | https://onlinecourses.swayam2.ac.in/aic20_sp32/preview |

TECHNICAL SEMINAR AND REPORT WRITING													
Course Code				25MCA208				Course Type			Seminar		
Teaching Hours/Week (L: T: P: S)				0:0:0:2				Credits			02		
Total Teaching Hours				15				CIE + SEE Marks			50		
Teaching Department: Master of Computer Applications													
Course Objectives:													
1.	To learn to prepare for a seminar on a technical topic and present it effectively using overhead projectors and presentation software.												
2.	To learn to prepare a well formatted and well-organized report with appropriate content using report preparation software.												
The student will prepare for a seminar on a technical topic and present it. A panel consisting of faculty members will evaluate the presentation and marks will be provided. The student will submit a report on the same. For the preparation of the presentation the student will refer to technical papers from journals. The presentation will be prepared using LaTeX tool.													
Course Outcomes: At the end of the course student will be able to													
1.	Review a technical topic												
2.	Present effectively												
3.	Help the students face placement interviews.												
4.	Prepare a well-organized report with appropriate content												
Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes													
	Program Outcomes→			1	2	3	4	5	6	7	8	PSO	PSO
	↓ Course Outcomes											1	2
	25MCA208.1			3	1	-	2	-	-	3	2	3	2
	25MCA208.2			3	1	-		-	-	-	2	3	2
	25MCA208.3			1	1	-	1	-	-	3	1	3	2
	25MCA208.4			-	-	-	-	-	-	-	-	3	2
												1: Low 2: Medium 3: High	
REFERENCE BOOKS:													
1.	Naomi Karten, “Presentation Skills for Technical Professionals: Achieving Excellence (Soft Skills for It Professionals)”- 23 Sept. 2010												
2.	Fink, A., 2009. Conducting Research Literature Reviews: From the Internet to Paper. Sage Publications												
3.	Day, R .A., 1992. How to Write and Publish a Scientific Paper, Cambridge University Press.												

FULL STACK WEB DEVELOPMENT USING MERN			
Course Code	25MCA301	Course Type	PCC
Teaching Hours/Week (L: T: P: S)	3:0:0:0	Credits	03
Total Teaching Hours	40	CIE + SEE Marks	50+50
Teaching Department: Master of Computer Applications			
Course Objectives:			
1.	Understand the MVC architectural pattern and its role in web development. Articulate what React is and why it is useful.		
2.	Utilize modern JavaScript features like ES6+ classes, modules, and arrow functions. Making use of controlled components, create and verify forms. Use a JavaScript package manager (either npm or Yarn)		
3.	Build reusable React components and implement state management effectively. Making HTTP requests to read or modify data.		
4.	Develop and integrate RESTful APIs using Express.js and MongoDB. Implement client-side routing using React Router.		
5.	Students able to create CRUD operations, routing mechanisms, and UI components.		
UNIT-I			
Introduction to the MERN stack:			04 Hours
Introduction, The MVC architectural Pattern. MERN Components React, Node.js, Express, MongoDB, Advantages and real-world applications of MERN Stack, understanding Isomorphic and types			
MERN Components			04 Hours
Advantages and real-world applications of MERN Stack. Understanding Isomorphic and types. Welcome to React- Obstacles and Roadblocks, Reacts Future			
UNIT-II			
Understanding React			04 Hours
Server setup, setting up a Development Environment -NVM, Node Js, Project, NPM, Express, Understanding JSX and it's compilation process, Separate Script File, Transform, Automate, React Library			
Web Server:			04 Hours
React Components-React classes, Composing components, passing data- using properties, property validation, using children's Dynamic, React Lifecycle Methods.			
UNIT-III			
Understanding React State:			04 Hours
React State – Setting state, Event handling, communicating from child to parent, Stateless components, Designing Components-state vs props			
React Components			02 Hours
component hierarchy communication, Introduction to React Context API.			
UNIT-IV			
Building RESTful APIs			05 Hours
REST , HTTP method as Actions, JSON, Express, Request Object, Response objects, Middleware, The list API , The create API, Using the LIST API, Using the Create API, Error Handling.			

Mongo DB:										05 Hours		
Mongo DB: Basics, Schema initialization , MongoDB Node.JS Driver, callbacks, promises, Reading from MongoDB, Writing to MongoDB												
UNIT-V												
Working with React Router and forms:										04 Hours		
Routing Techniques, Simple Routing, Route parameters, Route Query String, Programmatic Routing, Nested Routes, Browser history.												
Forms:										04 Hours		
Forms, Filter form, Get API, Edit page, Building UI Components and Forms, update API, Delete API.												
Course Outcomes: At the end of this course, the students will be able to:												
1.	Understand the concepts of MVC components and architecture and Implement node js, Express and MongoDB.											
2.	Use JSX transformation and automatic recompiling for better React development.											
3.	Develop React components and manage UI state effectively.											
4.	Implement State management and event handling in React applications.											
5.	Build Restful APIs, perform CRUD operations, and integrate UI components efficiently.											
Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes												
Program Outcomes→			1	2	3	4	5	6	7	8	PSO	PSO
↓ Course Outcomes											1	2
25MCA301.1			3	3	2	2	2	1	1	1	3	2
25MCA301.2			2	3	3	3	3	1	1	1	3	3
25MCA301.3			2	3	3	3	3	1	1	1	3	3
25MCA301.4			2	3	3	3	3	1	1	1	3	3
25MCA301.5			3	3	3	3	3	2	1	1	3	3
1: Low 2: Medium 3: High												
TEXTBOOKS:												
1.	Pro MERN Stack, Vasan Subramanian, 2019, ISBN-13(pbk):978-1-4842-2653-7											
2.	MERN Quick Start Guide, Eddy Wilson Iriarte Koroloiva, 2018, PACKT Publication, ISBN 978-1-78728-108-0											
REFERENCE BOOKS:												
1.	Full stack React, Anthony Accomazzo, Ari Lerner, Nate Murray, Clay Allsopp, David Gutman, and Tyler McGinnis, 2017.											

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING			
Course Code	25MCA302	Course Type	PCC
Teaching Hours/Week (L:T:P: S)	3:0:0:0	Credits	03
Total Teaching Hours	40	CIE + SEE Marks	50+50
Prerequisite	Mathematical Foundation for computer Applications		
Teaching Department: Master of Computer Applications			
Course Objectives:			
1.	To understand the advantages and types of Artificial Intelligence		
2.	To understand machine learning, the need for machine learning, its types and data preprocessing		
3.	To understand Regression Models, Decision Trees, Support Vector Machines, Naive Bayesian Learning and Evaluating Regression model performance.		
4.	To understand Clustering Techniques, K-NN, Dimensionality Reduction using Principal Component Analysis and Reinforcement Learning algorithms.		
5.	To learn deep learning concepts, Neural network, Gradient descent, Back Propagation, and Convolutional Neural Networks.		
UNIT - I			
Artificial Intelligence:			06 Hours
What is AI; Why Artificial Intelligence; Goals of AI; Advantages and disadvantages of AI; Applications of AI, Types of AI, Strategies for State Space Search (Representation of State Space as Graphs), Introduction to Heuristics			
UNIT - II			
Machine learning:			08 Hours
What is ML; Need for ML; Challenges and applications of ML; classification: Supervised, Unsupervised, Reinforcement learning, AI vs Machine learning, Data Preprocessing.			
UNIT - III			
Regression:			10 Hours
Simple, Linear Regression, Multiple Regression, Polynomial Regression, Logistic Regression, Decision Trees, Support Vector Machines, Naive Bayesian Learning, Evaluating Regression model performance.			
Unit - IV			
Clustering:			08 Hours
K-Means Clustering, Hierarchical Clustering, K-NN, Principal Component Analysis, Reinforcement Learning			
Unit - V			
Deep Learning and Artificial Neural Networks:			08 Hours
Introduction to deep learning, The Neuron, How do Neural network learn and work? Gradient descent, Back Propagation, ANN in python, Convolutional Neural Networks.			
Course Outcomes: At the end of the course student will be able to			
1.	Outline the fundamental concepts of Artificial Intelligence and its types.		
2.	Analyze the need for machine learning and data preprocessing		
3.	Apply Regression Models, Decision Trees, Support Vector Machines, Naive Bayesian Learning for data analysis and prediction.		

4.	Apply machine learning techniques like Clustering Techniques, K-NN, Dimensionality Reduction using Principal Component Analysis and Reinforcement Learning algorithms for data analysis and classification.
5.	Apply deep learning concepts like Neural network, Gradient descent, Back Propagation, and Convolutional Neural Networks for classification.

Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes

Program Outcomes→	1	2	3	4	5	6	7	8	PSO	PSO
↓ Course Outcomes									1	2
25MCA302.1	3	3	3	3	3	-	3	-	3	3
25MCA302.2	3	3	3	3	3	-	3	-	3	3
25MCA302.3	3	3	3	3	3	-	3	-	3	3
25MCA302.4	3	3	3	3	3	-	3	-	3	3
25MCA302.5	3	3	3	3	3	-	3	-	3	3

1: Low 2: Medium 3: High

TEXT BOOKS:

1.	Cosma Rohilla Shalizi, Advanced Data Analysis from an Elementary Point of View, 2015
2.	Tom M Mitchell, "Machine Learning", 1st Edition, McGraw Hill Education, 2017.
3.	Subramanya Bhat, Machine Learning – A Simplified Approach, 1 st Edition, Narosa Publishing House Pvt. Ltd., 2025
4.	Elaine Rich, Kevin K and S B Nair, "Artificial Intelligence", 3 rd Edition, McGraw Hill Education, 2017

REFERENCE BOOKS:

1.	Saroj Kaushik, Artificial Intelligence, Cengage learning, 2014
2.	Introduction to Machine Learning with Python - A Guide for Data Scientists (Muller Andreas)
3.	Data Science: Concepts and Practice - By Vijay Kotu, Bala Deshpande 2nd edition, Publisher -Morgan Kaufmann, 2018
4.	Allen B. Downey, "Think Python: How to Think Like a Computer Scientist", 2 nd Edition, Updated for Python 3, Shroff/O'Reilly Publishers, 2016
5.	Data Science for Dummies, Lillian Pierson, John Wiley & Sons, 21-Feb-2017

CLOUD COMPUTING AND BIG DATA ANALYTICS			
Course Code:	25MCA303	Course Type	PCC
Teaching Hours/Week (L:T:P:S)	3:1:0:0	Credits	04
Total Teaching Hours	50	CIE + SEE Marks	50+50
Prerequisites	Student must have some basic knowledge of Core Java, database concepts, and any of the Linux operating system flavors		
Teaching Department: Master of Computer Applications			
Course Objectives:			
1.	Understand the types, characteristics and challenges with digital data		
2.	Understand the concept of Big data Analytics and technology landscape		
3.	Understand the use of open source software framework called Hadoop		
4.	Understand different cloud computing service models and cloud security		
5.	Understand various technologies of cloud computing.		
UNIT - I			
Introduction to Big Data:			10 Hours
Classification of Digital Data, Structured Data, Sources of Structured Data, Ease of Working with Structured Data, Semi-Structured Data, Sources of Semi-Structured Data, Unstructured Data, Issues with “Unstructured” Data, How to Deal with Unstructured Data. Characteristics of Data, Evolution of Big Data, Definition of Big Data, Volume, Velocity, Variety, Challenges of Big Data, Other Characteristics of Data Which are Not Definitional Traits of Big Data, Traditional Business Intelligence (BI) versus Big Data, A Typical Data Warehouse Environment, A Typical Hadoop Environment, Coexistence of Big Data and Data Warehouse. Exploring the Big Data Stack			
Unit - II			
Introduction to Big Data Analytics :			10 Hours
Big Data Analytics, Classification of Analytics, Greatest Challenges that Prevent Businesses from Capitalizing on Big Data, Top Challenges Facing Big Data, Importance of Big Data Analytics, Terminologies Used in Big Data Environment, In Memory Analytics, In Database Processing, Symmetric Multiprocessor System, Massively Parallel Processing, Difference between Parallel versus Distributed Systems, Shared Nothing Architecture, Consistency, Availability, Partition Tolerance (CAP), Basically Available Soft State Eventual Consistency (BASE) Basic analytics, Advanced analytics, Operationalized analytics, Monetizing analytics. Modifying Business Intelligence Products to Handle Big Data, Studying Big Data Analytics			
Unit - III			
Hadoop Techniques:			10 Hours
Introducing Hadoop, RDBMS versus Hadoop, Hadoop versus SQL, Distributed Computing Challenges, Key Aspects of Hadoop, Hadoop Components. Features of Hadoop, Key Advantages of Hadoop, Versions of Hadoop, Hadoop 1.0, Hadoop 2.0, Overview of Hadoop Ecosystems, High Level Architecture of Hadoop, Hadoop Distributed File System, HDFS Daemons, Anatomy of File Read, Anatomy of File Write, Processing Data with Hadoop, MapReduce Daemons, MapReduce Example, Managing Resources and Application with Hadoop YARN, Limitations of Hadoop 1.0 Architecture, HDFS Limitation, Hadoop 2: HDFS, Hadoop 2 YARN: Taking Hadoop Beyond Batch, Hadoop Ecosystem. Introduction to NoSQL, NewSQL Uses, Features and Types, Disadvantages and Application of NoSQL,			

Unit - IV																																																																																							
Cloud Computing definition, Cloud service models (IaaS, PaaS & SaaS) :										10 Hours																																																																													
Cloud deployment models (Public, Private, Hybrid and Community Cloud), Private & Public Cloud Definition, Characteristics of Private Cloud, Private Cloud deployment models, Private Cloud Building blocks namely Physical Layer, Virtualization Layer, Cloud Management Layer, When to opt for Public Cloud, Public Cloud Service Models.																																																																																							
Virtualization																																																																																							
Basics of virtualization, Virtualization technologies, Server virtualization, VM migration techniques, Role of virtualization in Cloud Computing.																																																																																							
Unit - V																																																																																							
Cloud Tools and Applications:										10 Hours																																																																													
Aneka PaaS; Open stack: Introduction to open stack; Components of open stack; Amazon web services; Google AppEngine; Microsoft Azure; Scientific applications: Healthcare; Biology; Geo-Science, Business and Consumer applications: ARM & ERP; Productivity; Social networking.																																																																																							
Course Outcomes: At the end of the course student will be able to																																																																																							
1.	Classify the digital data																																																																																						
2.	Understand the top challenges facing big database																																																																																						
3.	Study components and advantages of Hadoop																																																																																						
4.	Understand Cloud deployment models																																																																																						
5.	Identify enabling technologies of cloud computing.																																																																																						
Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes																																																																																							
<table><tr><td>Program Outcomes→</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>PSO</td><td>PSO</td></tr><tr><td>↓ Course Outcomes</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td>1</td><td>2</td></tr><tr><td>25MCA303.1</td><td>-</td><td>3</td><td>-</td><td>-</td><td>3</td><td>-</td><td>3</td><td>3</td><td>1</td><td>1</td></tr><tr><td>25MCA303.2</td><td>-</td><td>3</td><td>3</td><td>3</td><td>3</td><td>2</td><td>3</td><td>3</td><td>2</td><td>1</td></tr><tr><td>25MCA303.3</td><td>-</td><td>3</td><td>3</td><td>3</td><td>3</td><td>-</td><td>3</td><td>3</td><td>3</td><td>3</td></tr><tr><td>25MCA303.4</td><td>-</td><td>3</td><td>-</td><td>3</td><td>3</td><td>2</td><td>3</td><td>3</td><td>2</td><td>3</td></tr><tr><td>25MCA303.5</td><td>-</td><td>3</td><td>2</td><td>2</td><td>2</td><td>3</td><td>3</td><td>3</td><td>1</td><td>2</td></tr></table>											Program Outcomes→	1	2	3	4	5	6	7	8	PSO	PSO	↓ Course Outcomes									1	2	25MCA303.1	-	3	-	-	3	-	3	3	1	1	25MCA303.2	-	3	3	3	3	2	3	3	2	1	25MCA303.3	-	3	3	3	3	-	3	3	3	3	25MCA303.4	-	3	-	3	3	2	3	3	2	3	25MCA303.5	-	3	2	2	2	3	3	3	1	2
Program Outcomes→	1	2	3	4	5	6	7	8	PSO	PSO																																																																													
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25MCA303.4	-	3	-	3	3	2	3	3	2	3																																																																													
25MCA303.5	-	3	2	2	2	3	3	3	1	2																																																																													
1: Low 2: Medium 3: High																																																																																							
TEXT BOOKS:																																																																																							
1.	Seema Acharya and Subhashini C: Big Data and Analytics, 2 nd Edition, Wiley India Pvt. Ltd, 2015																																																																																						
2.	Cloud Computing: Principles and paradigms By Raj Kumar Buyya, James Broberg, Andrezei M. Goscinski, 2013																																																																																						
REFERENCE BOOKS:																																																																																							
1.	Judith Hurwitz, Alan Nugent, Fern Halper, Marcia Kaufman: Big data for dummies																																																																																						
2.	Tom White: Hadoop – The Definitive Guide																																																																																						
3.	Chuck Lam: Hadoop in action																																																																																						
4.	Cloud Computing: A Practical Approach by J.Vette, Toby J. Vette, Robert Elsenpeter (Tata McGraw Hill)																																																																																						
5.	Virtualization Complete: Business Basic Edition" by Anil Desai (First Edition)																																																																																						
E Books / MOOCs/ NPTEL																																																																																							
1.	Big Data computing-nptel																																																																																						
2.	Cloud Computing and Big Data, C. Catlett, W. Gentzsch, L. Grandinetti, IOS Press, 2013																																																																																						

REACT NATIVE AND KOTLIN FOR MOBILE APPLICATION DEVELOPMENT			
Course Code	25MCA311	Course Type	PEC
Teaching Hours/Week (L: T: P: S)	3:0:0:0	Credits	03
Total Teaching Hours	40	CIE + SEE Marks	50+50
Teaching Department: Master of Computer Applications			
Course Objectives:			
1.	Assess the difference between native, hybrid and cross-platform application development		
2.	Develop applications based on user controls and apply the debugging and testing functionalities		
3.	Understand the fundamentals of Kotlin and Understand the variables, operators, control flow, and lambda.		
4.	Understand and Analyze Object Oriented capabilities and the UI components in Kotlin		
5.	Understand the concept of internal storage, external storage and SQLite concepts.		
UNIT-I			
Introduction			04 Hours
Different types of Mobile app development platforms: Native, Hybrid, Web applications, Introduction to React Native, React Native: Information flow, Architecture, Threading model. Setting up an environment for developing iOS and Android apps, Introduction to JSX, First Application with create-react-app and Expo, React lifecycle methods, The folder structure			
Basic Components			04 Hours
React Native Basic components: View, Layouts, Touch events, Accessibility, Text, StatusBar, Images, and media. Basic user interaction: Button, List View, Handling user input, TextInput, Additional controls			
UNIT-II			
Debugging and Testing React Native:			05 Hours
Debugging your React Native apps, Remote debugging, Logging, Inspecting React Native components. Testing: Introduction to the Jest testing framework, Snapshot testing your React Native components, working with functions, mocking modules			
Platform APIs:			03 Hours
Using Geolocation, Accessing the User's Images and Camera, Storing Persistent Data with AsyncStorage. Modules and Native Code.			
UNIT-III			
Android Introduction			02 Hours
Setting up Kotlin Environment (Android). Environment Familiarization: - User Interfaces, AVD, Emulator, Testing, Android Studio Code Editor. Android Architecture.			
Kotlin			06 Hours
Kotlin: - Kotlin & Java, Java to Kotlin, Kotlin & Android Studio, Semi-Colon in Kotlin. Kotlin Datatypes, Kotlin Variables, Variable Declaration, Type Annotation & Inference, Nullable, Safe Call Operator, Not Null Assertion, Nullable Type and let Function, Elvis Operator, Type Casting & Checking. Operators & Expression: - Expression, Assignment, Arithmetic, Augmented Assignment, Increment & Decrement, Equality, Boolean Logical, Range & Bitwise. Flow Control (Loops & condition). Functions & Lambda: - Type, Return Values, Parameters, Variable Number, Lambda, Higher Order.			

Basics Of OOP's, Inheritance & Sub classing, Activity Life Cycles, Activity State Changes, Save & Restore State.											
UNIT-IV											
Core Components and UI Handling										08 Hours	
Views, View Groups & Layouts, Android Event Handling, Multi Touch Event Handling, Fragments, Menus & Overflow Menus. Activities, Intents, Broadcast (Intents & Receivers), Content Providers, Manifest, Resource, Service - Thread & Async Task, Started & Bound Service.											
UNIT-V											
Storage										05 Hours	
Shared Preferences, Internal and External storage, Introduction to SQLite Database, CRUD operations using SQLite Database.											
Applications of android classes										03 Hours	
Introductions to Notifications, Using sound vibration and lights, Video Recording & Image Capture, Media Player.											
Course Outcomes: At the end of the course student will be able to											
1.	Differentiate between native, hybrid, and cross-platform mobile application development approaches based on performance, development complexity, and platform compatibility.										
2.	Design and implement user-interactive mobile applications using Kotlin and React Native, incorporating debugging and testing techniques to ensure performance and reliability.										
3.	Demonstrate understanding of Kotlin programming fundamentals including variables, data types, operators, control structures, functions, and lambda expressions.										
4.	Apply object-oriented programming principles in Kotlin to build modular mobile applications and implement UI components using Android's XML and programmatic layouts.										
5.	Implement mobile data persistence techniques using internal storage, external storage, Shared Preferences, and SQLite databases in Kotlin-based Android applications.										
Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes											
Program Outcomes→		1	2	3	4	5	6	7	8	PSO	PSO
↓ Course Outcomes										1	2
25MCA311.1		3	2	-	-	-	-	-	-	3	-
25MCA311.2		-	3	3	-	2	-	-	-	2	3
25MCA311.3		3	3	-	2	-	-	-	-	3	-
25MCA311.4		-	3	3	-	3	-	-	-	2	3
25MCA311.5		-	2	-	2	3	-	2	-	-	3
1: Low 2: Medium 3: High											
TEXTBOOKS:											
1.	React Native - Building Mobile Apps with JavaScript, Vladimir Novick, PACKT Publishing Limited, 2017										
2.	Building Native Mobile Apps with JavaScript, Bonnie Eisenman, Learning React Native, 2nd Edition, O'Reilly Media Inc., 2017										
3.	Kotlin for Android Developers: Learn Kotlin while developing an Android App, Antonio Leiva , CreateSpace Independent Publishing, 2016										
4.	Android Development with Kotlin, Marcin Moskala, Igor Wojda , Packt Publishing										

REFERENCE BOOKS:	
1.	Mastering React Native, Eric Masiello, Jacob Friedmann, Packt Publishing, 2017
2.	Programming Kotlin, Stephen Samuel, Stefan Bocutiu, Packt Publishing Limited, 2017
3.	Mastering Android Development with Kotlin, Milos Vasic, Packt Publishing Limited, 2017

ANGULAR AND SPRING BOOT			
Course Code:	25MCA312	Course Type	PEC
Teaching Hours/Week (L: T: P: S)	3:0:0:0	Credits	03
Total Teaching Hours	40	CIE + SEE Marks	50+50
Prerequisites :	Student must have basic knowledge of HTML, Java Script, Java		
Teaching Department: Master of Computer Applications			
Course Objectives:			
1.	Understand Single-page application architecture, the Building blocks of Angular and Components used in Angular		
2.	Understand Services and Dependency Injection in Angular, RxJS Observables, Routing and Navigation in Angular		
3.	Understand how to create Forms in Angular and HTTP-client Service in Angular		
4.	Understand Spring Core/DI-IOC, Spring MVC and Spring Boot		
5.	Understand creation of REST services in Spring, Spring Data JPA and creation of Microservices		
UNIT - I			
Single-page application architecture, Angular Building Blocks, Components			02 Hours
Single-page application architecture			
Single-page applications, Single-page application architecture, Introduction to TypeScript, Introduction to Angular, Building your first Angular app.			
Angular Building Blocks			03 Hours
Component architecture, Angular components, Understanding binding, Understanding directives, Pipes			
Components			03 Hours
Exploring components, Lifecycle hooks, Inter-component communication, View encapsulation, Change detection			
UNIT - II			
Services and Dependency Injection, RxJS Observables, Routing and Navigation			02 Hours
Services and Dependency Injection			
Introduction to services, Angular services in depth, Dependency injection			
RxJS Observables			03 Hours
Synchronous vs. reactive (asynchronous) programming, Event handlers, Observables, RxJS operators			
Routing and Navigation			03 Hours
Revisit single page application, Routing terminology, Sample application: Book-master/detail-demo, Route-params, Route order, Adding a redirect			
UNIT - III			
Forms in Angular, HTTP-client Service, Angular Modules and Standalone Components			03 Hours
Forms in Angular			
Overview of forms, Key differences, Data flow in forms, Form validations, Creating dynamic forms,			
HTTP-client Service			04 Hours
Overview of server communication, Requesting data from server, Configuring URL parameters, Sending data to the server, Handling request errors, Intercepting requests and responses			
Angular Modules and Standalone Components			02 Hours
Introduction to stand-alone components, Lazy loading, Bootstrapping with standalone component, Configuring dependency injection, Signals., Introduction to NgRx			

UNIT - IV											
Spring Core/DI-IOC, Spring MVC, Spring Boot										04 Hours	
Spring Core/DI-IOC											
Spring framework overview, DI/IOC container, XML configuration and annotation-based configurations, BeanFactory and ApplicationContext											
Spring MVC										02 Hours	
Introduction to Spring MVC, Understanding MVC architecture, Front controller design pattern, DispatcherServlet, Spring (Web) MVC architecture, Spring web components											
Spring Boot										02 Hours	
Java configuration, Understanding spring boot, Auto-configuration, Related annotations, Starters, Embedded containers, Externalized configuration											
UNIT - V											
Spring REST, Spring Data JPA, Microservices										02 Hours	
Spring REST											
Web services and REST introduction, REST architecture, Spring REST and related annotations, Spring REST, @RestController, Actions, Forwarding, Redirection											
Spring Data JPA										03 Hours	
Object relational mapping, Java Persistence API, JPARepository, @Repository, Configuration in application.properties											
Microservices										02 Hours	
Introduction to Microservices, Creating a Set of Cooperating Microservices, Deploying Our Microservices Using Docker, Adding Persistence, Developing Reactive Microservices											
Course Outcomes: At the end of the course student will be able to											
1.	Implement Single-page application architecture, the Building blocks of Angular and Components used in Angular										
2.	Implement Services and Dependency Injection in Angular, RxJS Observables, Routing and Navigation in Angular										
3.	Implement how to create Forms in Angular and HTTP-client Service in Angular										
4.	Implement Spring Core/DI-IOC, Spring MVC and Spring Boot										
5.	Implement creation of REST services in Spring, Spring Data JPA and creation of Microservices										
Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes											
	Program Outcomes→	1	2	3	4	5	6	7	8	PSO	PSO
	↓Course Outcomes									1	2
	25MCA312.1	3	2	2	3	3	3	2	3	3	2
	25MCA312.2	3	2	2	3	3	3	2	3	3	2
	25MCA312.3	3	2	2	3	3	3	2	3	3	2
	25MCA312.4	3	2	2	3	3	3	2	3	3	2
	25MCA312.5	3	2	2	3	3	3	2	3	3	2
1: Low 2: Medium 3: High											
TEXT BOOKS:											
1.	Full Stack Development with Angular and Spring Boot: Build scalable, responsive, and dynamic enterprise-level web applications Paperback – 18 July 2024 by Sangeeta Joshi										
2.	Hands-On Microservices with Spring Boot and Spring Cloud Paperback – 20 September 2019 by Magnus Larsson										

REFERENCE BOOKS:

1.	Learning Angular, by Aristeidis Bampakos
2.	Mastering Angular for Web Development: Build Dynamic, Scalable Web Apps with Angular Framework by Rafael Sanders
3.	Spring Start Here: Learn what you need and learn it well by Laurentiu Spilca
4.	Spring in Action, Sixth Edition 6th ed. Edition by Craig Walls
5.	Spring Boot: Up and Running: Building Cloud Native Java and Kotlin Applications 1st Edition by Mark Heckler
6.	Microservices with Spring Boot and Spring Cloud - Second Edition: Build resilient and scalable microservices using Spring Cloud, Istio, and Kubernetes 2nd ed. Edition by Magnus Larsson

UI/UX DESIGN & PRODUCT MANAGEMENT			
Course Code:	25MCA313	Course Type	PEC
Teaching Hours/Week (L:T:P:S)	3:0:0:0	Credits	03
Total Teaching Hours	40	CIE + SEE Marks	50+50
Teaching Department: Master of Computer Applications			
Course Objectives:			
1.	To adopt User centered design thinking and understand the User research methods.		
2.	Apply design thinking to empathise with users, define problem statements and ideate a product using wireframes.		
3.	Conduct Usability studies and learn fundamentals of Visual Design		
4.	Build a high fidelity prototype of your product, conduct user research, Test and iterate the design.		
5.	Understand the fundamentals of Product management.		
UNIT - I			
Fundamentals of UX Design:			04 Hours
Difference between UI and UX, Principles of Good Design, Design Sprints, Key Elements of UX. The UX Design Process: Stages of UX design Process: Research, Design, Prototyping, Testing, Implementation.			
UX research:			04 Hours
Types of User Research: Qualitative & Quantitative Integrating research into the design process, Choosing the right research method, Bias in UX research, Areas of Enquiry, Translating research into actionable insights.			
UNIT - II			
			04 Hours
Personas and User Journeys: Creating a Persona, Empathy Maps, Identifying user pain points, Using Personas in the Design Process, Creating User Stories and Scenarios, Problem statements, Hypothesis statements, Design ideation, Translating Requirements into Design Specifications Competitive audits.			
Interaction Design: Principles of Interaction Design, Designing User Interactions and Flows, Techniques for Interaction Design			
Build wireframes and Low Fidelity prototypes:			04 Hours
Difference Between Wireframes and Prototypes, User flow, Storyboarding, Paper wireframes, Digital wireframing using Figma, Avoiding deceptive patterns.			
UNIT - III			
			04 Hours
Visual Design: Key Principles of Visual Design, Typography, Color Theory, Images, Creating a Cohesive Visual Language.			
Accessibility in UX Design: Importance of Accessibility in UX Design, WCAG (Web Content Accessibility Guidelines), WCAG Levels of Compliance, Americans with Disabilities Act (ADA), Tools and Techniques for Designing Accessible Experiences.			
			04 Hours

Usability Testing: Types of Usability Testing, Planning and Conducting Usability Tests.
A/B Testing: Principles of A/B Testing, Designing and Running A/B Tests, Using A/B Test Results to Improve UX

UNIT - IV

High fidelity Prototype : **04 Hours**

Gestures and motion, Accessibility considerations, Build a high-fidelity prototype using Figma.

Maintaining and Evolving UX: **04 Hours**

Key Principles of Continuous UX Improvement, Gathering User Feedback Post-Launch, Iterating on Design Based on User Feedback and Analytics

UNIT - V

Product Management: **05 Hours**

Understanding Product, Product Manager Role and day-to-day Responsibilities, Product Development Lifecycle, Product Vision & Strategy Foundations, Product Requirements Document (PRD), Lean Product Development, Agile, Scrum, Kanban, Competitive and Market Analysis, Roadmapping, Minimum Viable Product.

Product Analytics: **03 Hours**

Acquisition metrics, Activation Metrics, Engagement Metrics, Retention Metrics, Revenue Metrics, Metrics Framework: HEART Metrics Framework, AARRR (Pirate) Metrics Framework

Course Outcomes: At the end of the course student will be able to

1.	Apply user centered design thinking to understand users through User Research.
2.	Use design thinking to empathise with users, define problem statements and ideate a product using wireframes.
3.	Conduct Usability studies and understand fundamentals of Visual Design
4.	Design a high fidelity prototype of a product by conducting user research.
5.	Understand how to build products, evaluate & implement them.

Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes

Program Outcomes→	1	2	3	4	5	6	7	8	PSO	PSO
↓ Course Outcomes									1	2
25MCA313.1	1	3	3	3	3	3	3	3	3	3
25MCA313.2	1	3	3	3	3	3	3	3	3	3
25MCA313.3	2	3	3	3	3	3	3	3	3	3
25MCA313.4	2	3	3	3	3	3	3	3	3	3
25MCA313.5	1	3	3	3	3	3	3	3	3	3

1: Low 2: Medium 3: High

TEXT BOOKS:

1. **The Ultimate UX Designer's Handbook: Master User-Centered Design, Use AI to Your Advantage, Build a Portfolio, Balancing Multiple Roles and Responsibilities ... to Excel in Your Role - By Valentina Stenhamn, Mixon Stenham, 2024**

2.	The Product Book -How to become a great Product Manager, by Product School, Josh Anon and Carlos Gonzalez De Villaumbrosia, 2nd Edition
REFERENCE BOOKS:	
1.	The Design of Everyday Things - Don Norman,by ABC Creations
2.	About Face: The Essentials of Interaction Design by Alan Cooper, Robert Reimann, David Cronin, Christopher Noessel
3.	Elements of User Experience, by Jesse James Garrett, Published by New Riders
4.	Lean UX: Designing Great Products with Agile Teams 2nd Edition,by Jeff Gothelf & Josh Seiden
E Books / MOOCs/ NPTEL	
1.	Google UX Design Professional Certificate, Google Career Certificates, Coursera
2.	Become a Product manager, Course on UDEMY, Instructor: Cole Mercer, Evan Kimbrell
3.	IBM Product Manager Professional Certificate, Coursera by John Rofrano, Daniel C. Yeomans, Matt Versdahl and Rav Ahuja.

HEALTH CARE ANALYTICS

Course Code	25MCA321	Course Type	PEC
Teaching Hours/Week (L: T: P: S)	3:0:0:0	Credits	03
Total Teaching Hours	40	CIE + SEE Marks	50+50
Prerequisite	Mathematical Foundation for computer Applications Data Warehousing and Data Mining		

Teaching Department: Master of Computer Applications

Course Objectives:

1.	Understand the role that data analytics plays in healthcare operations.
2.	Be familiar with the characteristics of healthcare data- associated data systems- and analytics tools.
3.	Understand the organizational needs and process to enact healthcare data analytics
4.	Understand the Biomedical Information Mining and clinical applications
5.	Mining of statistical model and visual analytics for health care.

UNIT-I

Fundamentals of Healthcare Analytics **06 Hours**

Components of healthcare analytics, Data and information, Measurement, Metrics and Indicators. From data to knowledge. Analytics building blocks. Descriptive, predictive, and prescriptive analytics. Healthcare analytics applications, an introduction.

Medical Images and Analytics **04 Hours**

Overview of Digital Imaging and Communications in Medicine (DICOM) standards. Biomedical image analysis, biomedical Imaging modalities, object detection, image segmentation, image registration- feature extraction.

UNIT-II

Mining of sensor data in healthcare **08 Hours**

Challenges of healthcare data analysis. Biomedical signal analysis, types of biomedical signals, correlation analysis.

UNIT-III

Genomic Data and Analytics **07 Hours**

Genomic data analysis for personalized medicine, genomic data generation, Microarray data, Next Generation Sequencing, Public Database, Predictive models for integrating clinical and genomic data, introduction, issues and challenges in integrating clinical and genomic data, different types of integration.

UNIT - IV

Biomedical Information Mining and Social Media Analytics **06 Hours**

Electronic Health Records (EHR), components of EHR, coding systems, benefits and challenges of EHR. HL7(Health Level 7). Natural language processing, core NLP Components, Morphological Analysis, Lexical Analysis, Syntactic Analysis, Semantic Analysis, Data Encoding, mining information from clinical text- challenges of processing clinical reports, clinical applications.

Social media analysis for healthcare **03 Hours**

Social media analysis for detection and tracking of infectious disease. Social Media Analysis for Public Health Research.

UNIT - V

Advanced Healthcare Data Analytics **04 Hours**

Basic statistical prediction models, Alternative clinical prediction models, Survival models, Evaluation and validation. Temporal data mining for healthcare data, Association analysis, Temporal pattern mining.

Visual analytics for healthcare										02 Hours	
Visual analytics for clinical workflow- visual analytics for clinicians and patients.											
Course Outcomes: At the end of the course student will be able to											
1.	Explain the fundamental concepts of healthcare analytics, including data measurement, analytics methodologies, and applications in healthcare decision-making										
2.	Analyze biomedical images and sensor-based healthcare data using signal processing, object detection, and feature extraction techniques										
3.	Evaluate genomic data and its integration with clinical information for predictive modeling and personalized medicine										
4.	Apply data mining, NLP, and social media analytics to extract insights from healthcare records and public health data.										
5.	Develop and validate healthcare predictive models using statistical techniques, temporal data mining, and visual analytics for clinical decision support.										
Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes											
	Program Outcomes→	1	2	3	4	5	6	7	8	PSO	PSO
	↓ Course Outcomes									1	2
	25MCA321.1	-	3	2	1	1	-	2	1	3	2
	25MCA321.2	-	3	2	1	1	-	2	1	3	2
	25MCA321.3	-	3	2	1	1	-	2	1	3	2
	25MCA321.4	-	3	2	1	1	-	2	1	3	2
	25MCA321.5	-	3	2	1	1	-	2	1	3	2
1: Low 2: Medium 3: High											
TEXTBOOKS:											
1.	C.K. Reddy and C.C. Aggarwal, Healthcare Data Analytics 2015.										
2.	Christo ElMorr and Hossam Ali-Hassan, Analytics in Healthcare A Practical Introduction, 2019.										
REFERENCE BOOKS:											
1.	Adam Bohr and Kaveh Memarzadeh, “Artificial Intelligence in Healthcare”, Academic Press, 2020.										
2.	H. Yang and E.K. Lee, Health care Analytics: From Data to Knowledge to Healthcare Improvement, 2016.										
3.	Richard Bibb, Dominic Egg beer and Abby Paterson, Medical Modelling, “The Application of Advanced Design and Rapid Prototyping Techniques in Medicine”, 2 nd Edition, 2015.										
E Books / MOOCs/ NPTEL											
1.	E-Book: Healthcare Analytics Made Simple.										
2.	BIG DATA ANALYTICS FOR HEALTHCARE-source: Source: www.coursera.org										
3.	http://www.hl7.org/										
4.	HL7 Messaging Standard Version 2.8.2 - http://www.hl7.org/implement/standards/product_brief.cfm?product_id=185										

GENERATIVE AI AND PROMPT ENGINEERING

Course Code	25MCA322	Course Type	PEC
Teaching Hours/Week (L: T: P: S)	3:0:0:0	Credits	03
Total Teaching Hours	40	CIE + SEE Marks	50+50
Prerequisite	Basic understanding of machine learning and programming in Python.		

Teaching Department: Master of Computer Applications

Course Objectives:

1.	To understand the evolution and principles behind Generative AI models.
2.	To implement deep learning-based generative models using TensorFlow.
3.	To apply LSTM and Transformer-based models for text and media generation.
4.	To explore prompt engineering techniques for interacting with LLMs.
5.	To design, tune and evaluate effective prompts using ethical practices.

UNIT-I

Fundamentals of Generative AI: **08 Hours**

Overview of Generative AI: discriminative vs generative models, Historical evolution: RBMs, VAEs, GANs, Mathematical foundations: probability, latent variables, Use cases: image synthesis, music, chatbots, TensorFlow setup, GPU usage, reproducibility

UNIT-II

Image Generation with VAEs and GANs : **08 Hours**

Variational Autoencoders (VAEs) and implementation, Generative Adversarial Networks (GANs): architecture and training, StyleGAN, pix2pix, deepfakes, Image-to-image translation and transfer

UNIT-III

Text, Music & Transformer-based Generation : **08 Hours**

Sequence models (RNN, LSTM) for text/music generation, Transformers and Attention mechanism, Fine-tuning pre-trained models (BERT, GPT-2), Creative AI: MuseGAN, game-play AI, hybrid architectures

UNIT - IV

Prompt Engineering Fundamentals : **08 Hours**

Introduction to Prompt Engineering, Types: zero-shot, one-shot, few-shot prompts, In-context learning and CoT (Chain-of-Thought) prompting, Role-based prompts and structure templates

UNIT - V

Advanced Prompting and Evaluation : **08 Hours**

Prompt tuning and optimization, RAG (Retrieval-Augmented Generation) and prompt chaining, Handling hallucination and ambiguity, Evaluation metrics (BLEU, ROUGE), ethics and safety

Course Outcomes: At the end of the course student will be able to

1.	Understand the foundational concepts of Generative AI and model categories.
2.	Implement VAEs and GANs for image generation using TensorFlow.
3.	Apply sequence models and transformers for text and media generation.
4.	Design and apply effective prompt engineering techniques.
5.	Evaluate, tune, and ethically manage prompt-based GenAI systems.

Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes											
Program Outcomes→	1	2	3	4	5	6	7	8	PSO	PSO	
↓ Course Outcomes									1	2	
25MCA322.1	3	2	--	--	--	--	--	--	3	--	
25MCA322.2	3	3	2	2	--	--	--	--	3	2	
25MCA322.3	2	3	3	3	2	--	--	--	3	3	
25MCA322.4	2	2	3	3	3	--	2	--	2	3	
25MCA322.5	3	2	3	3	3	--	2	2	3	3	
1: Low 2: Medium 3: High											
TEXTBOOKS:											
1.	Joseph Babcock, Raghav Bali, 'Generative AI with Python and TensorFlow 2', Packt Publishing, 2023.										
2.	Dennis Rothman, 'Prompt Engineering for Generative AI', O'Reilly Media, 2023.										
REFERENCE BOOKS:											
1.	Anirudh Koul, Siddha Ganju, Meher Kasam, 'Generative AI with LangChain, LLMs, and ChatGPT', Packt, 2024.										
2.	Tom Taulli, 'Generative AI: Navigating the Future of Artificial Intelligence', Wiley, 2023.										
3.	OpenAI Documentation – https://platform.openai.com/docs										
4.	Hugging Face Documentation – https://huggingface.co/docs										

NATURAL LANGUAGE PROCESSING			
Course Code	25MCA323	Course Type	PEC
Teaching Hours/Week (L:T:P:S)	3:0:0:0	Credits	03
Total Teaching Hours	40	CIE + SEE Marks	50+50
Teaching Department: Master of Computer Applications			
Course Objectives:			
1.	To introduce the fundamentals of NLP and provide an understanding of text preprocessing techniques such as tokenization, stemming, lemmatization, and handling noisy text.		
2.	To familiarize students with word representations and similarity techniques, including n-grams, TF-IDF, and word embeddings for semantic analysis.		
3.	To develop expertise in probabilistic language models, including Hidden Markov Models (HMM), Part-of-Speech (POS) tagging, and Named Entity Recognition (NER).		
4.	To explore syntactic parsing and machine translation, covering various parsing techniques and state-of-the-art transformer-based models such as BERT and GPT.		
5.	To apply NLP techniques in real-world applications, including information extraction, sentiment analysis, event detection, and model deployment.		
UNIT - I			
Introduction to Natural Language Processing:			04 Hours
History, Scope, and Challenges of NLP, Applications of NLP in Real-World Scenarios, NLP Pipeline and its Phases, Cross-lingual and Multilingual NLP			
Text Processing & Morphological Analysis:			04 Hours
Regular Expressions for Text Processing, Tokenization (Word, Sentence, Subword), Text Normalization (Lemmatization, Stemming), Handling Noisy Text (Stopword Removal, Lowercasing), Processing Indian Languages in NLP			
UNIT - II			
Spelling Correction & Text Similarity:			04 Hours
Minimum Edit Distance, N-Grams – Counting Words in Corpora, Simple (Un-smoothed) N-Grams, Applications of Spelling Correction (Autocomplete, Search)			
Word Representation & Embeddings:			04 Hours
Word Meaning Representation (Vector Semantics), Cosine Similarity for Measuring Word Similarity, TF-IDF: Weighing Terms in Documents, Introduction to Word Embeddings (Word2Vec, fastText)			
UNIT - III			
Language Models & Sequence Labelling:			04 Hours
Applications of Language Modeling (Speech Recognition, Chatbots), Named Entity Recognition (NER)			
Part of Speech (POS) Tagging:			04 Hours
English Word Classes & Tag Sets, Hidden Markov Models (HMMs) for POS Tagging, Viterbi Algorithm for Sequence Prediction			

UNIT - IV											
Syntactic Parsing										05 Hours	
Grammars and Syntax Structure, Parsing Strategies: Top-Down vs. Bottom-Up Tree Parsing, Depth-First vs. Breadth-First Parsing, Chart Parsers & Transition Networks											
Machine Transformer Models:										03 Hours	
Pre-trained Transformer Models (BERT, GPT, T5)											
UNIT - V											
NLP Applications:										04 Hours	
Information Extraction, Relation Detection and Classification, Sentiment Analysis, Text Summarization											
Temporal & Event Processing:										04 Hours	
Temporal Information Extraction, Event Detection and Classification, Template Filling for Event Processing											
Course Outcomes: At the end of the course, student will be able to											
1.	Understand the fundamentals of Natural Language Processing (NLP) and basic text preprocessing techniques.										
2.	Learn to perform finite-state morphological parsing, detect and correct spelling errors, implement n-gram models and understand Hidden Markov Models (HMM).										
3.	Implement text similarity and word representation techniques for NLP applications.										
4.	Develop language models and perform Part-of-Speech (POS) tagging using probabilistic methods										
5.	Apply NLP techniques for information extraction, event processing, and sentiment analysis, and understand their real-world applications										
Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes											
	Program Outcomes→	1	2	3	4	5	6	7	8	PSO	PSO
	↓ Course Outcomes									1	2
	25MCA323.1	3	3	3	3	-	-	3	3	3	3
	25MCA323.2	3	3	3	3	-	-	3	3	3	3
	25MCA323.3	3	3	3	3	-	-	3	3	3	3
	25MCA323.4	3	3	3	3	-	-	3	3	3	3
	25MCA33.5	3	3	3	3	-	-	3	3	3	3
1: Low 2: Medium 3: High											
TEXT BOOKS:											
1.	Jurafsky, D. and J. H. Martin, "Speech and language processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition", 2 nd Edition, Prentice Hall, 2008										
2.	Allen, James, "Natural Language Understanding", 2 nd Edition, Benjamin/Cumming, 1995										
REFERENCE BOOKS:											
1.	Steven Bird, S., Klein, E., Loper, E, "Natural Language Processing with Python-Analyzing Text with the Natural Language Toolkit", O'Reilly Media, 2010										
2.	Grant SIngersoll, Thomas S. Morton, and Andrew L. Farris," Taming text: how to find, organize, and manipulate it" Manning Publications Co., 2013										

3.	Feldman Ronen, and James Sanger,” The text mining handbook: advanced approaches in analyzing unstructured data”, Cambridge university press,2007
4.	Christopher D Manning, and Hinrich Schütze,” Foundations of statistical natural language processing”, MIT press, 1999
E Books / MOOCs/ NPTEL	
1.	http://tamingtext.com/2013/01/11/taming-text-print-copies-now-available/
2.	Youtube Lecture for Indian Language Processing: Natural Language Processing for Indic Languages – SciPy India 2020 https://web.stanford.edu/~jurafsky/slp3/

INDUSTRIAL AND MEDICAL IoT			
Course Code	25MCA331	Course Type	PEC
Teaching Hours/Week (L:T:P:S)	3:0:0:0	Credits	03
Total Teaching Hours	40	CIE + SEE Marks	50+50
Prerequisite	Data Communication and Networks		
Teaching Department : Master of Computer Applications			
Course Objectives:			
1.	Understanding the fundamental characteristics, physical and logical design, communication models, and relevant protocols of the Internet of Things		
2.	Understanding of the principles, architectures, and enabling technologies of IIoT and Industry 4.0, enabling them to analyze and apply these concepts in industrial settings		
3.	Students will understand how medical IoT and WBAN technologies can be used in healthcare		
4.	To equip students with the foundational knowledge and practical skills to understand, design, and program IoT endpoints using sensors, actuators, Raspberry Pi, and Arduino		
5.	Students will gain a broad understanding of IoT applications in homes, cities, industries, and healthcare		
UNIT - I			
Introduction to IoT:			04 Hours
Characteristics of IoT			
Physical Design of IoT:			
Things in IoT, IoT protocol			
Logical Design of IoT:			
Functional blocks of IoT, Communication models & APIs			
Developing IoTs:			04 Hours
IoT design methodology, Case study on IoT system for home automation, Case study on IoT system for weather monitoring			
UNIT - II			
Introduction and Architecture of IIoT & Industry 4.0			04 Hours
IIoT and Key Differences: IoT vs IIoT, Prerequisites of IIoT			
Industry 4.0: Design requirements, Drivers, Smart business perspective			
Architecture of IIoT: Categorization of reference architecture in IIoT, Industrial Internet Reference Architecture (IIRA) basics, framework			
Technologies of IIoT:			04 Hours
Off-Site Technologies: Necessity of Cloud Computing, Cloud Computing and IIoT, Cloud Computing Services in IIoT			
Fog Computing for IIoT: Applications and solutions			
On-Site Technologies:			
Augmented Reality: History, Categorization, Applications			
Virtual Reality: History, Categorization, Application			
Big Data and Advanced Analytics: Acquisition, Storage, Necessity of Data Analytics			

UNIT - III											
Introduction to Medical IoT										03 Hours	
Healthcare 4.0 & Smart Hospitals, MIIoT Devices: Wearables, Smart Implants, Remote Patient Monitoring											
Role of IoT in Wireless Body Area Networks (WBAN):										06 Hours	
Introduction to WBAN, Hardware architecture of biomedical sensors, Methods for deployment of biomedical sensors, Data dissemination of sensory information, Requirements of BMSs in WBAN, Applications of Medical Sensors, Data Dissemination Protocols in WBAN, Routing protocols, Medium access control protocols, Data Dissemination Architecture using Healthcare-IoT, Open Issues in WBAN											
UNIT - IV											
IoT Physical Device & Endpoint :										03 Hours	
Sensors, actuators, and smart objects: Definition and role of sensors in IoT, Classification of sensors, Characteristics of sensor, Types of sensors. Actuators: Definition and role of actuators in IoT. Smart Objects: Definition and characteristics of smart objects.											
Basic Building Blocks of an IoT Device										06 Hours	
Introduction to Raspberry Pi: Board details, Overview of Raspberry Pi models, Programming Raspberry Pi with Python: Python basics for IoT, GPIO programming using the RPi.GPIO, Interfacing sensors and actuators with Raspberry Pi using Python., Reading sensor data and controlling actuators, Working with external libraries											
Introduction to Arduino Board: Programming Arduino devices, C/C++ basics for Arduino, Arduino programming structure (setup(), loop()),Digital and analog I/O programming, Interfacing sensors and actuators with Arduino, Using Arduino libraries for sensors and actuators.											
UNIT - V											
Applications of IoT, IIoT and MIIoT :										06 Hours	
Applications of IoT: Case Study on Home Automation, Case Study on Smart Parking, Case Study on Smart Irrigation.											
Applications of IIoT: Smart Manufacturing, Smart Factory: Characteristics, Technologies used in Smart Factories											
Applications of MIIoT: Remote Patient Monitoring, Smart Hospitals											
Course Outcomes: At the end of the course student will be able to											
1.	Articulate the core concepts of IoT, including its characteristics, physical and logical designs, and the role of various communication protocols.										
2.	Able to analyze and apply the principles, architectures, and technologies of IIoT and Industry 4.0 to solve industrial challenges.										
3.	Understand how to use medical IoT and WBAN to create healthcare solutions, and they will be able to discuss the benefits and challenges of these technologies.										
4.	Learn to connect sensors and actuators, program Raspberry Pi and Arduino, and understand how IoT devices work										
5.	IoT is used in real-world situations like smart homes, hospitals and industries and will be able to understand and design these systems.										
Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes											
	Program Outcomes→	1	2	3	4	5	6	7	8	PSO	PSO
	↓ Course Outcomes									1	2
	25MCA331.1	3	1	1	1	1	1	1	1	2	3
	25MCA331.2	2	2	1	2	1	1	1	1	2	3
	25MCA331.3	2	2	1	2	1	1	1	1	2	3

	25MCA331.4	3	3	2	3	3	2	1	2	2	3
	25MCA331.5	3	2	3	2	2	1	2	2	2	3
1: Low 2: Medium 3: High											
TEXT BOOKS:											
1.	Vijay Madiseti, Arshdeep Bahga, “Internet of Things: A Hands -On Approach”										
2.	Valentina Emilia Balas et al., Internet of Things in Healthcare and Wellbeing, 2020										
3.	Introduction to Industrial Internet of Things and Industry 4.0" by Sudip Misra, Chandana Roy, and Anandarup Mukherjee, 2021										
REFERENCE BOOKS:											
1.	IoT Fundamentals: Networking Technologies, Protocols, and Use Cases for the Internet of Things" by David Hanes										
2.	IoT in Healthcare Systems: Applications, Benefits, Challenges, and Case Studies, 2024, Piyush Kumar Shukla, Aditya Patel, Prashant Kumar Shukla , Prashant Parashar										
3.	Hugh Boyes*, Bil Hallaq, Joe Cunningham, Tim Watson “The industrial internet of things (IIoT): An analysis framework”. Published by Elsevier										
E Books / MOOCs/ NPTEL											
1.	https://www.splunk.com/en_us/data-insider/what-is-the-internet-of-medical-things-iomt.html#iomt-and-telehealth										

.NET FRAMEWORK AND C#			
Course Code	25MCA332	Course Type	PEC
Teaching Hours/Week (L:T:P:S)	3:0:0:0	Credits	03
Total Teaching Hours	40	CIE + SEE Marks	50+50
Teaching Department: Master of Computer Applications			
Course Objectives:			
1.	Understand the .NET framework.		
2.	Understand the concept of class hierarchy and method overloading. Overriding abstract classes and interfaces exception handling I/O stream classes.		
3.	Understand the use of events and delegates Use of multi-threading.		
4.	Understanding multithreading and I/O stream concepts.		
5.	Data access with ADO.NET		
UNIT-I			
Introduction to the .NET Platform			03 Hours
Benefits of .NET Framework Architecture of .NET Framework 4.0 Components of .NET Framework 4.0: CLR CTS Metadata and Assemblies .NET Framework Class Library Windows Forms			
Introducing C#			04 Hours
Need of C# C# Pre-processor Directives Creating a Simple C# Console Application Identifiers and Keywords. Data Types Variables and Constants: Value Types Reference Types Type Conversions Boxing and Unboxing Variables and Constants. Operator Precedence Using the ?? (Null Coalescing)Operator Using the :: (Scope Resolution) Operator and Using the is and as Operators			
UNIT-II			
Classes Objects and Structures			02 Hours
Classes and Objects: Creating a Class Creating an Object Using this Keyword Creating an Array of Objects Using the Nested Classes Defining Partial Classes and methods Static Classes and Static Class Members Structures: Syntax of a struct and Access Modifiers for structs Enumerations.			
Arrays and Strings			01 Hours
Arrays Multidimensional arrays jagged arrays Assigning array references Foreach loop Strings of operator overloading			
Methods			01 Hours
Controlling access to class members Pass objects to methods Using ref and out parameters Using a variable number of arguments Returning objects Method overloading Constructors Garbage collection and destructors Overloading constructors.			
Properties Object- Oriented Programming Concepts			03 Hours
Read-only Property Static Property Accessibility of accessors and Anonymous types. Indexers Encapsulation: Encapsulation using accessors and mutators Encapsulation using Properties. Extension methods. Polymorphism: Compile time Polymorphism/ Overloading Runtime Polymorphism/ Overriding. Abstraction: Abstract classes Abstract methods. Interfaces: Syntax of Interfaces Implementation of Interfaces Interfaces and Inheritance			

UNIT-III	
Delegates Events and Exception Handling	05 Hours
Delegates: Creating and using Delegates Multicasting with Delegates. Events: Event Sources Event Handlers Events and Delegates Multiple Event Handlers. Exception Handling: System exception class Exception handling fundamentals Consequences of an uncaught exception Using multiple catch statements Catching all exceptions Nested try blocks Throwing an exception Using finally Commonly used exceptions.	
Namespaces the Preprocessor and Assemblies	03 Hours
Namespaces Preprocessor Assemblies and the internal access modifier; Runtime Type ID Reflection and Attributes: Runtime type definition Reflection Using Reflection Attributes Using the built-in attributes.	
UNIT - IV	
Multithreaded Programming	03 Hours
Multithreading Fundamentals Thread class Determining the end of Thread IsBackground property Thread priorities Synchronization Thread communication using Wait() Pulse() PulseAll() UsingMethodImplAttribute Suspending resuming stopping threads Determining threads state.	
Working with Collections, Building Components	02 Hours
Collections overview, Collection interfaces, Dictionary entry structure, General-purpose collection class, Storing bits and BitArray, Specialized collections, Accessing a collection using an enumerator. Introduction to components, Overview of C# component, IComponent, Simple component, Overriding Dispose(), Employing the using statement, Containers.	
Using I/O	02 Hours
C#'s I/O built in streams Stream classes Console I/O FileStream and Byte-Oriented file I/O Character based file I/O Redirecting the standard streams Reading and writing binary data Using memory stream Using StreamReader and StreamWriter.	
Unit-V	
Graphical User Interface with Windows Forms	04 Hours
Handling Mechanism Another Way to Create Event Handlers Locating Event Information.Introduction Windows Forms Event Handling: A Simple Event- Driven GUI Visual Studio Generated GUI Code Delegates and Event- Control Properties and Layout Labels TextBoxes and Buttons GroupBoxes and Panels CheckBoxes and RadioButtons ToolTips Mouse-Event Handling Keyboard-Event Handling	
Database Access with ADO.NET	04 Hours
Data Access with ADO.NET: The need for ADO.NET Faces of ADO.NET Role of ADO.NET Data Providers Understanding the ADO.NET Namespaces Types of System.Data Examining the DataColumn type Examining the DataRow Type Details of the DataTable Building a complete DataTable Understanding the DataView type Understanding the role of the DataSet Expressing relations using the DataRelation type Building a simple Test Database Selecting a Data Provider Types of the System.Data.OleDb Namespace Working with the Connected Layer of ADO.NET Working with the OleDbDataReader Inserting updating and deleting records using OleDbCommand Executing a stored Procedure using OleDbCommand Disconnected layer and the OleDbDataAdapter type Working with the SQL Data Provider.	

Web Development Using .Net										03 Hours																																																																														
Web Applications Using MVC.: Understanding the MVC Pattern Creating Controllers and Actions Routing Mechanisms.																																																																																								
Creating Controllers and Actions: Guides on defining controllers and their action methods to handle HTTP requests.																																																																																								
Routing Mechanisms: Details how routing works in ASP.NET Core MVC to map URLs to controller actions.																																																																																								
Course Outcomes: At the end of the course student will be able to																																																																																								
1.	To understand .net framework and C# program structure																																																																																							
2.	To perform polymorphism and to use abstract classes																																																																																							
3.	Use memory stream string reader and string writer, delegates and events																																																																																							
4.	To create multiple threads communication																																																																																							
5.	To understand .net framework and C# program structure																																																																																							
Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes																																																																																								
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TEXTBOOKS:																																																																																								
1.	Herbert Schildt, The Complete Reference C#, TATA McGraw-Hill edition																																																																																							
2.	Andrew Troelsen, C# and .NET Platform, Apress,																																																																																							
REFERENCE BOOKS:																																																																																								
1.	Ben Albahari, Peter Drayton and Brad Merril, "C# Essentials".																																																																																							
2.	"Microsoft C# Language Specifications", WP Publishers and Distributors Pvt. Ltd.																																																																																							
E Books / MOOCs/ NPTEL																																																																																								
1.	The little ASP.NET core book by Natte barbettini.																																																																																							
2.	C# 7.0 in a Nutshell: The Definitive Reference																																																																																							

DIGITAL AND SOCIAL MEDIA MARKETING			
Course Code:	25MCA333	Course Type	PEC
Teaching Hours/Week (L:T:P:S)	3:0:0:0	Credits	03
Total Teaching Hours	40	CIE + SEE Marks	50+50
Teaching Department: Master of Computer Applications			
Course Objectives:			
1.	Understand the concept of Digital marketing and Content Marketing.		
2.	Describe critical components of Social Media Marketing and digital advertising.		
3.	Understand how to market your product using email communications and social media influencers.		
4.	Apply Search Engine optimization techniques to improve the digital presence.		
5.	Apply key metrics in Digital Marketing for tracking and measuring success		
UNIT - I			
Introduction:			04 Hours
Understanding Digital Marketing, Traditional Marketing V/s Digital Marketing. Digital Marketing Funnel, Digital Marketing Techniques,Digital Marketing Channels,Digital Marketing Objectives, Digital Marketing Strategies.			
Content Marketing Framework			04 Hours
Techniques of Well-Developed Content Marketing, Full Marketing Funnel for content marketing, Create Intent-Based Content, Ascension Focused Content Marketing, Segmented Content Marketing, Complete Content Marketing is Cross Channeled,Smart Content Development Tool, Content Marketing Plan, Tactics to Measure Content Marketing Success			
UNIT - II			
Digital Advertising			04 Hours
Paid Traffic, Best Sources for Paid Traffic, How to Implement Digital Advertising Easily, Concepts of Evaluating Your Target Audience, Five Elements of High Performing Advertisement Campaigns, How to Implement and Scale Your Ad Campaign, Metrics to Measure Your Digital Advertising Campaign			
Critical Components of Social Media Marketing:			04 Hours
Social Listening, Social Influencing, Social Networking, Social Selling, Social Media Success Metrics			
UNIT - III			
E-mail Marketing			04 Hours
Roles of E-mail Marketing, Methods of Well-Executed E-mail Marketing, Types of Effective Marketing E-mails, The Phases of E-mail Marketing, The Metrics to Measure E-mail Marketing			
The art of influencer marketing			04 Hours
Why use influencers for marketing, Influencer types, The rise of the virtual influencer, Selecting social media influencers,Collaborating with social media influencers.			
UNIT - IV			
SEO strategy for a digital presence			08 Hours

The importance of search engines, Working with search engine algorithms, The ethics of search engine optimisation, Competitor and market research, Ethical and sustainability considerations, How search engines work, Implementing search engine optimisation, Keyword clustering and cannibalisation, Measuring SEO strategy efforts.

UNIT - V

Key Metrics in Digital Marketing for Tracking and Measuring Success

04 Hours

Why Data Is Important, Categories of Metrics in Digital Marketing, Measuring Your Digital Marketing Metrics Successfully, Digital Marketing Metrics You Need to Track and Measure, Traffic Metrics, Conversion Metrics, Revenue Metrics

Conversion Rate Optimization (CRO)

04 Hours

Conversion Rate Optimization (CRO), Why Is CRO So Important? ,How to Perfectly Execute CRO, The Metrics

Course Outcomes: At the end of the course student will be able to

1.	Describe the concept of Digital marketing and Content Marketing
2.	Identify critical components of Social Media Marketing and digital advertising.
3.	Summarize different social media and marketing communication methods.
4.	Apply Search Engine optimization techniques to improve the page rankings.
5.	Analyze digital marketing using different metrics for tracking and measuring success

Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes

Program Outcomes→	1	2	3	4	5	6	7	8	PSO	PSO
↓ Course Outcomes									1	2
25MCA333.1	-	3	3	3	-	3	-	3	3	3
25MCA333.2	-	3	3	3	-	3	-	3	3	3
25MCA333.3	-	3	3	3	-	3	-	3	3	3
25MCA333.4	3	3	3	3	-	3	-	3	3	3
25MCA333.5	3	3	3	3	-	3	-	3	3	3

1: Low 2: Medium 3: High

TEXT BOOKS:

1. “Digital Marketing Essentials: You always wanted to know”, Vibrant Publishers, 2020
2. Heinze, Aleksej; Fletcher, Gordon; Cruz, Ana; Fenton, Alex (2024-08-12T22:58:59.000). Digital and Social Media Marketing: A Results-Driven Approach (p. xiii). Taylor & Francis 3rd Edition

REFERENCE BOOKS:

1. Andrew T. Stephen, Yakov Bart, “Social Media Marketing- Principles and Strategies”, Stukent, 2015
2. Alex Fenton, Mostafa Mohamad, Ashley Jones “Digital and Social Media Marketing: A Results-Driven Approach”, 2nd Edition, Rottledge.

E Books / MOOCs/ NPTEL

1. Digital Marketing - By Shainesh G | Indian Institute of Management Bangalore (IIMB), NPTEL, 2025

FULL STACK WEB DEVELOPMENT USING MERN LAB										
Course Code			25MCA304			Course Type			PCC	
Teaching Hours/Week (L: T: P: S)			0:0:4:0			Credits			02	
Total Teaching Hours			26			CIE + SEE Marks			50+50	
Teaching Department: Master of Computer Applications										
Course Objectives:										
1.	Understand the need for server side rendered apps.									
2.	Use a JavaScript package manager (either npm or Yarn).									
3.	Making HTTP requests to read or modify data.									
4.	Students can understand how callbacks and MongoDB function.									
5.	Students able to create CRUD operations, routing mechanisms, and UI components.									
List of Experiments										
1.	Design a resume of a job aspirant using React components like Classes and Functions. Style the resume by applying CSS.									
2.	Build Student Registration Portal using entities like component, State and Props									
3.	Design an employee Management system using RESTFULL APIs in React.									
4.	Deploy connectivity between React and Node Application for Inventory Management system.									
5.	Demonstrate Node Js Application to perform CRUD operation for online Book Cart.									
6.	Development of an application using RESTFULL API's									
Course Outcomes: At the end of this course, the students will be able to:										
1.	Develop dynamic and interactive user interfaces using React by leveraging JSX transformation, state management, event handling, and efficient UI component integration for modern web applications.									
2.	Understand and implement full-stack web development by applying the concepts of MVC architecture, utilizing Node.js, Express, and MongoDB for backend development, and building RESTful APIs with CRUD operations									
Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes										
Program Outcomes→	1	2	3	4	5	6	7	8	PSO	PSO
↓Course Outcomes									1	2
25MCA304.1	3	3	3	3	3	2	1	1	3	3
25MCA304.2	2	3	3	3	3	1	1	1	3	2
1: Low 2: Medium 3: High										
REFERENCE BOOKS:										
1.	Vasan Subramanian – Pro MERN Stack: Full Stack Web App Development with Mongo, Express, React, and Node, 2 nd Edition, Apress, 2019.									
2.	Anthony Accomazzo, Ari Lerner, Nate Murray, Clay Allsopp, David Gutman, Tyler McGinnis – Full Stack React: The Complete Guide to ReactJS and Friends, Fullstack.io, 2017.									
3.	Alex Banks, Eve Porcello – Learning React: Modern Patterns for Developing React Apps, 2 nd Edition, O'Reilly Media, 2020.									
4.	Mario Casciaro, Luciano Mammino – Node.js Design Patterns: Master Best Practices to Build Modular and Scalable Server-Side Web Applications, 3 rd Edition, Packt Publishing, 2020.									
5.	Shivakumar Babji – Full Stack Web Development: HTML, CSS, JavaScript, Bootstrap, and PHP, Independently Published, 2023.									
6.	Alok Kumar – Practical Full Stack Machine Learning. BPB Publications. 2021.									

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING LAB											
Course Code				25MCA305			Course Type			PCC	
Teaching Hours/Week (L: T: P: S)				0:0:4:0			Credits			02	
Total Teaching Hours				26			CIE + SEE Marks			50+50	
Prerequisite				Foundation for computer Applications							
Teaching Department: Master of Computer Applications											
Course Objectives:											
1.	To understand and implement data preprocessing techniques.										
2.	To understand and implement Supervised and Unsupervised Machine Learning Algorithms.										
3.	To understand and implement Deep Learning Algorithms.										
4.	To understand and implement performance evaluation techniques for the model built.										
List of Experiments											
1.	Write basic programs (Conditional, branching, looping, Methods and modules, classes and objects, numpy, pandas, sklearn) using Python										
2.	Write Programs to demonstrate supervised and unsupervised learning										
3.	Write a program to demonstrate the working of the decision tree based ID3 algorithm. Use an appropriate data set for building the decision tree and apply this knowledge to classify a new sample.										
4.	Build an Artificial Neural Network by implementing the Back propagation algorithm and test the same using appropriate data sets.										
5.	Write a program to implement the Naive Bayesian classifier for a sample training data set stored as a .CSV file. Compute the accuracy of the classifier, considering few test data sets.										
6.	Assuming a set of documents that need to be classified, use the Naive Bayesian Classifier model to perform this task. Built-in Java classes/API can be used to write the program. Calculate the accuracy, precision, and recall for your data set.										
7.	Write a program to construct a Bayesian network considering medical data. Use this model to demonstrate the diagnosis of heart patients using standard Heart Disease Data Set. You can use Java/Python ML library classes/API.										
8.	Apply k-Means algorithm to cluster a set of data stored in a .CSV file. You can add Java/Python ML library classes/API in the program.										
9.	Write a program to implement k-Nearest Neighbor algorithm to classify the iris data set. Print both correct and wrong predictions. Java/Python ML library classes can be used for this problem.										
10.	Implement the non-parametric Locally Weighted Regression algorithm in order to fit data points. Select appropriate data set for your experiment and draw graphs										
Course Outcomes: At the end of the course student will be able to											
1.	Apply data preprocessing techniques, supervised and unsupervised machine learning algorithms.										
2.	Apply deep learning algorithms and implement performance evaluation techniques.										
Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes											
Program Outcomes→		1	2	3	4	5	6	7	8	PSO	PSO
↓ Course Outcomes										1	2
25MCA305.1		3	3	3	3	3	-	3	-	3	3
25MCA305.2		3	3	3	3	3	-	3	-	3	3
1: Low 2: Medium 3: High											

REFERENCE BOOKS:	
1.	Cosma Rohilla Shalizi, Advanced Data Analysis from an Elementary Point of View, 2015
2.	Tom M Mitchell, “ Machine Learning”, 1 st Edition, McGraw Hill Education, 2017.
3.	Elaine Rich, Kevin K and S B Nair, “Artificial Intelligence”, 3 rd Edition, McGraw Hill Education, 2017
4.	Introduction to Machine Learning with Python - A Guide for Data Scientists (Muller Andreas)
5.	Allen B. Downey, “Think Python: How to Think Like a Computer Scientist”, 2 nd Edition, Updated for Python 3, Shroff/O‘Reilly Publishers, 2016

MINI PROJECT											
Course Code	25MCA306	Course Type	PRJ								
Teaching Hours/Week (L:T:P: S*)	0:0:4:9	Credits	04								
* Self Learning											
Total Teaching Hours	48	CIE + SEE Marks	50+50								
Teaching Department: Master of Computer Applications											
Course Objectives:											
1.	To provide the student the experience of developing quality software solutions by involving in all the stages of the software development life cycle.										
2.	To learn to work in a team and present the work effectively.										
The objective of the MCA mini project work is to provide the student with the experience of developing quality software solutions by involving in all the stages of the software development life cycle like requirements engineering, systems analysis, systems design, software development, testing strategies and documentation with an overall emphasis on the development of reliable software systems. The primary emphasis of the project work is to understand and gain the knowledge of the principles of software engineering practices.											
Mini project can be assigned to an individual student or to a group having not more than 3 students. The mini project has to be carried out during the third semester of the MCA program. The CIE marks awarded for the Mini-project work, shall be based on the evaluation of the work done in every stage of the software development life cycle. A student has to submit a project report at the end of the semester. Contribution to the Mini-project and the performance of each group member shall be assessed individually in the semester end examination (SEE) conducted at the department											
Course Outcomes: At the end of the course student will be able to											
1.	Apply the principles of software engineering practices and develop a correct software for a real world problem.										
2.	Create the project and defend the work effectively.										
Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes											
	Program Outcomes→	1	2	3	4	5	6	7	8	PSO	PSO
	↓ Course Outcomes									1	2
	25MCA306.1	3	3	3	3	3	3	3	3	3	3
	25MCA306.2	3	3	3	3	3	3	3	3	3	3
1: Low 2: Medium 3: High											
REFERENCE BOOKS :											
1.	Roger S. Pressman — Software Engineering – A Practitioner’s Approach, 7 th Edition, Mc Graw-Hill International Edition										
2.	Ali Bahrami, “Object oriented systems development using the unified modelling language”, 1 st Edition, McGraw-Hill										
3.	John Sonmez, “Soft Skills: The software developer's life manual”										
4.	Ian Sommerville, —Software Engineeringll, 9 th Edition, Pearson Education Asia										

INTERNSHIP													
Course Code				25MCA401			Course Type			INT			
Teaching Hours/Week (L: T: P: S)				0:0:40:0			Credits			04			
Total Teaching Hours				160 Hours (4 Weeks)			CIE + SEE Marks			50+50			
Teaching Department: Master of Computer Applications													
Course Objectives:													
1.		To learn to integrate theory and practice and develop work habits and attitudes necessary for job success											
2.		To develop communication, interpersonal and other critical skills in the job interview process and build a record of work experience.											
All the students admitted to the MCA course shall have to undergo mandatory internship of 4 weeks during the vacation of II/III semester. A University examination shall be conducted during IV semester and the prescribed credit shall be included in IV semester. Internship shall be considered as a head of passing and shall be considered for the award of degree. Those, who do not takeup/complete the internship shall be declared fail and shall have to complete during subsequent University examination after satisfying the internship requirements.													
Course Outcomes: At the end of the course student will be able to													
1.		Integrate theory and practice and develop work habits and attitudes necessary for job success											
2.		Develop communication, interpersonal and other critical skills in the job interview process and build a record of work experience.											
Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes													
Program Outcomes→				1	2	3	4	5	6	7	8	PSO	PSO
↓ Course Outcomes												1	2
25MCA401.1				3	2	3	2	3	2	2	2	3	2
25MCA401.2				3	3	2	3	3	3	3	3	3	2
1: Low 2: Medium 3: High													

MAJOR PROJECT			
Course Code	25MCA402	Course Type	PRJ
Teaching Hours/Week (L: T: P: S)	0:0:40:0	Credits	15
Total Teaching Hours	600 Hours (15 weeks)	CIE + SEE Marks	100+200
Teaching Department: Master of Computer Applications			
Course Objectives:			
1.	to learn to develop quality and reliable software solution individually by involving in every phase of the software development life cycle, prepare a project report, a technical paper and present it effectively.		
2.	To learn to work in the industry as a professional.		
The Master of Computer Applications (MCA) programme prepares the students to take up positions as Systems Analysts, Systems Designers, Software Engineers, Programmers and Project Managers in any field related to information technology. The MCA students have been imparted with comprehensive knowledge and skills in core areas of computer science courses with equal emphasis on the theory and practical. The students pursuing MCA at NMAMIT, Nitte are encouraged to spend at least six months working on a project preferably in a software industry or any research organization.			
The objective of the MCA project work is to develop quality software solutions. During the development of the project, a student should involve in all the stages of the software development life cycle like requirements engineering, systems analysis, systems design, software development, testing strategies and documentation with an overall emphasis on the development of reliable software systems. The primary emphasis of the project work is to understand and gain the knowledge of the principles of software engineering practices, so as to participate and manage large software engineering projects in future.			
The project work should be carried out individually. The project work needs to be carried out with due care and should be carried out with all seriousness by the students. Topics selected should be complex and large enough to justify as a MCA project and 6 months of work. The project should be genuine and original and should not be copied from anywhere else. If found copied, the project report will be rejected and necessary action will be taken. The project work should compulsorily involve software development.			
Before finalizing a project, a student is required to discuss the details of the project with his/her internal guide and proceed only on approval from the internal guide.			
It is also mandatory for students to prepare a technical paper on the project and after the paper being scrutinized by their respective internal guides, publish the paper in a scientific journal.			
Course Outcomes: At the end of the course student will be able to			
1.	Develop quality and reliable software solution effectively by applying software engineering concepts.		
2.	Prepare a project report and give effective presentation		

Course Outcomes Mapping with Program Outcomes & Program Specific Outcomes

Program Outcomes→	1	2	3	4	5	6	7	8	PSO	PSO
↓ Course Outcomes									1	2
25MCA402.1	3	3	3	3	3	3	3	3	3	3
25MCA402.2	3	3	3	3	3	3	3	3	3	3

1: Low 2: Medium 3: High

REFERENCE BOOKS:

1.	Roger S. Pressman —Software Engineering – A Practitioner’s Approach, 7 th Edition, Mc Graw-Hill International Edition
2.	Ali Bahrami, “Object oriented systems development using the unified modelling language”, 1 st Edition, McGraw-Hill
3.	John Sonmez , “Soft Skills: The software developer's life manual”
4.	Ian Sommerville, Software Engineering, 9 th Edition, Pearson Education Asia